Hawaiʻi's Electricity Future: Solar Reform, Enhanced Geothermal, and the JERA LNG Proposal

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Abstract

This is a preliminary release, pre-v1.03, open for public comment; we ask that comments and suggestions arrive by September 15, 2026 (tentative; the repository README carries the current date). After the comment period and our responses, the report will be locked as version 1, and further suggestions will be directed to the v2 regional-grid model. This report supersedes the authors' 2026 working paper; the changes relative to it are documented in docs/CORRECTIONS.md. All dollar figures are real 2024 US$, present value as of 2027. Scenario results are at 0.1% optimization tolerance, with a handful of degenerate cells at 0.15% pending tighter re-solves (docs/HARD_CELLS.md) and the EGS cost sensitivity carried as a documented capital reprice (docs/SOLVER_NOTES.md).

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Executive Summary

This report asks what Hawaiian Electric and Hawaiʻi should build to keep Oʻahu's lights on through 2050 as the grid moves to 100 percent clean power, and what it will cost. We solve a capacity-expansion model, the planning tool utilities and regulators use to schedule decades of construction, several hundred times. Each run varies oil prices, solar and battery costs, land-use rules, the Waiau Repower, and the JERA LNG proposal. Every input traces to a public source or a labelled author assumption, and the full model, data, and code are public. The base case follows current federal law, including the storage and geothermal tax credits (48E) that survive the 2025 budget reconciliation act; we report a no-credit sensitivity for the case those credits are denied or repealed. Five findings organize the results. The two largest levers are both about how Hawaiʻi builds solar — on open land and on rooftops. The choices that dominate public debate matter less.

Finding 1. The largest lever is the price Hawaiʻi pays to build solar and batteries. Solar delivered here costs far more than on the mainland. The evidence points mainly to soft costs: long procurement cycles, permitting, interconnection queues, and customer acquisition. The size of the gap is hard to measure precisely. If solar and battery costs stay 50 percent above our baseline for the whole period, total system cost rises by about \$2.1 billion; at 70 percent above, by about \$2.8 billion. Our baseline already includes a 20 percent Hawaiʻi premium, so these cases correspond to roughly 1.8 and 2.0 times the mainland benchmark, near what recent procurement awards imply (Section 2.3). The lever is large because a big solar buildout happens on every pathway, with or without new fuel plants and largely even without the clean-energy mandate, so the price per installed watt multiplies across every gigawatt built. Reforms that close even part of the gap are worth more than any fuel decision in this report, and they are within the State's own control (Section 2).

Finding 2. The second-largest lever is rooftop solar and storage, and most of its value is available with today's technology and behavior. Oʻahu's rooftops carry about 793 MW of solar today, added at about 42 MW per year over 2020–2024. In 2024 the installed base reached the level Hawaiian Electric's 2016 plan projected for 2034 (Appendix A.13). We measure what this fleet does to the grid directly from utility demand records: each installed megawatt removes about 0.61 MW of midday grid demand, and home batteries shift about 45 percent of their storage capacity into the evening hours each day (Appendix A.11). Because rooftop supply substitutes almost one-for-one for utility-scale solar, its growth determines how much open land is needed for grid-scale solar. Limited further growth of rooftop solar leads to about 4,100 MW of utility-scale solar on about 20,400 acres by 2050; continuing the current trend requires only about 3,600 MW on about 18,000 acres. Removing the sellback limits could grow rooftop capacity much faster: pay households and businesses the full value of all the power they export, even when it exceeds the value of what they draw from the grid, instead of today's limited credits. If this doubled the installation pace, rooftop capacity would reach about 2,100 MW by 2050 and cut the utility-scale land requirement toward 15,000 acres (Section 2.7). That trajectory is an illustration. No one knows how much capacity would follow a tariff no one has yet offered, and even this path uses only about half of the mapped rooftop potential. It shows the direction and the scale: rooftop growth substitutes for open land.

The grid does not need sophisticated coordination to capture this value at today's scale. We tested this directly: we solved the same system twice — once with rooftop batteries following today's simple pattern, charging at midday and discharging in the evening, and once with the identical batteries dispatched optimally. At the base adoption path the difference is $0.03 billion over 2027–2050, about two-hundredths of a cent per kilowatt-hour. Predictable behavior is nearly as good as perfect coordination. But the value of coordination grows steeply with adoption. With 2.1 GW of distributed solar (the accelerated trajectory the sellback reform of Section 2.8 is designed to unlock), optimal dispatch of the rooftop batteries is worth about $0.23 billion (0.17 cents per kilowatt-hour) — a reason to build real-time price exposure into that reform from the start.

Finding 3. No new fuel plant pays for itself. If LNG comes to Oʻahu, its value is in the power plants the island already has. JERA proposes a new 500 MW plant fed by imported liquefied natural gas. We solve the proposal at both ends of JERA's own cost range and focus mainly on the middle. The low end is their construction estimate, which excludes items such as insurance, customs, design allowance, and contingency; the high end is their own +20 percent sensitivity, which restores them. At that midpoint the bundle costs $0.75 billion more than building no new fuel plant at reference oil prices, roughly half a cent per kilowatt-hour (Appendix A.1), within a range of $0.54 to $0.96 billion at that oil price. The gap is positive at every oil price tested: it widens to $1.63 billion on the market's low oil path and $1.21 billion on its high path (Table ES.1). The more useful comparison is between the plant and the fuel: whatever advantage LNG offers comes from the fuel, not the plant. The same modern plant burning today's fuel oil costs even more, while LNG burned in existing plants saves money. JERA's new plant, even in its best case — its lowest capital estimate, with solar at twice mainland cost — roughly breaks even. Converting the independent Kalaeloa plant alone saves $0.39 billion, and converting Hawaiian Electric's Kahe 5 and 6 and its Campbell Industrial Park (CIP) combustion turbine as well saves nearly three times as much, with no new construction. The savings hold up under conservative accounting: charging Hawaiian Electric's entire 2016 multi-island conversion budget ($450 million in today's dollars) against just the three Oʻahu units still leaves about $0.60 billion; adding a further $260 million of 2016 pipeline plans on top still leaves at least $0.34 billion (Sections 4.3, 4.7). Rooftop growth widens the gap against new plants: more rooftop supply leaves less demand for a new plant to serve.

The two LNG questions turn out to depend on oil prices in opposite ways, and separating them clarifies the choice. Converting existing plants is a fuel-switching decision: its value is the price gap between fuel oil and gas, multiplied by the volume burned. Because the proposed LNG contract is indexed to oil at a rate that nearly matches how fuel oil itself tracks crude, that gap stays near $6.60 per million Btu whether oil is cheap or expensive. What varies across oil worlds is how much fuel is left to switch and how much clean investment cheap gas defers, so conversion savings peak on the central oil paths and fade at both extremes (Section 4.1) — but they never depend on the per-unit gap being right. Building a new plant is an efficiency decision: its value is the fuel saved by a more efficient machine, which is worth more when fuel costs more. That value falls short at both ends of the range. When oil is cheap the plant runs hard but saves little, because the fuel it displaces is cheap. When oil is expensive solar and storage are a better option, and the plant runs at just 19 percent of capacity in 2035. So the LNG case, to the extent there is one, rests on switching fuel in plants Oʻahu already owns, and that case does not need an oil-price forecast to be right. Considerations outside the cost model point the same way. The proposal commits Oʻahu to a single supplier for two decades: twenty years of FSRU operation, under supply-contract terms not yet public that are likely written to protect the supplier, so the quoted price is best read as a floor. It defers roughly a decade of clean-energy deployment and likely increases greenhouse gas emissions. And it strands Hawaiian Electric's generation assets, a loss that falls on the utility's already fragile finances if capital recovery is disallowed, or on its customers if it is allowed (Sections 4 and 8).

Finding 4. Under current law, Enhanced Geothermal is in the least-cost build. Enhanced Geothermal, a newer form of geothermal power that creates its own underground reservoir rather than relying on natural steam, enters the cheapest build under the federal geothermal tax credit: the model develops the full ~100 MW of identified Oʻahu resource in the base case, lowering system cost by about $0.56 billion. At the optimistic cost projection the saving roughly doubles; at the pessimistic projection the model builds nothing and loses nothing, so the downside is bounded. Enhanced Geothermal is therefore part of the cheapest way to run the island, provided a first demonstration project proves the resource works here (Section 3) and development proceeds in partnership with Native Hawaiian communities (Section 3.5).

Finding 5. The constraints many people worry about mostly do not bind. Start with land. Every pathway that meets the 2045 mandate, including JERA's, builds nearly the same solar on nearly the same 20,400 acres, differing mainly in timing. The assumed eligible inventory exceeds what gets built, and flat-land reserves stand behind it — cropland that current law allows under a special use permit, military parcels, closed golf courses and quarries, and the built environment itself. The LNG question is largely separate from the land-use question: Hawaiian Electric's and the State Energy Office's own studies, like ours, find that LNG does little or nothing to change the ultimate solar buildout (Sections 2.5 and 2.6). Next, reliability. Every scenario keeps the lights on at every modeled hour, across weather drawn from the 2007–2008 record, including its single hardest day of low sun, weak winds, and an evening peak, plus a required reserve margin. Oʻahu does not need new thermal generation for reliability: the existing fleet, plus the planned 99 MW Puʻuloa engine plant, suffices. Some claim a high-renewable grid still needs enough firm capacity to meet peak demand. It does not, because storage moves energy within the day: firm resources need to cover only demand net of solar and wind averaged over the day, far below the momentary peak. It also helps that demand is generally lower on low-sun days (Section 5). Last, the mandate itself. Keeping the 2045 clean-energy requirement costs about $0.25 billion, roughly two-tenths of a cent per kilowatt-hour, because the cheapest path runs ahead of the requirement through 2040 anyway and builds 92 percent of the solar even if the requirement is removed. Abandoning it saves about $0.41 billion (about 0.3 cents per kilowatt-hour) if the replacement is a new gas plant and import terminal, and up to about $0.7 billion if existing plants are converted to LNG instead, the cheaper route Finding 3 identifies. We tested whether a new, more efficient plant earns its keep in that no-mandate world when the model may choose freely: it declines the 2030 plant entirely and adds at most about 250 MW around 2045, as the converted units age out, for a saving within the solve tolerance (Section 4.8). Either way, capturing this modest saving requires abandoning the clean-energy mandate, committing to a long-term contract, hoping (awkwardly) that rooftop and grid-scale solar don't overperform, and hoping the contract price, or better, holds throughout. That last hope looks dubious: today's spot price (JKM, the Asian spot-LNG marker, July 2026) hovers around $21–22/MMBtu, nearly double the reference contract price we assume (Section 4.8).

Also decisive. The Waiau Repower, Hawaiian Electric's proposal to rebuild its old Waiau oil units, raises system cost by $1.38 to $1.49 billion at every oil price tested, and every bundle that contains it inherits that penalty (Section 6). (This exceeds the literal cost of the plant because the cost of capital is higher than our discount rate.) The repower's six small units carry a resiliency argument the model leaves unpriced: no unit in v1 ever suffers a forced outage. Section 6.3 sizes that argument and invites comment on it. Pricing outages would cut hardest against the single-site 500 MW plant. If a new plant is built at all, 500 MW is larger than the system wants: a 375 MW version does modestly better in every configuration tested, and converting existing units beats both at JERA's quoted costs (Sections 4.1, 4.3, 4.7). One caution: JERA attributes its attractive quote partly to scale, so a smaller plant might not get the same price per megawatt.

Table ES.1 — Total 2027–2050 system cost by trajectory and oil price (present value, billions of 2024$; difference vs. no new fuel plant in parentheses; current-law base case, 0.1% solve tolerance). JERA rows show the midpoint of the bare construction estimate (bare-EPC: engineering, procurement, and construction, excluding contingency and similar items) and the +20% capital case; the band spans the two (Figure ES.1). The conversion row is shown at reference oil, the only oil path solved on this base-rooftop trajectory; Figure 4.3 solves the conversions across all four oil paths and three solar-cost levels on the trend-rooftop trajectory, where the saving peaks on the central paths and fades at both extremes (Section 4.1). The conversion row nets out the full 2016 conversion-program estimate ($450M in 2024$) as conversion capital; its bracketed value also adds the entire 2016 onshore-pipeline package ($260M), an extreme charge by construction (Section 4.7).

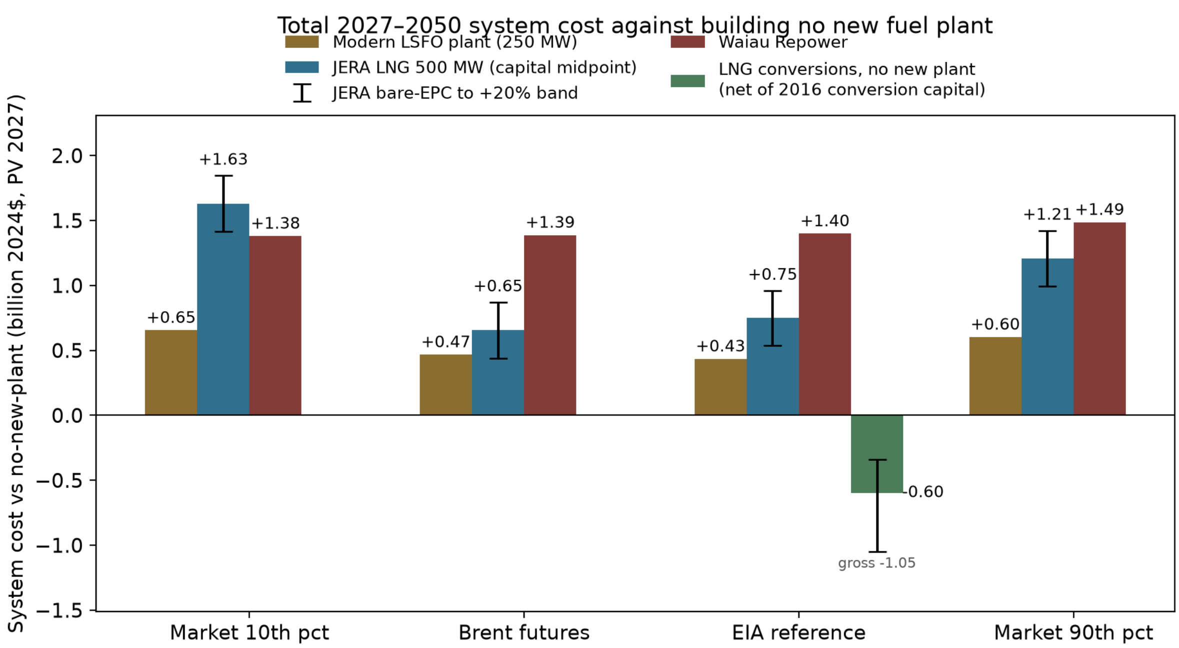

Figure ES.1 — Total 2027–2050 system cost of each proposed investment, measured against building no new fuel plant. Each bar is the present value of total system cost (billion 2024$, discounted to 2027) for a pathway including the named investment, minus the no-new-plant path solved under the same oil-price case. Oil cases: the market's 10th percentile, the Brent futures strip, the EIA-anchored reference, and the market's 90th percentile (Appendix A.14). Whiskers on the JERA bars span the plant's bare-EPC capital to +20 percent; the bar sits at the midpoint. The conversions bar (shown at reference oil; Figure 4.3 spans all four oil paths) is LNG conversion of existing plants with no new plant, net of a $0.45 billion conversion-capital charge benchmarked to HECO's 2016 program; its whiskers run from the gross fuel saving to a stricter bound that adds the 2016 onshore pipeline package. Bars above zero cost more than building no new plant.

Trajectory	Market 10th pct	Brent futures	EIA reference	Market 90th pct
No new fuel plant	21.16	24.30	25.83	27.32
LNG conversions, no new plant — net of conversion capital	—	—	25.24 (−0.60) [*−0.34*]	—
Modern LSFO plant (250 MW)	21.81 (+0.65)	24.77 (+0.47)	26.27 (+0.43)	27.93 (+0.60)
JERA LNG 500 MW — midpoint	22.79 (+1.63)	24.95 (+0.65)	26.58 (+0.75)	28.53 (+1.21)
...JERA band [bare-EPC, +20%]	[*+1.42, +1.84*]	[*+0.44, +0.87*]	[*+0.54, +0.96*]	[*+1.00, +1.42*]
Waiau Repower only	22.54 (+1.38)	25.69 (+1.39)	27.23 (+1.40)	28.81 (+1.49)
Waiau + LSFO plant	23.27 (+2.12)	26.23 (+1.93)	27.71 (+1.88)	29.41 (+2.09)
Waiau + JERA LNG — midpoint	24.28 (+3.12)	26.43 (+2.13)	28.06 (+2.23)	30.02 (+2.69)

Limits of inference. These findings are conditional on the cost trajectories, fuel-price regressions, sample-day reliability design, and institutional assumptions documented in the appendices, and on the menu of options tested. The model is a single-zone representation of Oʻahu: it does not price transmission upgrades or the locational value of distributed resources — a tradeoff the zonal grid model now under development will address (Section 8).

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1. Why this report now

1.1 The policy moment

Three electricity decisions are before the Hawaiʻi Public Utilities Commission and the State. They are treated as separate proceedings, but each shapes the costs the others will determine.

The first is the **Waiau Repower**, Hawaiian Electric's plan to replace six retiring oil-fired steam units at Waiau with six new fuel-flexible **simple-cycle combustion turbines** totaling about 253 MW, running on at least 51 percent renewable fuel at commissioning, 75 percent by 2040, and 100 percent by 2045 under the PUC order (Docket 2025-0211, Decision and Order 42411, March 2026). Hawaiian Electric's amended cost estimate was $1.155 billion; the Commission capped recoverable cost near the original $847 million bid (an absolute ceiling of $931.7 million), leaving roughly $220–310 million of the stated cost exposed to shareholders, depending on where allowed recovery lands within the cap. Section 6 develops both the system-cost and the cost-recovery questions.

The second is **JERA's proposal** to build a 500 MW combined-cycle power plant in Kapolei fueled by imported liquefied natural gas, delivered through a floating storage and regasification unit (FSRU) moored offshore Barbers Point, adjacent to Campbell Industrial Park. The proposal specifies twenty years of FSRU operation; the supply contract's term and volume provisions have not been made public. The publicly reported investment is about $2 billion in total, roughly 75 percent for the plant and 25 percent for the import infrastructure [ENR 2026; JERA proposal, March 17 2026]. Governor Green's office has cited the proposal's potential to reduce residential bills by about 20 percent; Section 1.2 examines that claim, and Section 4 the full system comparison. The Pacific market context matters for a commitment of this length: Japan's LNG demand has been declining and JERA is among the world's largest LNG resellers. In March 2026 JERA terminated a 20-year supply contract with Commonwealth LNG before construction began (Section 4.6).

The third is **retail wheeling** under 2025 Act 266, now in late-stage PUC implementation, a market-design reform that bears directly on the delivered cost of new solar, which Section 2 identifies as the largest lever in this report.

1.2 The "20 percent bill cut" claim

Roughly half of a residential bill pays for generation. The other half pays for wires, meters, and programs that cost the same under any fuel. In the near term the 2026 conflict in Iran and the Persian Gulf has raised oil prices, so fuel can temporarily run above half of generation cost. Over the planning horizon, fuel is about 37 percent of cost in the early years at reference oil prices, declining toward 8–10 percent by 2050 as renewables displace combustion (Appendix A.4). Even if every dollar of fuel spending captured LNG's actual price advantage over low-sulfur fuel oil — roughly $5 per million Btu, on a base of about $17 at reference oil prices (Table 1.1) — the realistic ceiling is a few percent of the bill, declining toward one percent. The 20 percent figure is unreachable under any reading of the same data. Section 4 tests the more generous "bundled" interpretation directly, with the same conclusion.

Table 1.1 — LSFO and LNG delivered cost at three paths of Brent, the benchmark world oil price, 2030 (2024$). LSFO uses the post-2024 R3 contract regression (slope 0.7388, intercept $37.30/bbl, 6.22 MMBtu/bbl); LNG is the HSEO/FGE-style contract price plus the regasification charge ($1.31/MMBtu at modeled utilization). Both fuels are priced off the same crude, so the gap between them barely moves across a Brent range of $23 to $145: the saving is about $5.2/MMBtu whatever oil does. Section 4.5 draws out what follows.

Long-run oil path	Brent ($/bbl, 2030)	LSFO ($/MMBtu)	LNG delivered ($/MMBtu)	Δ (LNG − LSFO)
Market 10th percentile	23	8.77	3.57	−5.20
Brent futures	58	12.87	7.64	−5.23
EIA reference	90	16.69	11.43	−5.26

Long-run oil path	Brent ($/bbl, 2030)	LSFO ($/MMBtu)	LNG delivered ($/MMBtu)	Δ (LNG − LSFO)
Market 90th percentile	145	23.26	17.96	−5.30

A five-dollar-per-MMBtu saving on the fuel is real. Whether it justifies the terminal, the contract, and the plant, against an alternative that burns steadily less fuel of any kind, is the system question Section 4 answers.

Table 1.1 and the fuel-price regressions use the R3 LSFO contract regression (Roberts 2026 brief) applied to the four Brent paths of Appendix A.14; values in 2024$.

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2. Solar and storage: the largest lever

2.1 What the model finds

At the baseline cost basis — NREL ATB 2024 Moderate for solar and battery, plus a 20 percent Hawaiʻi premium, with the battery's co-location saving taken from NREL's own PV-plus-battery hybrid — total 2027–2050 system cost is $25.83 billion at reference oil prices. Holding solar and battery costs 50 percent above that baseline for the full horizon raises system cost to $27.94 billion (+$2.1B); 70 percent above raises it to $28.65 billion (+$2.8B). The baseline already includes the 20 percent Hawaiʻi premium, so these cases correspond to roughly 1.8× and 2.0× the mainland ATB benchmark; the 1.5× case approximates the effective cost level implied by recently approved contracts (Section 2.3). We present evidence that the premium Hawaiʻi actually pays for grid-scale solar stems mainly from exceptionally high soft costs — categories that respond to policy reform.

For scale: the Waiau Repower decision moves system cost by about $1.4 billion; the LNG-versus-no-new-plant decision moves it by about $0.8 billion at its reference-oil midpoint. Solar-and-storage procurement reform is worth several times any fuel decision in this report.

2.2 Where the premium sits: soft costs

Hawaiʻi solar carries a documented premium over mainland deployments. The premium is concentrated in soft costs — the length of the procurement cycle, permit fees and timing, interconnection-queue position, customer-acquisition cost, and the risk premia developers attach to an RFP-and-PPA process with a history of post-award cancellations and delays. These are cost categories that respond to regulatory and market-design reform. Hardware follows global manufacturing trends Hawaiʻi cannot influence.

Two clarifications set the limits of this claim. First, the mainland benchmark is not hardware alone. NREL's utility-scale PV cost model, which underlies the ATB figures used here, builds capital cost from the ground up and already includes soft costs, among them permitting, inspection and interconnection, customer acquisition, developer overhead, and profit (NREL PV System Cost Benchmark, Ramasamy et al., Q1 2023, NREL/TP-87303, CAPEX component table). Bringing Oʻahu near mainland cost therefore means narrowing Hawaiʻi's excess soft cost, not removing a category every market carries. Second, the size of that excess is difficult to measure directly, because Hawaiʻi procurement records do not report these categories separately. The evidence below is correspondingly indirect. It shows that the fundamentals that could justify a large premium — hardware, labor, and land — do not appear to do so, leaving the process as the residual explanation. And the details of that process make it easy to see why it inflates costs. The retail-wheeling reform mandated by 2025 Act 266, now in late-stage implementation, could address the soft-cost gap directly by replacing the multi-stage procurement process with transparent market access to avoided-cost pricing. The pricing and interconnection details underlying the wheeling tariff will be essential. The evidence:

2.3 The cost-fundamentals evidence, in detail

The evidence in this section supports two claims: Hawaiʻi's residential solar costs look ordinary once system size is accounted for, and the large premium sits in utility-scale procurement, where it grew sharply through the recent inflation. One table summarizes it; the blocks that follow carry the sources and adjustments.

Evidence	Hawaiʻi	Mainland	Reading
Residential installed \$/W (2026)	\$3.14 Honolulu	\$2.11–2.39 Phoenix/ Houston/LA	lower edge of the national band; gap within normal state variation
Utility PPA, 2018–20 vintage (levelized 2024\$)	~7.7 ¢/kWh	~3.1 ¢/kWh	a premium, but both markets near lows
Utility PPA, 2024–26 vintage	~19.5 ¢/kWh (Mahi reprice)	~6.5 ¢/kWh	escalation mostly Hawaiʻi-specific
Common global shock	~+12 ¢/kWh	~+2.5–3.5 ¢/kWh	the excess is the procurement channel
Ground rent	a few tenths of a ¢/kWh	—	no scarcity rents; land is not the binding cost

Residential installed cost. EnergySage Marketplace (February–May 2026) puts the Honolulu average at \$3.14 per watt installed (\$29,233 before incentives on a typical 9.3 kW system), against \$2.11 in Phoenix, \$2.19 in Houston, and \$2.39 in Los Angeles. Tesla's published all-in residential pricing — designed to be comparable across U.S. markets — runs about \$2.27–2.82 per watt nationally; on that pricing Hawaiʻi runs slightly above California and below New York and Massachusetts. SolarReviews reports the Hawaiʻi residential average at \$2.82 with a \$2.14–3.20 range. LBNL's *Tracking the Sun* (2024) reports a national 20th–80th percentile band of roughly \$3.20–5.50 per watt (2023\$; ≈\$3.3–5.7 in 2024\$) and state fixed-effect spreads of roughly \$2 per watt: Honolulu sits at the lower edge of the national band, and its ~\$1/watt gap to Phoenix is within normal state-to-state variation. Hawaiʻi households also install smaller systems for the same annual load, so the total cost of a typical Honolulu system is comparable to one in Phoenix or Houston in absolute dollars.

Utility-scale procurement. Hawaiian Electric's Stage 1 awards (approved 2018–19) delivered four-hour solar-plus-storage on Oʻahu at 8–12 cents per kilowatt-hour (nominal award prices;

≈10–14 cents in 2024\$): Hoʻohana, Mililani I, Waiawa Phase 1, AES West Oʻahu. The contracts approved since 2024 price the same service at 21–23 cents: Mahi, a 2020-vintage project whose amended contract (Docket 2025-0414) roughly tripled its originally executed price, and Puʻuloa Solar, a new selection under review (a solar-plus-storage project, distinct from the 99 MW Puʻuloa engine plant of Section 4.6). Mainland prices also rose substantially over this period, roughly doubling from their 2019/20 lows: matched-configuration medians moved from about 3.1 to 6.5 cents per kilowatt-hour (LBNL *Utility-Scale Solar*, 2025 data file; levelized PPA prices in constant 2024 dollars; Box 2.1). On the same levelized 2024-dollar basis, Hawaiʻi's matched-configuration executions of 2018–2020 carried a median near 7.7 cents per kilowatt-hour, and the amended Mahi contract is roughly 19.5 cents. Because modules, batteries, and capital are priced in global markets, the common shock added roughly the same number of cents everywhere: about 2.5 to 3.5 cents per kilowatt-hour on the mainland at matched configuration, against roughly 12 cents for the Mahi repricing of an identical project. The additional escalation is specific to Hawaiʻi's procurement channel.

The clearest mechanism is the interaction of fixed nominal prices with slow interconnection. A Hawaiʻi PPA fixes its price in nominal dollars at award, with no adjustment between award and interconnection (Section 2.8). Award-to-operation has run four to seven years on Oʻahu: Mililani I about four, AES West Oʻahu about five, Hoʻohana seven, and two Stage 2 projects were still under construction six years after award. When inflation spiked in 2022 and stayed high and uncertain, each year of delay eroded the real value of an award. Developers respond by bidding high enough to absorb the risk, renegotiating, or walking away: of the 2020 Stage 2 round, Kupehau, Mehana, and Barbers Point were cancelled, and Mahi survived by repricing.

The technology itself continued to get cheaper. IRENA's Renewable Cost Database records a 90 percent decline in the global weighted-average levelized cost of utility-scale solar between 2010 and 2024, to about 4.4 cents per kilowatt-hour, and a 93 percent decline in battery storage costs over the same period. Its 2026 study of firm renewable power finds that solar sized with storage to deliver continuous supply fell about 30 percent in cost between 2020 and 2025 at high-quality sites, in real 2025 dollars, and projects further declines through 2035 (IRENA, *24/7 Renewables: The economics of firm solar and wind*, 2026, vendored in sources/).

The rise in United States prices over the same years was not a rise in the cost of the equipment. IRENA gives four reasons U.S. solar costs sit above China's: financing costs three to five percentage points higher, "reflecting higher perceived investment risk and more limited access to long-term, low-cost capital"; balance-of-system and infrastructure costs raised by "higher labour costs, more complex permitting regimes, and grid-connection requirements"; a pace of grid expansion and interconnection that "constrains deployment and adds project-level cost"; and added risk premiums. It notes that in many U.S. markets "interconnection charges are borne directly by project developers rather than socialised across the broader system." The interconnection backlog on the mainland reached about 2.6 terawatts at the end of 2023, 95 percent of it solar, wind, and storage, and the typical project now takes about five years to connect, against under two a decade earlier (LBNL, *Queued Up*, 2024). Rising real interest rates added to all of it, and they raise the cost of solar and storage by more than they raise the cost of a fuel-burning plant: a plant with no fuel recovers almost all of its cost as capital, and capital cost rises with the interest rate.

Hawaiʻi has the highest utility-scale solar prices in the country. Its grid is small, and it does not have a large interconnection backlog. Its interconnection and procurement process is slow and costly, and it applies to a system that already carries an unusually large amount of distributed solar. The procurement and interconnection costs that raised prices across the United States are larger in Hawaiʻi, and that is the reason its prices are highest.

Box 2.1 — How large is Hawaiʻi's utility-solar premium today? Careful measurements give answers from about 1.8 to 2.2 times mainland cost, and the spread comes mostly from how the comparison is set up, not from disagreement about Hawaiʻi. Three choices drive it.

What is compared. The model needs the unsubsidized cost of installed capital, benchmarked to NREL's ATB — that basis is what our multipliers mean. Most public discussion instead compares contract (PPA) prices, which fold in financing, configuration, and, most important, federal subsidies.

How subsidies enter. PPA prices are bid net of federal tax credits, and the credits differ across the two markets in a way that inflates the apparent gap. Recent mainland utility solar typically elects the production tax credit — a flat 2.75 to 3 cents per kilowatt-hour (§45Y), which covers a large share of a cheap mainland contract. Hawaiʻi's battery-heavy hybrids plausibly claim the 30 percent investment credit on both components instead (storage has no production-credit option, and the production credit favors low-cost, high-output projects — the opposite of Hawaiʻi's configurations), and because Hawaiʻi's prices are much higher, federal support covers a far smaller share of them. Comparing subsidized prices therefore overstates the underlying premium; adding the credits back to both sides shrinks the gap substantially. Configuration compounds this: Hawaiʻi hybrids pair four-to-five-hour batteries sized near 100 percent of PV capacity, far larger than the typical mainland hybrid, so comparisons that skip battery matching overstate the gap again. The mainland figures in this section are configuration-matched (battery at least 90 percent of PV capacity, four hours or longer) for that reason.

When. The nominal premium roughly doubled after 2020 (the table above), so any estimate mixing contract vintages picks its own answer.

On a consistent basis — unsubsidized, configuration-matched, and vintage-matched — the procurement record implies roughly 1.4 to more than 2.2 times mainland cost. Our 1.5× sensitivity (about 1.8× mainland) sits well above what the Phase 1 and Kauaʻi contracts imply; the State Energy Office's 2.154 multiplier sits near the top. The differences across these points are within the measurement noise of a handful of contracts. No conclusion in this report depends on the point estimate: results are solved at 1.2, 1.8, and 2.04 times the 2024 ATB baseline for unsubsidized solar and batteries, and stated across that span. Since 2024, battery costs — a disproportionately large share of a Hawaiʻi installation's cost — have fallen about 30 percent, making these multiples somewhat conservative.

One asymmetry deserves note: we apply no Hawaiʻi premium to any other resource — including offshore wind — which appears consistent with the assumptions of HSEO and Hawaiian Electric. The assumption is untested in both directions: there is no local procurement record to estimate an offshore premium from, and none of the process factors behind the solar premium — procurement cycles, permitting, interconnection — is specific to solar. If those factors persist, a resource with no construction record here and floating-platform engineering would be at least as exposed to them (Section 4.5). Where a single number is needed, Hawaiʻi currently pays roughly double the mainland price to deploy utility-scale solar — and the evidence of Section 2 is that this figure reflects process rather than geography.

Land and ground rent. Recent University of Hawaiʻi solar RFPs and contemporary utility-scale leases price ground rent at a few tenths of a cent per kilowatt-hour delivered — a trivial share of project cost. If developable land were the binding constraint, lease rates would show scarcity rents. We see none.

Federal incentives. The analysis credits no solar ITC (assumed phased out) and no state credits. Storage and geothermal credits were not phased out like the solar and wind credits, so the base case retains the 30 percent federal storage credit (48E) for construction beginning through 2033, phasing to zero for starts after 2035. This schedule — battery capital ×0.70 for 2027–2035 vintages, full price after — is in the base case; a no-credit sensitivity, which removes it, raises the no-new-plant system cost by about \$0.6 billion at reference oil (the cost of losing the credit to the foreign-entity rules or repeal). The expiring credit also reshapes the buildout the way an expiring credit should: the model pulls roughly 2,600 MWh of storage from the 2040s into 2035 and moves about 430 MW of co-located solar into 2030–2035, capturing the credit before it lapses.

The solar and wind tax credits are nearly expired, so it is worth asking what the State of Hawaiʻi has already lost by not streamlining interconnection of solar and batteries sooner. Even the most pessimistic land-availability estimates from the Hawaiʻi State Energy Office put grid-scale solar potential on Oʻahu at roughly 2 GW. At a 25 percent capacity factor, that capacity would deliver about 4.4 TWh per year.[1] That is roughly two-thirds of Hawaiian Electric's current sales, and more than all of its current oil-fired generation. More would still be needed over the long run, to displace the independent Kalaeloa plant and to meet demand growth.

Suppose that capacity had been built at 10 cents per kilowatt-hour, the price of the 2019 awards. The avoided fuel cost at reference Brent is about 17 cents per kilowatt-hour (Table 1.1 prices at the fleet's heat rates). The savings would have been roughly \$0.3 billion per year. At 5–6 cents, the price implied by mainland soft costs, the savings exceed \$0.5 billion per year. Both figures are conservative. The utility's filed blended energy cost in July 2026 is 27 cents (Hawaiian Electric monthly energy cost adjustment filings, July 2026), and displaced energy avoids more than fuel alone. Countries with streamlined interconnection and similar labor costs have seen lower solar prices than the U.S. mainland, even before subsidies.

The lost savings are an incentive for reform, and a warning of what failing to reform will keep costing (Section 2.8). There is also a small chance of reviving recently cancelled solar projects in time to use the federal credits. Under the 2025 federal tax law, a revived project qualifies if it was under construction before July 2026 or in service by the end of 2027. The window is tight but not closed. A "connect and manage" interconnection rule, described below, is the kind of reform that could open it.

[1] *Footnote: the 25 percent capacity factor is conservative for this counterfactual. Hourly profiles for the model's 54 land tranches come from Matthias Fripp's Switch-Hawaiʻi data pipeline, which runs NSRDB*

satellite irradiance through NREL's PVWatts single-axis-tracking model (github.com/switch-hawaii/data). Tranche capacity factors range from about 15 to 28 percent AC. The best 2 GW of tranches average about 27 percent, and a 2 GW program would use the best land first. The model's solved builds average 23–24 percent only because they are much larger — 3,600 to 4,100 MW of utility solar by 2050, depending on the rooftop trajectory — and reach into poorer tranches. These figures are themselves conservative for less-restricted siting: leeward Class B/C cropland enters only through the capped 10 percent draw, and other high-irradiance land to the west is excluded as military or discretionary. The roughly 21 percent in national fleet data reflects the existing, largely fixed-tilt vintage.

2.4 Interconnection

The 2024 Integrated Grid Planning filing reports average study-completion times of 24–30 months for utility-scale projects. For a grid of Oʻahu's size and topology, studies of that length are difficult to justify technically. Yet, in practice, interconnection has taken twice as long or more (four to seven years). And unlike the mainland, the State does not have a huge interconnection queue to help rationalize the delay. A connect-and-manage approach — construction proceeding on a preliminary feasibility determination, with dispatch managed against actual conditions — is standard practice in ERCOT and Great Britain and would shorten the queue without weakening reliability review. Soft-cost reform lowers the price of each project. Queue reform raises the rate at which projects arrive. The two are complements (Section 2.8).

Texas is the clearest illustration of how far this can go. Generators there interconnect under connect-and-manage at a fraction of the cost and time of the invest-and-connect regions, without the network-upgrade charges that dominate elsewhere, which places its interconnection soft costs at or below the mainland average (Utility Dive, "Can ERCOT show the way to faster and cheaper grid interconnection?", 2023; ERCOT led all U.S. operators in interconnection volume in 2024, with Berkeley Lab attributing the lead in part to connect-and-manage). Texas also carries a very large interconnection queue, but for reasons unrelated to the cost of the process itself: a surge of speculative generation and large-load data-center requests, much of which has not submitted enough information to be studied. A long queue under cheap, open access is a different situation from the slow, costly one Oʻahu faces. The comparison suggests Hawaiʻi's soft costs are higher than the interconnection task itself requires, though only Texas has pushed them this low, and doing so depends on dispatch rules that are clear, fair, and audited (Section 2.8).

The Hawaiʻi premium sits in process. Hardware arrives at world prices plus a slight transport premium and tariff, if not produced domestically; what Hawaiʻi adds is time, queues, and risk premia, all of which policy can reduce (Section 2.8).

2.5 Does Oʻahu have enough land?

How much land the build needs. The model's least-cost build reaches about 4,100 MW of utility-scale solar by 2050, with rooftop systems on the conservative trajectory of Section 2.7 (about 1,000 MW by 2050); faster rooftop growth cuts the utility build to about 3,600 MW on the trend trajectory. How much land a megawatt needs has fallen steadily as panels have grown more efficient. The conventional figure is about six acres per megawatt for tracking solar, counting the full site — roads, setbacks, and buffers. On direct array area alone (row spacing included, the rest of the site not), the 2019 U.S. medians were 4.2 acres per MW-DC (5.6 per

MW-AC) for tracking plants and 2.9 (3.6) for fixed-tilt, which packs about half again more capacity per acre (Bolinger and Bolinger, *Land Requirements for Utility-Scale PV*, Lawrence Berkeley National Laboratory, 2022; vendored in sources/). Those medians are seven years old, and the same study measured tracking power density rising 43 percent over 2011–2019, mostly from module efficiency; even at half that pace, today's tracking plants sit near five acres per MW-AC — and the trend runs through the 2030s and 2040s, when most of the buildout modeled here occurs. We use five acres per megawatt-AC throughout — the same basis as the model's capacity variables — and state the judgment: probably optimistic for a project built in the next few years, plausibly pessimistic for one built around 2045; one number for a quarter-century of construction splits that difference. The density choice shapes the solution. The model selects parcels in merit order, and 41 of the 54 land tranches are built to their caps in the least-cost 2050 build, so a worse acreage requirement shrinks every tranche's cap and pushes construction outward into steeper, costlier, lower-capacity-factor land. At five acres per megawatt, the 2050 build needs about 20,400 acres (about 18,000 on the trend rooftop trajectory).[6] The model assumes tracking solar throughout, because that is what the optimization selects. If land constraints were ever to bite, the next step is parcel-level characterization and a wider technology menu — fixed-tilt and agrivoltaic configurations (for instance double-sided vertical panels that capture early and late sun) use land differently and would open more potential than assumed here; both require changes to model structure and are planned for v2 ("What we will do next").

[6] *Footnote: an external benchmark for the land claim. Covelli, Virgüez, Caldeira, and Lewis (2024, Applied Energy 375: 124054) find that a least-cost system meeting 100 percent of O'ahu's hourly averaged demand from wind, solar, batteries, and hydrogen storage alone would occupy 265 km², 17.1 percent of the island, against about 83 km² (20,400 acres, roughly 5 percent) for the least-cost build here. Their supplementary tables locate the difference. Wind is 186 of their 265 km²: 1,026 MW of turbines charged at 44.7 acres per megawatt, the full farm area including spacing. The paper notes that turbine foundations occupy under 2 percent of that area, and the rest remains available for farming or grazing. Our framework instead caps onshore wind at 150 MW under the county setback ordinance and meets firm needs with enhanced geothermal and existing thermal capacity on renewable fuel. The solar blocks are nearly the same size: 78 km² there (3,485 MW of fixed-tilt at 5.5 acres per megawatt), 83 km² here (about 4,100 MW of tracking at 5 acres per MW-AC). Their solar factor is also dated: it comes from NREL's 2013 land-use survey (Ong et al. 2013, NREL/TP-6A20-56290) for installations under 20 MW, and the successor study's 2019 fixed-tilt median (Bolinger and Bolinger 2022, cited above) is about 3.6 acres per MW-AC, which would shrink their solar block by a third. Their capacity factors are computed from horizontal irradiance (GHI/1,000 W/m²), so a tracking configuration would also reduce the megawatts of solar and storage they need. Theirs serves all demand from utility-scale plants; our trajectories put a further 1,000–2,120 MW on rooftops by 2050. The paper reaches the same conclusion about substitution: shifting its mix from wind toward solar cuts its land use by more than half at about four cents per kilowatt-hour of added cost (their Fig. S16).*

What the screen admits. The screen's rule is fully documented. It admits agricultural- and country-zoned land only; subtracts Class A soils, golf courses, road buffers, and (via the zoning filter) military installations; caps prime Class B/C land at 10 percent per cluster — the as-of-right limit quantified below — and admits all Class D/E and non-agricultural land. Terrain enters through a graduated screen: land to 15 percent slope builds at reference cost, 15–20 percent at a 5 percent premium, 20–30 percent at a 10 percent premium, and slopes above 30 percent are excluded. Screening and clustering yield 5,451 MW of buildable capacity — 27,256 acres at five acres per megawatt — across 54 tranches (18 areas in three slope bands), 91 percent of it on marginal or non-agricultural soils. The selected B/C parcels are not physical site choices; they are a quasi-random draw representing the acreage current law allows, and any similarly sized set would serve the analysis equally well. The slope premiums matter in practice: about a third of the least-cost build lands above 15 percent slope, so a flat-land-only reading of the inventory would misstate both the acreage and where it sits.

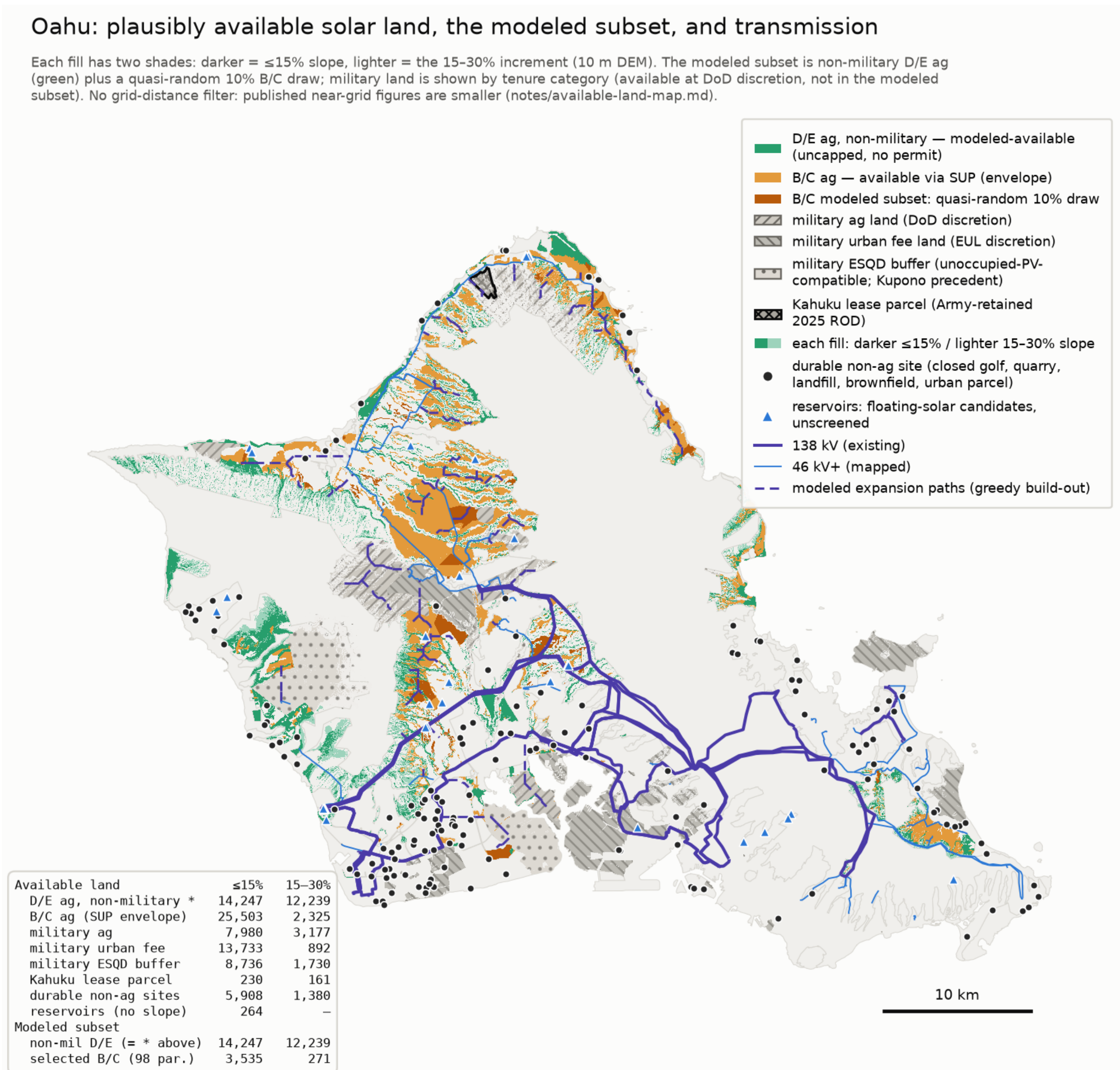

Figure 2.3 — Plausibly available solar land, the modeled subset, and the transmission network. Land plausibly available for utility-scale solar, colored by tenure and legal pathway: uncapped Class D/E agricultural land (modeled as available), the Class B/C envelope reachable by special use permit together with the quasi-random 10 percent draw the model uses, military lands by discretion category, durable non-agricultural sites (closed golf courses, quarries, landfills, brownfields), and reservoir surfaces (floating-solar candidates). Within each category, darker fill marks slopes of 15 percent or less and lighter fill 15–30 percent. Purple lines are the existing

138 kV and mapped 46 kV-plus network; dashed lines are modeled expansion paths. The inset table gives acreage by category and slope band; the modeled subset totals about 30,300 acres.

The inventory beyond the screen. The map and the model describe the same inventory at two levels of detail. The map's modeled subset totals about 30,300 acres — roughly 26,500 acres of Class D/E agricultural land, which carries no statutory cap, plus a 3,800-acre draw of Class B/C parcels. Around it, the map documents what the screen excludes by construction, each category with its own institutional path and none assumed in the base inventory: about 25,500 acres of Class B/C cropland under 15 percent slope that current law admits only through special use permits (the modeled draw takes just 3,500 of those acres); roughly 30,000 acres of military-controlled land, in categories from plausible (explosive-safety buffers compatible with unoccupied solar, following the Kupono precedent) to discretionary; about 5,900 acres of durable non-agricultural sites — closed golf courses, quarries, landfills, brownfields; and reservoir surfaces suited to floating solar. Beyond the mapped categories sit industrial, brownfield, and other disturbed lands inside the urban district, which the screen never examines, and two categories under parcel-by-parcel investigation: federal landholdings (tenure, mission constraints, and the 2029 state-land lease questions require careful treatment) and closed golf acreage (which the screen subtracts regardless of operating status).

Steep slopes. Building at scale on 15–30 percent grades is less settled practice than the cost premiums alone suggest: tracker foundations, grading, and stormwater management on those grades carry engineering and permitting questions a capacity-expansion model does not resolve, and some of that terrain may prove harder to develop than a 5–10 percent adder implies. Two observations bound the concern. First, the hardware tolerance is real but finite: standard single-axis trackers handle grades to roughly 15 percent with modest cost increases (companion study slope-cost review), and 92 percent of the mapped B/C envelope — 25,503 of 27,828 acres — lies at or below that grade, though terrain binds unevenly (about 65 percent of Class C acreage exceeds a 5 percent grade, against about 40 percent for Classes A and B). Second, flatter land the screen excludes — the categories above — stands in reserve. If steep-slope construction disappoints, the substitution runs toward those flatter categories rather than toward more total land; the question the map settles is physical availability, not siting certainty for any particular parcel.

What the land records show. The claim that suitable land is unavailable sits uneasily with the documented structure of the rules. The companion land study quantifies the statutory cap on prime Class B/C soils — as-of-right development limited to the lesser of 10 percent of a parcel or 20 acres (2014 Act 55) — directly at the parcel level (its notes/cap-quantification.md, run July 2026). Under current law, as-of-right B/C eligibility on O'ahu is about 3,600 acres, and the binding element is the hard 20-acre cap, not the 10 percent share. Raising the share to 20 percent while keeping the 20-acre cap adds only about 1,100 acres; dropping the hard cap at the existing 10 percent nearly triples eligibility, to about 9,400 acres; both changes together yield about 15,700 acres, a 4.3-times increase. That figure can look surprising against the island's roughly 34,400 acres of agricultural-district B/C soil, but the cap is a share of each parcel's *total* acreage, not of its B/C acreage, and B/C soil concentrates in very large parcels — the largest tenth hold 91 percent of it — so a 20 percent parcel share reaches nearly half the B/C total. The same concentration means relaxing the cap mostly transfers development option value to large landowners, a point the companion study develops.

Above the cap, B/C development is *unlimited* through a Special Use Permit, conditional on an agricultural lease at 50 percent or more below fair-market rent plus decommissioning security. The record shows the route workable when the economics support it: of the eight applications to come before the Land Use Commission, seven were approved unanimously without intervenors; the eighth, on Maui, is under review. And the economics generally do support it: for much of this land the solar value far exceeds the agricultural value, so even a minimal second income stream makes dual use attractive — while advancing the state's goals of keeping agricultural land productive and growing local food supply. The exception is land whose agricultural value exceeds its solar value, notably high-value seed crops; but seed crops require large buffers against cross-fertilization, and those buffers might themselves be profitably planted in panels. Our 10 percent baseline is therefore likely conservative: much of the B/C deployment it represents would proceed as agrivoltaic projects under the special-use pathway, and some large parcels could build to the as-of-right cap with no permit at all — though at five acres per megawatt a 20-acre site is about 4 MW, below the 5 MW minimum today's PPA terms require, a procurement conflict Section 2.8 (reform 2) addresses. Setback rules severely constrain onshore wind on Oʻahu; we have found no case of a blocked solar project on the island.

Ownership, terrain, and the grid. Ownership of the eligible land is unconcentrated — HHI $\approx$ 560, 44 percent government-held, no pivotal private owner — so land market power does not rescue the scarcity argument either. An agrivoltaic standard giving solar the same as-of-right dual-use pathway wind enjoys under HRS §205-4.5 would streamline the process while keeping, and possibly growing, the land in farming (Section 2.8, reform 4). The constraint the companion study finds actually binding is not acreage but transfer capacity: roughly 70 percent of the screened utility-solar resource sits north of the island's transmission necks, and moving a multi-gigawatt northern build south requires bounded corridor upgrades on the order of \$10–200 million — small against the plant decisions in view and unpriced in this single-zone model; the nodal model of v2 will quantify them.

The screen errs in both directions, and the errors partially offset. Some acreage inside the screen will prove undeliverable in practice — owners who decline to lease, sites behind transmission corridors not yet built, projects communities decline to accept, and parcels the available data mischaracterize. But the screen also excludes the plausibly viable reserves cataloged above — the roughly 30,800 acres of agricultural-district B/C soil beyond the as-of-right cap, the military categories, the durable non-agricultural sites, the reservoir surfaces, and the urban-district lands it never examines. None of these is counted in the 27,256 eligible acres, which itself exceeds the roughly 20,400 acres the least-cost build actually uses.

Physical acreage is sufficient; the binding questions are pace, process, and terms. The companion study of Oʻahu land availability and the political economy of the land-use rules — the legislative record of HRS §205-2/§205-4.5, cap counterfactuals, ownership, terrain, and grid proximity — is public at github.com/mikejrob/solar-wind-landuse and carries the full detail behind this section and the next. It is a living inventory: fixes, parcels, and local knowledge are welcome there.

2.6 What the fuel choices change about land, and when

Section 2.5 shows the acreage exists. This section asks a different question: how the fuel and generator choices on the table change the land the island actually uses, and when.

Every mandate-compliant pathway builds nearly the same solar with and without LNG. The JERA LNG path reaches nearly the same 2050 level as the no-new-plant path — a difference under one percent, and in this solution the LNG path ends slightly *higher*. What changes is timing and what fills the gap: by 2035 the JERA path has built about 860 MW less solar (roughly 4,300 acres less land then in use), with gas-fired generation supplying the difference until the mandate closes the gap by 2045 (Figure 2.1).

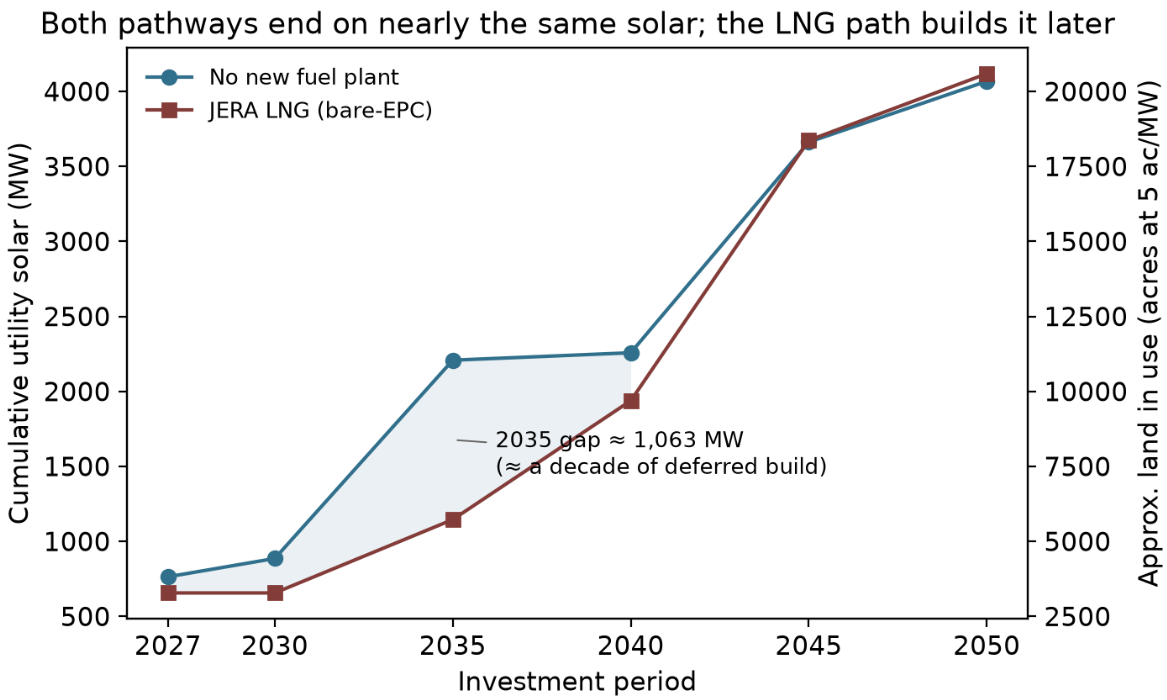

Figure 2.1 — Cumulative utility-scale solar on the no-new-plant and JERA LNG pathways. Cumulative installed utility-scale solar (MW, left axis; approximate land in use at five acres per megawatt, right axis) by investment period, at reference oil prices. The pathways end near the same total; the shaded area marks the build the LNG path defers, roughly a decade's worth centered on 2035.

Figure 2.2 shows the generation mix over time for four pathways. On the least-cost path (panel a), oil declines through the 2030s and is largely gone by 2045, with utility-scale solar the dominant source; the path's renewable share runs ahead of the renewable portfolio standard (RPS) milestones through 2040, a point Section 2.7 returns to in pricing the mandate. The JERA path (panel b) shows what the plant actually does: LNG displaces most oil from 2030 to 2044, and it also displaces solar — utility-scale capacity grows visibly later than in panel a, converging only as the mandate closes. Blocking Enhanced Geothermal (panel c) removes the geothermal band and backfills with oil and solar. Accelerated rooftop growth (panel d) roughly doubles the distributed band by 2050 and shrinks the utility-scale build. One timing caveat: the model builds its 100 MW of EGS in the first period and runs it at 0.93 capacity factor from 2027, because nothing in the model constrains geothermal development time; a realistic demonstration-and-permitting timeline shifts that band's start into the 2030s without changing the 2050 mix

(Appendix A.7).

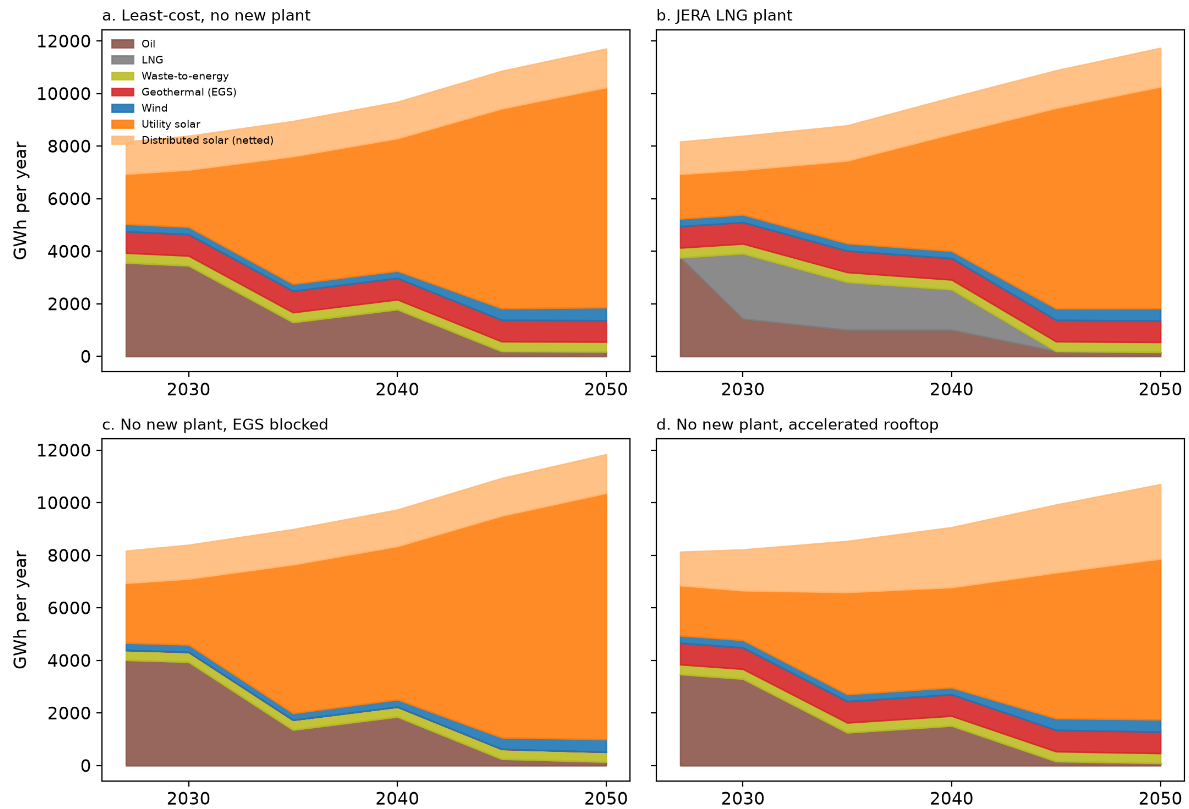

Figure 2.2 — Generation mix over time on four solved pathways. Annual generation (GWh in a model-weighted typical year) by source, at reference oil prices: (a) least-cost with no new fuel plant, (b) the JERA LNG plant, (c) no new plant with geothermal unavailable, (d) no new plant with accelerated rooftop adoption. Thermal output is split between oil and LNG in proportion to each period's fuel use. "Distributed solar (netted)" is the grid-visible rooftop contribution of the pathway's adoption trajectory: about 1,000 MW by 2050 in panels a–c, about 2,120 MW in panel d. The model can build geothermal earlier than permitting likely allows, so the timing of the early EGS band is optimistic.

The State's own study shows the same pattern within its scenarios. In HSEO's generation tables, adding LNG changes its solar hardly at all — the LNG case carries 93–100 percent of the no-LNG case's solar in every year (95 percent cumulatively). LNG substitutes for oil and, later, for imported renewable fuels in HSEO's analysis; there is little substitution for solar there either. HSEO assumes much less grid-scale solar, in part because it inherits the same restrictive land cap from the utility's planning: about 22,000 acres, described as "approximately 90% of the technically feasible land" (study pp. 6–7), and in part because its scenarios use only about a quarter of the land available under that cap. That cap is one choice from a menu. NREL's study for Hawaiian Electric (Grue et al. 2020, updated 2021, archived in this repository) reports twelve Oʻahu land screens spanning 561 to 13,965 MW, and the utility's pick, PV-Alt-1 (3,810 MW on about 24,800 acres), is a middle case. Our Section 2.5 screen is architecturally the same case — Class A soils out, B/C cropland at 10 percent, graduated slopes to 30 percent, military lands out — and finds a footprint within 10 percent of NREL's; most of the difference between our 5,451 MW and NREL's 3,810 MW potential is packing density — five acres per megawatt against NREL's 6.5 — rather than a different land judgment. The official plans and this report largely agree about the land.

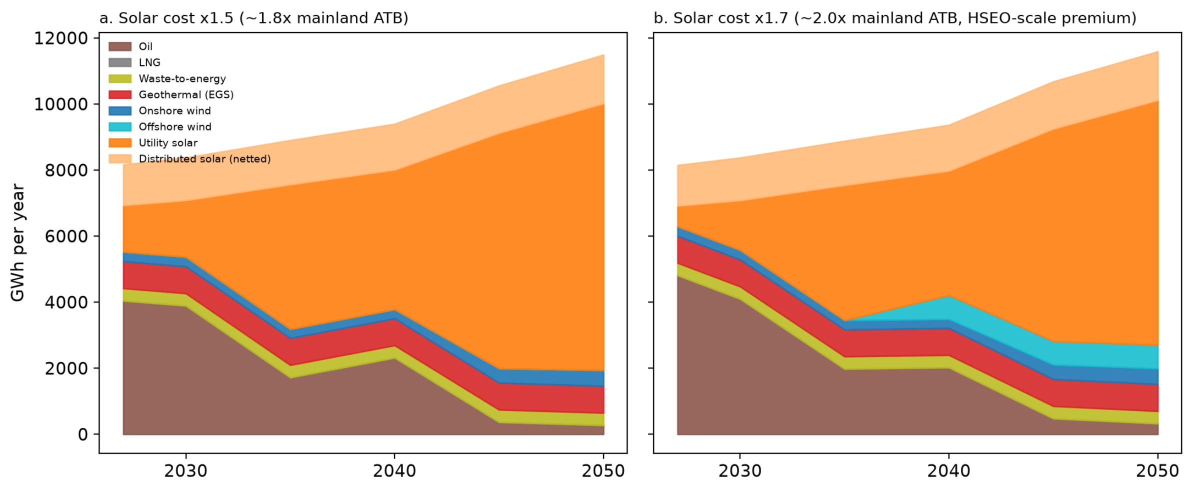

Figure 2.4 — Least-cost generation pathways when utility solar carries HSEO-scale cost premiums. Annual generation (GWh in a typical year) by source on the no-new-plant path at reference oil prices, with utility-solar capital at 1.5 times our baseline (about 1.8 times the mainland ATB benchmark, left) and 1.7 times (about 2.0 times mainland, close to HSEO's 2.154 Hawaiʻi multiplier, right). Offshore wind enters only in the right panel — about 200 MW, the model's minimum block, from 2040. Utility-scale solar remains the largest source in both panels.

Where official plans project less utility solar than we do, the difference is made up with more expensive options. HSEO applies a larger Hawaiʻi cost penalty to solar and none to offshore wind; at those prices its model uses only about a quarter of the land its own screen allows. Its Oʻahu results tables (Reference low-cost case, study p. 218; the tables run to 2045) carry about 915 MW of utility-scale solar, about a quarter of the 3,700 MW our least-cost path builds by the same year and of what their own land screen says is available. Its distributed-solar assumption — 1,514 MW when offshore wind is available, 1,902 MW when it is not — is consistent with the current trend; the pair brackets our trend rooftop trajectory and approaches our accelerated one. The remaining gap is filled by 400 MW of offshore wind and, from 2045, roughly 650 MW of combustion turbines burning imported hydrogen. Our own model pivots similarly to offshore wind when solar costs more than twice the ATB baseline while offshore wind carries no Hawaiʻi premium (Section 4.5 examines the HSEO study's methodology directly). Hawaiian Electric's Integrated Grid Plan similarly presents a base scenario "integrating nearly 3,000 MW of solar and storage by 2050" and a land-constrained scenario that shifts part of that toward offshore wind, rooftop solar, and firm renewables. The land-constrained scenario is the utility's plan of record (IGP Supplemental Response, Docket 2018-0088, Nov 14 2023 — quote p. 10; plan composition pp. 13, 21, 62; public filing).

Two facts bear on those substitutions: offshore wind does not appear as a near-term resource in HSEO's current planning materials (HSEO Ocean Energy Fact Sheet, Oct 2025, energy.hawaii.gov/wp-content/uploads/2025/10/HSEO-Ocean-Energy-Fact-Sheet.pdf), and our model selects it only when solar is made very expensive, near HSEO's assumed premium. The crux, therefore, is whether the Hawaiʻi premium is solar-specific or reflects soft costs of procurement, which would afflict offshore wind just as much. If the problem with solar is soft costs, and reform can hold the premium near 20 percent over mainland costs (solar remains the least-cost path well above that), the solar still gets built and most of the land is still used, even if the clean-energy mandate is abandoned; Section 4.8 prices that case directly.

2.7 Rooftop solar and storage

Distributed solar is Oʻahu's other large solar resource, and how fast it grows changes how much utility-scale land the island needs. The island's rooftops carry about 793 MW today, installed at about 42 MW per year over 2020–2024 despite the 2015 end of retail net metering and shrinking export credits. Section A.11 describes how we estimate what this fleet does to demand from the metered record: each installed MW removes about 0.61 MW of midday grid load, batteries move about 0.45 MWh per installed MWh into the 19:00–22:00 window each day, and about a quarter of rooftop generation now serves demand that never crosses the meter.

We run three installed-capacity trajectories (A.12, A.13). The conservative one, about 1,000 MW by 2050, roughly continues the recent installation rate in gross terms; the trend one reaches about 1,560 MW; the accelerated one, about 2,120 MW, is an illustrative projection of the response to unlimited sellback at avoided cost, the reform this report develops in Section 2.8, with new installs pairing two megawatt-hours of storage per megawatt (a 6.5 kW system with one 13.5 kWh battery, already a typical configuration). Each step of rooftop growth displaces utility-scale build: 2050 utility solar falls from about 4,100 MW (conservative) to 3,600 (trend) to about 3,000 (accelerated), or from about 20,400 acres of land to roughly 15,000.

The built environment is a reserve currently closed by policy. If utility-scale land binds, this is where much of the substitution could go, alongside the flatter-land reserves of Section 2.5 (offshore wind is another possibility). The model carries 4,062 MW of rooftop potential (canopies over parking would add more; we do not yet count them here, but will in the next zonal grid model). Even our accelerated scenario, at about 2,100 MW by 2050, uses just over half of that potential — and the accelerated trajectory itself is an illustration. We do not know how much capacity would be added if sellback rules were relaxed; it could be less than we have penciled in, or a great deal more. Its purpose is perspective: to show how far rooftop growth can substitute for open land. Hawaiian Electric reports that 49 percent of Oʻahu single-family homes now carry rooftop systems, with customer-sited capacity across its territory approaching 1.2 GW (Hawaiian Electric, 2025–26 releases),[3] growth achieved under restrictive tariffs. That share should be read as an upper bound on household penetration: it is a count of systems; many customers added a second system under successive export tariffs rather than expanding a grandfathered one, and some systems sit on multi-family or non-residential roofs. The share of single-family households with any solar is therefore somewhat lower — and many roofs that already have solar can expand their capacity.

Current tariffs let distributed systems offset their own bills but compensate exports well below avoided cost and prohibit surplus credits for providing energy to others, which suppresses the investment that would fill these surfaces. Rooftop and canopy solar is also a partial escape from the soft-cost problem itself: Honolulu rooftop pricing is near mainland levels (Section 2.3), and larger commercial-scale installations would improve its economies further. Among the levers in this report, liberalizing distributed-solar tariffs is arguably the easiest. Concern about the land requirement is itself an argument for the rooftop reform: every megawatt added on a roof or canopy reduces the utility-scale build and the land it needs.

[3] *Footnote: distributed-capacity totals differ across sources in ways consistent with rating conventions. Hawaiian Electric's quarterly data (June 30, 2026) reports 79,347 O'ahu PV systems without stating a rating basis; EIA's monthly utility data (May 2026), which is AC, shows about 645 MW of O'ahu customer-sited PV across net-metered and later tariffs, while the permit-record series behind this report reaches 793 MW by mid-2025. System counts agree across the sources to within half a percent, and the capacity gaps match a typical DC-to-AC inverter loading ratio (about 1.14 on residential systems at matched dates), so the likely explanation is that the utility and permit series are DC nameplate while EIA reports AC — but neither Hawaiian Electric nor the permit records state the basis. The 1.2 GW figure here is on that probable DC basis; EIA's AC-basis statewide small-scale estimate is about 1.0 GW. Appendix A.11 explains why the report's estimates and projections are unaffected by this ambiguity.*

One caveat points to future work. The present model represents O'ahu as a single zone, so it credits distributed resources with none of their locational value: generation and storage at the point of consumption can defer transmission and distribution upgrades that a remote utility-scale buildout requires. That saving — potentially a meaningful offset to distributed solar's higher installed cost — is invisible here. The zonal grid model under development for the next edition is designed to assess exactly this tradeoff.

What better scheduling of rooftop batteries is worth. We measure this with a paired experiment. Both runs carry the identical rooftop fleet — same panels, same batteries, same capital, same load — and differ in exactly one thing: whether the batteries follow today's observed pattern (charging from midday output, discharging through the evening) or are dispatched by the optimizer alongside the rest of the system. The cost difference is therefore the value of coordination itself, free of the accounting choices that otherwise separate the demand-side and supply-side representations of rooftop solar. On the conservative path the difference is \$0.03 billion over 2027–2050 (about 0.02 cents per kilowatt-hour); on the trend path \$0.01 billion; on the accelerated path — 2.1 GW of distributed solar with roughly 2,800 MWh of customer batteries by 2050 — \$0.23 billion, or 0.17 cents per kilowatt-hour. All six cells are solved at the 0.1 percent tolerance, so the paired differences are not tolerance artifacts. The investment plan is similarly robust: solved either way, 2050 utility solar moves by 25 to 200 MW across the three trajectories.

Two readings follow. At current adoption, predictable evening-shifted behavior is nearly as good as perfect dispatch; the grid can plan around it, and nothing in the near-term case for rooftop solar depends on smarter coordination. But the value of coordination rises steeply with adoption: with 2.1 GW of distributed solar — the scale the Section 2.8 tariff reform could create — it is worth roughly eight times its value at today's adoption, and approaches the entire cost of the 2045 mandate (below). One caveat bounds these figures: they price battery scheduling only. Letting household consumption itself respond to prices — demand response proper — is not modeled anywhere in this report, and prior work finds such response is worth several times more on a grid like this one, so the full value of real-time pricing likely exceeds the battery figure.[4]

[4] *Footnote: responding to prices costs households effort, and that cost does not appear in our present-value*

accounting. But rooftop customers already spend that effort. They manage their demand to get the most from their panels and batteries under today's tariffs: buying from the grid costs much more than selling to it; some customers (NEM-plus, self-supply) are curtailed if they backfeed, and so face use-it-or-lose-it timing; many charge their vehicles strategically. That management is costly, and it is in large part what produces the load shapes we model. Real-time pricing would redirect the same effort toward system value — arguably a lighter burden than the tariff rules it would replace.

The cost of the mandate. With the 100 percent RPS constraint removed and the model choosing freely, the no-new-plant least-cost path still reaches about 3,740 MW of utility solar by 2050 — 92 percent of the mandated build — and oil's share of grid supply falls under 5 percent by 2050 on economics alone. What the mandate buys is pace at the end: without it the model keeps about 22 percent oil in 2040 and 10 percent in 2045, retiring it over the following decade as solar and storage costs decline. The system-cost difference is about $0.25 billion in present value, roughly 0.2 cents per kilowatt-hour (both cells at the 0.1 percent solve tolerance). Achieved efficiently, the mandate is cheap insurance on pace; the paths converge to nearly the same place either way. Figure 4.4 sets the no-mandate path beside HECO's and HSEO's published plans.

2.8 Implications for procurement reform

Five reforms are within the Commission's and Legislature's authority and bear directly on the solar-cost lever.

1. **Pricing reform.** Retail wheeling under 2025 Act 266 replaces the embedded risk premia of the RFP-and-PPA process with transparent market access to avoided-cost pricing. It reduces per-watt delivered cost directly, and it is the largest single lever in this report. Whether wheeling delivers depends on getting the pricing right. The price is favorable for a reason peculiar to Hawaiʻi: because the marginal generator is oil, the utility's own short-run avoided cost — set by the fuel burned in existing plants — already runs well above the cost of new solar, so paying avoided cost is enough to attract large projects without any subsidy. On the mainland, where the marginal fuel is cheap gas, an avoided-cost offer sits below solar's cost and draws nothing; here it sits far above it. Wheeling under Act 266 is one route to that price. A second needs no legislation at all: the Commission could require the utility to purchase independent renewable output at avoided cost directly — a standing offer that, on an isolated grid with no competitive wholesale market, federal law (PURPA) arguably already compels, and that today reaches only installations of 100 kW or less.

2. **Procurement reform.** Several standard PPA terms add cost or delay and could be changed. Contracts fix the Unit Price in nominal terms for the contract term; through Stage 2 (2019) price escalation was expressly prohibited. The Stage 3 RFP (2022) added a one-time inflation adjustment — capped at 10 percent, indexed to the GDP deflator, and refined in the later IGP RFP to a 15 percent combined cap with a tariff-cost adjustment — made after the 2021–23 cost spikes stranded earlier fixed-price awards. But the adjustment window closes at PUC approval, so delay through the interconnection queue afterward still erodes the real price the utility pays, weakening its incentive to move projects quickly. Because the utility controls the interconnection queue yet bears no cost from delay, aligning its incentive with speed would help — for instance, a performance-based penalty for interconnection times beyond a defined standard, tied to the metrics Hawaiʻi's PBR frame-

work already tracks, so the party that sets the pace also bears a cost when the pace is slow. Transparent selection of winning bids by an independent third party retained by the Commission, rather than by the counterparty that also owns the grid, might reduce the risk premia developers attach to the process; the change sits within the Commission's own competitive-bidding framework and requires no legislation. Alternatively, the legislature could force Hawaiian Electric to divest its generation assets and refocus its business on grid balancing and delivery services, reducing conflicts of interest with independent generators (Senate Bill 3326). Zero-degradation clauses, which require developers to guarantee no output decline, could be relaxed to a realistic allowance of 0.3–0.5 percent per year. Where feasible, a simplified take-or-pay for capacity would lower financing cost. And a streamlined track for small installations of 5 MW or less would open Class B and C agricultural land that current terms foreclose: at the five-acre-per-megawatt density of Section 2.5, the as-of-right cap on that land — the lesser of 10 percent of a parcel or 20 acres — tops out near 4 MW, so PPA terms requiring more than 5 MW are a de facto prohibition on exactly the projects the cap permits. Denser fixed-tilt layouts or continued panel-efficiency gains could lift a 20-acre site past the threshold; a lower minimum would help, but a streamlined, fast-connect exception for small installations removes the conflict directly. Small installations could be offered avoided-cost pricing with one-to-two-hour storage rather than the usual four-hour standard.

3. **Interconnection reform.** Queue reform, ideally built around connect-and-manage (Section 2.4): a project interconnects on a preliminary feasibility determination and is dispatched against actual grid conditions, rather than waiting years for the network-upgrade studies that dominate interconnection cost and time elsewhere. It lowers soft costs, shortens the timeline over which any cost reduction is captured, and reduces risk for developers. Making it work in a single-utility setting where no market arbitrates curtailment requires dispatch rules that are clear, fair, and independently audited, so that a project accepting managed curtailment knows in advance how and when it will be curtailed.

4. **Land-use reform, built to complement agriculture.** One precedented, community-forward option: develop a **menu of pre-characterized agrivoltaic systems** for Class B and C land — specific, named configurations pairing solar with compatible crops, grazing, or managed groundcover — that, if adhered to, could proceed without a discretionary Special Use Permit, paralleling the as-of-right dual-use pathway wind already enjoys under HRS §205-4.5. The qualifying standards could be *stricter* than today's special-use permit conditions where it matters — verified agricultural production, coverage and height limits, decommissioning security, viewshed and cultural-site protections — so the pathway streamlines procurement without weakening protections. The Hawaiʻi Department of Agriculture is the natural body to develop and certify the qualifying systems, in partnership with farmers, ranchers, and the communities where projects would sit. Standards designed with those communities from the outset, rather than presented to them, are the difference between social license and a litany of siting fights.

   The same reform carries a public-safety co-benefit. **Unused, unmanaged former agricultural land is a serious fire hazard.** Fallow agricultural land now comprises roughly 40 percent of all agricultural land in Hawaiʻi, about a quarter of the state's land area, much of it carrying tall, unmanaged nonnative grasses and trees on former sugar and pineapple plantations. Total area burned statewide has increased more than fourfold since the early twentieth century, rising “in direct correlation with the decline of plantation agriculture” (Trauernicht et al. 2015; Bond-Smith, Bremer, Burnett, Trauernicht, and Wada, UHERO, 2023). Plantations also once supplied the on-the-ground presence for fire detection and response that these lands have since lost. Returning them to managed, productive use, including agrivoltaics with maintained groundcover and grazing, reduces fuel loads and the fragmentation of firebreaks while producing energy and food.

5. **Distributed-solar tariff reform.** Allow unlimited sellback at real-time avoided-cost pricing. Current tariffs permit only own-bill offsets and below-avoided-cost export compensation, which idles the rooftop and canopy potential documented in Section 2.7. Reform does three things at once: it grows clean generation on roofs and canopies that need no new land, it gives rooftop owners a reason to dispatch their batteries when the grid most needs the power, and it lets household demand respond to prices. It remains the easiest reform on this list — it requires building nothing and rezoning nothing — and it would unambiguously reduce system costs and pollution. The gains compound with scale: at today’s adoption, battery timing is worth little, but with 2.1 GW of distributed solar — the trajectory this tariff could put Oʻahu on — it is worth about $0.23 billion (Section 2.7), and the demand-response benefit, which our modeling does not count, comes on top.

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3. Enhanced Geothermal is in the least-cost build

Enhanced geothermal (EGS) cost cases are 6.2 / 10 / 14.7 $M/MW at 2030, before the federal credit. These assumptions represent the optimistic DOE GeoVision trajectory, a compromise reference, and the ATB 2024 Conservative profile.[2] Under current law, the geothermal tax credit applies, lowering the effective cost of a 2027–2035 build by 30 percent, and at that credited reference cost the model builds the full identified resource in the base case.

EGS cost case (credited)	System cost, no LNG ($B)	EGS built	Saving vs no-EGS
Option off / none	26.40	0 MW	—
High ($14.7M/MW gross)	~26.40	0 MW	~$0
Reference ($10M/MW gross)	25.83	100 MW	$0.56B
Low ($6.2M/MW gross)	~25.4	100 MW	~$1.0B

At reference cost — the base-case assumption — Enhanced Geothermal is part of the cheapest build: the no-new-plant baseline of $25.83 billion already contains 100 MW of it, and blocking it would raise that baseline by $0.56 billion. At the optimistic cost, the saving roughly doubles. At the pessimistic cost, the model builds nothing and loses nothing, so the downside is bounded. Because Enhanced Geothermal builds all-or-nothing at its resource cap and its dispatch does not change with its capital cost, the sensitivity is a capital reprice off the reference and blocked cases; see docs/SOLVER_NOTES.md in the repository.

Enhanced Geothermal also saves land. Against its no-EGS counterpart, the solved base case's 100 MW block displaces 394 MW of utility solar — about 2,000 acres at five acres per MW — along with 145 MW (1,250 MWh) of storage. And if the developable resource on Oʻahu proves larger than the roughly 100 MW modeled here, both cost and land requirements fall further.

[2] *Footnote: the reference case is a documented judgment call sitting between a DOE-referenced ~$9M/MW and ATB 2024 Moderate ~$12M/MW; sources and reasoning are in the repository conventions file. The cost-case labels (6.2/10/14.7 $M/MW) are gross of the federal tax credit; the model itself applies the 30 percent credit to 2027–2035 builds under current law, so the reference case solves at an effective $7.0M/MW for those vintages.*

3.0 Background

Hawaiʻi Island operates a 38 MW conventional geothermal plant, Puna Geothermal Venture, in the Puna district on Kīlauea's Lower East Rift Zone, in service since 1993, drawing on a naturally permeable reservoir fed by the active volcanic system. Oʻahu has no such reservoir, because its volcanic system is older and quiescent, but it has hot dry rock at depth. Enhanced Geothermal Systems (EGS) extract that heat: drill two wells to commercially hot rock (typically 2–5 km), fracture the rock between them, circulate water, and run the heat through a generator. The technique is proven at pilot scale. DOE's FORGE site demonstrated commercial-scale stimulation and circulation in 2024, and Fervo Energy's Cape Station in Utah is expected to deliver the first commercial-scale EGS power to a U.S. grid in late 2026. At multi-plant scale it remains pre-commercial. NREL's reV framework applied at 2.5 km depth with standard accessibility filters identifies on the order of 100 MW of potentially developable EGS resource on Oʻahu across roughly a dozen indicative sites; this report models it as a single 100 MW block.

3.1 Cultural context and community engagement

Hawaiʻi has a long and controversial relationship with geothermal power. The Puna plant on the Big Island has been a source of concern for the surrounding community and Native Hawaiian cultural practitioners across three decades — the spiritual relationship to Pele, hydrogen-sulfide emissions, siting, and the 2018 Kīlauea eruption, when lava reached the plant site and forced a shutdown of more than two years. EGS differs technically in ways that matter for that conversation: it does not tap the volcanically fed reservoirs that carry hydrogen sulfide, mercury, and radon; it operates as a closed loop; and the candidate Oʻahu sites sit on the old, quiescent system far from any active rift. Those differences address the specific health concerns; the questions of consent and relationship to ʻāina remain. Any Oʻahu EGS pathway should be developed in partnership with Native Hawaiian community members, cultural practitioners, and lineal descendants of the affected ahupuaʻa from the earliest planning stages — covering siting, water, monitoring protocols, and benefit-sharing. The cost asymmetry justifies the demonstration economically; community partnership is a prerequisite.

3.2 The conditional structure: when it pays and when it does not

The cost cases and results are in the table above (Section 3 opening). The value also depends on solar's trajectory: EGS is worth the most in exactly the scenarios where solar deployment underperforms. If procurement reform stalls and the effective premium runs at 50–80 percent, EGS savings grow and even higher-cost EGS enters the build. EGS functions as insurance against a failure of solar-procurement reform. It is inexpensive when reform succeeds and valuable when it stalls.

3.3 What EGS does for the system

When built, the ~100 MW of flat zero-carbon baseload displaces 394 MW of utility solar and 145 MW (1,250 MWh) of storage in the solved base case and, in LNG-forced scenarios, displaces the most expensive LNG dispatch hours. Reliability is preserved at every modeled hour in either case; what changes is the build mix and its cost.

3.4 The downside if the technology disappoints

A demonstration plant generates baseload power for decades regardless of which cost trajectory materializes. For a 10 MW first-of-a-kind demonstration at 50 percent federal cost share (FORGE-successor programs plus the geothermal provisions that remain in force): gross capital ≈ \$100M at the reference trajectory and \$147M at the high; Hawaiʻi's net exposure ≈ \$50M / \$74M; and the present value of 30 years of delivered energy at a conservative 7 cents per kilowatt-hour covers most or all of that exposure: at an 85 percent capacity factor a 10 MW plant delivers about 75 GWh per year, worth \$5.2M annually, with a 30-year present value of \$102M at the 3 percent social discount rate or \$72M at 6 percent (Appendix A.7). The downside is bounded and modest. The upside is the \$0.56–1.0 billion above. Project-development risks include drilling-induced seismicity — managed under standard traffic-light protocols that scale back or halt injection as monitored seismicity crosses preset magnitude thresholds (Majer et al. 2012, *Protocol for Addressing Induced Seismicity Associated with Enhanced Geothermal Systems*, U.S. DOE Geothermal Technologies Office; Fervo applies such a protocol at Cape Station) — along with water handling, permitting, and community acceptance.

3.5 What a demonstration requires

Four prerequisites: a federal demonstration pathway (FORGE successor plus tax provisions still in force); deeper site characterization than the public record provides (itself federally fundable); a PPA framework that can accommodate EGS lead times and drilling risk; and drilling workforce partnerships. A 2034–36 first-of-a-kind window is plausible if initiated in 2027–28, the same window in which the JERA and late-2030s resource decisions will be made. Federal support exists today and may not persist, which argues for moving early.

4. The thermal question and the JERA proposal

Oʻahu's steam fleet is old and inefficient: full-load heat rates run 10,000–11,000 Btu per kilowatt-hour, and average operating heat rates run higher still, since units cycle and spend hours part-loaded. A modern combined-cycle plant runs near 6,900. Stated either way — roughly a third to 40 percent less fuel per kilowatt-hour, or half again to two-thirds more electricity per unit of fuel — the efficiency gain is what makes any new-plant proposal attractive at first glance. This section tests whether that gain justifies new construction, on which fuel, at what size, and whether the cheapest use of LNG involves a new plant at all.

4.1 The trajectory comparison

Table ES.1 carries the headline matrix. Three results:

The Waiau Repower is uneconomic in every case. It adds +\$1.38 to +\$1.49 billion on its own, and every bundle containing it inherits the penalty. This result is robust to every sensitivity in the report.

A right-sized plant costs less than the proposal. At the midpoint of JERA's cost range, a 375 MW version comes in \$0.38 billion above no-new-plant against \$0.75 billion for the 500 MW version — smaller, but still a cost increase. One caution attaches. We price the smaller plant at the same dollars-per-kilowatt as the 500 MW proposal, while JERA attributes its attractive unit cost partly to scale (proposal p. 17), so the 375 MW figure may understate what a smaller plant would actually cost.

The JERA-500 versus no-new-plant comparison favors no new plant, and the margin moves with solar costs. At the midpoint of JERA's capital range the bundle is \$0.75 billion more expensive at reference oil (band +0.54 to +0.96), \$0.65 billion on the futures path, \$1.63 billion at the market's low oil path, and \$1.21 billion at its high path — a U-shape explained in Section 4.7. Where the fuel-price mapping has to extrapolate, at the low end of the market band, it flatters new gas capacity rather than penalizing it (Appendix A.14); the bundle costs more in every case regardless. The margin is sensitive to the solar premium in the direction one would expect: firm gas capacity substitutes for solar-plus-storage, so the more Hawaiʻi pays for solar and storage, the better LNG looks. At our baseline premium (20 percent over mainland ATB) the result is the modest LNG penalty above; if deployment costs stay near today's procurement-implied levels (effective premiums of roughly 80–104 percent, the 1.5× and 1.7× sensitivities), the margin closes to roughly a tie, with lower or higher oil prices or the ATB Advanced solar cost path breaking the tie slightly in favor of solar. If the storage tax credits are denied and oil prices remain moderate, the comparison slightly favors the JERA proposal.

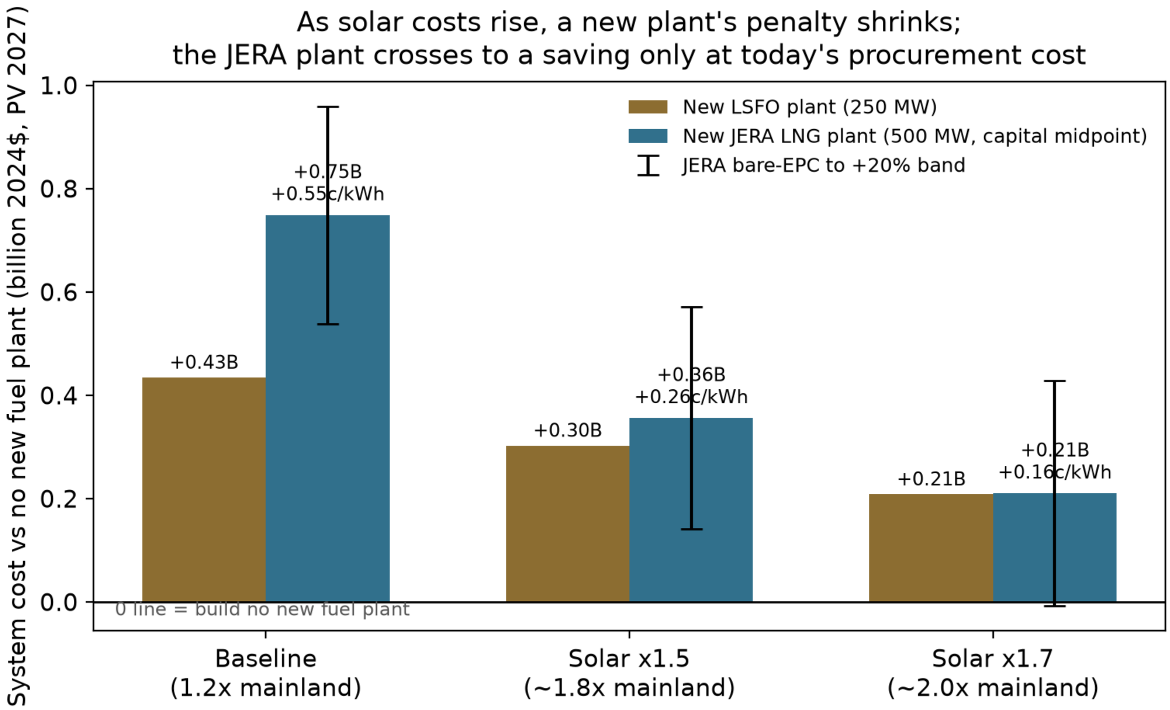


Figure 4.2 — New-plant options against no new plant as solar costs rise. Present-value system cost (billion 2024$, discounted to 2027) of a new 250 MW LSFO plant and the 500 MW JERA LNG plant, each minus the no-new-plant path, at reference oil prices, with utility solar and battery capital at our baseline (1.2 times mainland ATB), 1.5 times (about 1.8 times mainland), and 1.7 times (about 2.0 times mainland). JERA whiskers span bare-EPC to +20 percent capital. The cents-per-kWh labels divide the dollar difference by 135.6 TWh of discounted delivered energy. Bars above zero cost more than building no new plant.

Figure 4.2 puts numbers on this for the reader who expects solar to stay expensive. If solar deployment stays at today's procurement cost (the 1.5× case, about 1.8× the mainland benchmark), the JERA plant's midpoint penalty narrows to +$0.35 billion; at the 1.7× case (about 2.0× mainland) it narrows to +$0.21 billion, reaching break-even only at the bare-EPC capital cost. A new LSFO plant stays more expensive than no new plant at every solar cost, from +$0.43 billion at the baseline to +$0.21 billion at 1.7×. Two things bound how much weight this deserves. The near-break-even appears only under solar costs we argue are a policy choice, not a fixed condition (Section 2); and a new plant is in any case the costliest way to use LNG, since converting existing plants captures the same fuel saving more cheaply (Section 4.7), so even the 1.7× figure understates how favorable LNG becomes through conversion if high solar costs persist.

The whole picture, over both uncertainties at once. Figure 4.3 sets the three thermal commitments against building no new plant over the two dimensions that actually move the answer: what oil does, and what Oʻahu pays for utility solar. Reading it takes one rule — red costs more than building no new plant (or converting existing to LNG), blue saves.

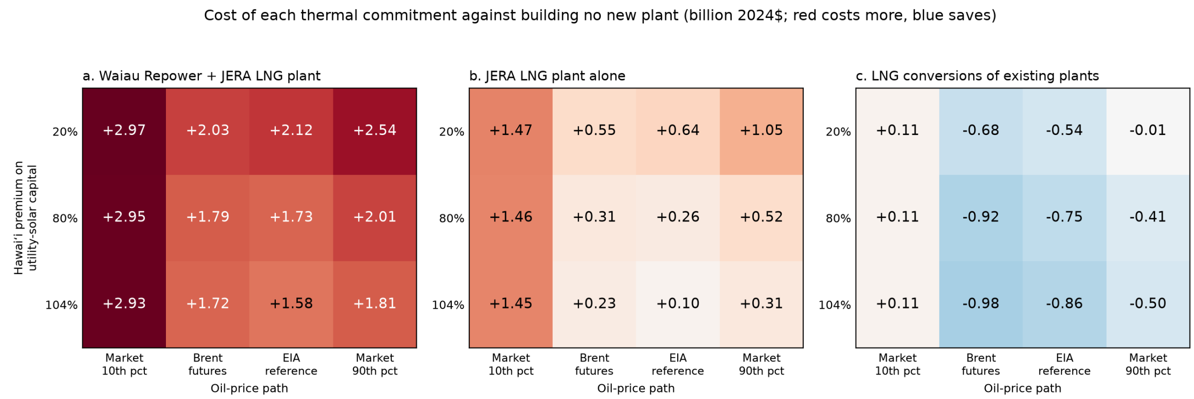


Figure 4.3 — Cost of each thermal commitment against building no new plant, across oil prices and solar costs. Each cell is the present value of total 2027–2050 system cost (billion 2024$, discounted to 2027) for the named commitment, minus the no-new-plant path solved on the same oil path and the same solar cost. Columns are the four Brent paths of Appendix A.14; rows are the Hawaiʻi premium on utility-solar capital, from the 20 percent premium used throughout the report to the 104 percent premium implied by the State Energy Office's cost multiplier. All cells use the trend rooftop trajectory and leave enhanced geothermal available. Conversion cells carry the $0.45 billion conversion-capital charge of Section 4.7. Red cells cost more than building no new plant (or converting existing to LNG), blue cells save; blank cells are not yet solved. Differences smaller than about $0.13 billion are within the solver tolerance of the cells they are drawn from.

Conversions across the oil paths. The conversions panel is the only one of the three with any blue cells, and the multi-oil solves give its saving a definite shape: largest on the two central oil paths, fading toward both extremes. Net of the $0.45 billion conversion-capital charge, converting existing plants saves $0.54 to $0.68 billion on the central paths at the baseline solar premium, and $0.74 to $0.99 billion at the higher premiums. At the market's high path the saving nearly vanishes at the baseline premium (−$0.01 billion) but persists at $0.40 to $0.49 billion under the higher premiums, and at the market's low path the conversions become a small net cost of about $0.11 billion at every solar premium (both cells behind that figure are 0.1 percent solves, so it carries roughly ±$0.04 billion of solver tolerance).

The shape follows from the cost accounting. Committing to conversions means paying two charges that do not depend on where oil goes: about $0.52 billion of LNG import-infrastructure fixed costs over the 2030–2044 terminal window, plus the conversion capital. The price gap between oil and gas is also nearly flat across the oil paths — both fuels track Brent at almost the same per-unit rate (Section 4.7) — so what varies is how the rest of the system responds. At the low path nothing else changes: cheap oil had already suppressed the clean buildout, so the only benefit is the fuel bill, which falls about $0.83 billion — not quite enough to cover both charges. On the central paths cheap gas does double duty: it cuts the fuel bill by a similar amount and lets the model defer $0.8 to $0.9 billion of solar and storage capital that sits right on the margin of being built. At the high path the roles reverse. Expensive oil pushes the no-conversion system to build its way out of fuel — by 2040 it carries about 450 MW more solar and more storage than the converted system — so its costs are contained by substitution toward solar rather than by fuel switching. Conversions skip that extra clean capital but keep burning fuel at high prices, and the larger fuel bill gives back most of what the avoided capital saved, leaving the net saving near zero at the baseline solar premium. The higher premiums make that substitution dearer, which is why conversions hold more of their value in the bottom rows. The conversion case is therefore strongest exactly where the market puts the middle of the oil distribution.

4.2 The JERA capital cost

The proposal (17 March 2026, p. 30) prices the plant at $1,510M for 500 MW ($3,020/kW, ~2026$) and the import infrastructure at $460M ($200M FSRU, $10M onshore pipeline, $250M fixed infrastructure; proposal p. 30). The plant figure's footnote reads: "Including EPC, Procurement, Construction, Installation, Capital Spares and Freight; Excluding Customs and duties, Insurance, design allowance and contingency"; page 35 adds imported equipment to the exclusions. The publicly reported ~$2 billion total (about 75 percent plant) matches these figures [ENR 2026]. The proposal's own downside case (p. 29) tests capital 20 percent higher. We solve every JERA scenario at both costs and present the midpoint with the band. We consider this fair and still conservative: across 401 electricity projects studied, three in four exceeded their cost estimates, with thermal plants averaging about 13 percent overruns (Sovacool, Gilbert & Nugent 2014). The band's +20 percent top therefore sits above the thermal-plant average, on an estimate that itself excludes contingency.

The quoted price deserves comment in the current market. Gas-turbine supply chains are strained: Lazard's June 2025 review reports the cost of a new combined cycle at a ten-year high, with recently observed market quotes of $2,400–2,600/kW for mainland projects entering service after 2028 and turbine shortages driving long lead times (Lazard LCOE+ June 2025, pp. 4, 8, vendored in `sources/`). Hawaiʻi construction carries its own premium on top. Against that backdrop, $3,020/kW for a Hawaiʻi CCGT is an attractive price, if JERA can deliver it. For reference, HECO's 2016 planning assumptions priced a small (152 MW) unit at $4,050/kW (nominal; ≈$3,900/kW in 2024$), roughly a third above the proposal's figure before any post-2020 escalation; JERA attributes the difference to scale and its own procurement. The delivered-cost question (what is actually built, at what price, under what contract) is among the most decision-relevant unknowns in the proposal, and it is why our headline treats JERA's own +20 percent case as the top of the band rather than an extreme. The commercial structure is similarly open. The proposal describes an independent-power-producer arrangement seeking "offtake certainty" (pp. 25, 31) with a 40-year revenue-requirement framing (p. 29), but does not specify the offtake contract's terms, including who bears completion and performance risk or how fuel-price risk passes through. The fuel price we carry ($11.4/MMBtu delivered, about $10.1 before the regasification charge) is itself well below current Pacific spot and Qatar-linked term markets, and Section 4.6 discusses why a contract floor at that level should not be read as a ceiling.

An asymmetry in the fuel modeling formerly favored the JERA plant, and has been removed. Every existing unit on the system, and our own LSFO-plant comparator, carries a part-load fuel curve: average heat rates rise when a unit runs below full output (Kalaeloa averages about 9.5 MMBtu/MWh at its minimum load against a 5.7 incremental rate; the modeled LSFO plant about 8.5 against 6.6). The JERA plant's fuel use was instead modeled as strictly proportional to output — 6.92 MMBtu/MWh at every load — so it paid no part-load penalty even though it runs well below full output for much of its life. The plant now carries a curve derived from the operating record of 408 comparable combined-cycle units on the mainland, and a minimum load raised from 30 to 62.5 MW per block (Appendix A.8; `sources/epa_cems/`).[5]

[5] *Footnote: this refinement was prompted by JERA, whose July 2026 memo on the withdrawn paper pressed the distinction between a plant's full-load heat rate and the average rate it achieves in operation. The point was well taken and applies to the modeling as well as the presentation, and we thank them for it. The revised curve barely moves cost: the JERA premium over no-new-plant changes by less than $0.01 billion in every configuration tested (for example +$0.535 to +$0.538 billion at 500 MW bare-EPC, +$0.958 to +$0.959 billion at the +20 percent capital case). What it moves is dispatch. Facing a real part-load penalty and a higher minimum load, the optimizer runs the plant less and buys the displaced energy from solar and storage instead, so LNG imports over the horizon fall about 16 percent (293 to 246 million MMBtu) and the plant's combustion-CO_2 advantage over the no-new-plant path widens from about 0.5 to 1.3 Mt. Because the methane threshold scales with imports and with that advantage, the leak rate at which LNG's greenhouse edge disappears roughly triples (Appendix A.10).*

4.3 Plant or fuel? Where the LNG advantage comes from

The proposal combines two things that need not go together: a modern, efficient plant and a cheaper fuel. Their effects can be separated. If the advantage lies in the fuel, it can be captured by burning LNG in the plants Oʻahu already has. If it lies in the plant, the same plant can be built to burn the low-sulfur fuel oil Oʻahu already imports. Our scenarios span both choices at once — LSFO against LNG, and the existing fleet against a new combined-cycle plant — so each can be read directly. The table shows total-system-cost differences against building no new fuel plant, in billions of 2024$ at reference oil:

	Burns LSFO (today's fuel)	Burns LNG (terminal built)
Existing plants only	baseline (0)	−0.39 (Kalaeloa converted) to −1.05 (also Kahe 5 & 6, CIP CT); −0.60 net of the full 2016 conversion-program charge
New 500 MW combined cycle	+1.24	+0.54 bare-EPC; +0.75 at the capital midpoint

Conversion capital is set to zero in the solved conversion cells; the −0.60 net figure charges the entire 2016 conversion-program estimate ($450M in 2024$) against the converted units as a capital adjustment, the same treatment as the EGS cost sensitivity. Details and caveats in Section 4.7.

Reading down the left column isolates the plant. Capital is effectively a wash between the LSFO and LNG versions of the plant (our LSFO combined cycle carries $2,900/kW against JERA's $2,863/kW in 2024$), so the LSFO plant is JERA's plant on today's fuel. It raises system cost, and raises it more the larger it is: +$0.43 billion at 250 MW, +$0.80 billion at 375, +$1.24 billion at 500. The new plant is more efficient, with a full-load heat rate near 6.9 MMBtu/MWh against roughly 8.6 for Kalaeloa's combined-cycle units and 9.7 for Kahe's newest steam units. The saving is one minus the ratio of heat rates: about 20 percent against Kalaeloa, 29 percent against Kahe's newest units, and 35–40 percent against the oldest steam units, whose rates in average operation run near 11. Whether a saving of that size repays $2,900/kW of capital depends on how much the plant runs, so the answer has to come from the solved scenarios rather than the heat rates alone. The answer is no: in every case we ran — four oil paths, three rooftop trajectories, solar premiums up to 104 percent, both land screens — the system is cheaper without the new LSFO plant than with it, by margins from +$0.43 billion at the baseline premium down to +\$0.21 billion at the 104 percent case (250 MW, reference oil). The efficient plant does beat some alternatives: it undercuts the JERA bundle outright when oil is cheap, and every Waiau-bundled option throughout. But in no solved scenario is building it optimal against the options that build no plant at all, because the system's cheapest response to expensive fuel on this grid is to displace it with solar and storage, and only secondarily to burn it more efficiently.

Reading across the top row isolates the fuel. Contract-priced LNG lands at about \$11.4/MMBtu delivered, regasification included, against LSFO's \$16.7 at reference Brent (Table 1.1): roughly a third cheaper per unit of heat, an edge as large as the new plant's heat-rate advantage and available (up to conversion feasibility and cost) without building anything but the terminal. Burned in the existing fleet alone it saves \$0.39 billion through Kalaeloa and \$1.05 billion when Kahe 5 and 6 and the CIP combustion turbine convert as well. The model even routes gas through the CIP turbine, the least efficient unit on the island, at 11.7 MMBtu/MWh: with a fuel-price gap this large, plant efficiency is a secondary consideration.

The fuel carries the advantage. On its own the plant raises system cost; the fuel is what lowers it. Given the fuel, adding JERA's plant to a Kalaeloa conversion still costs more than converting alone. Converting the rest of the existing fleet instead roughly triples the saving (-1.05 total) for almost no new capital, because converted capacity supplies the same service at a higher heat rate with near-zero capital cost. This also explains the divergence from HSEO's decomposition, in which the new plant's efficiency is about half the benefit (Section 4.5): HSEO's counterfactual keeps burning oil in old steam units through 2044, so efficiency has a large margin to harvest; in a model free to substitute solar and storage, that margin is competed away and only the fuel-price channel survives.

The reverse question, whether to prefer an LSFO plant available at the LNG plant's price, has the same answer with the sign reversed. On fuel cost alone the LSFO plant loses by \$0.47 billion at matched 500 MW size and the capital midpoint (\$0.68 billion at bare-EPC). What the LSFO version buys instead is structure: no import terminal, no decades-long take-or-pay commitment, fuel from the existing in-state supply chain with its established biofuel transition path, and no single-supplier exposure. Whether that structure is worth the fuel premium at reference oil (less at low oil, more at high) is a judgment the Commission can now make with the tradeoff stated in dollars.

4.4 LNG was tried a decade ago and abandoned. What changed?

Hawaiʻi has been here before. In 2016 Hawaiian Electric held a signed 20-year agreement with FortisBC to import 800,000 tonnes of LNG per year from the Tilbury facility in British Columbia beginning in 2021. The program was expressly contingent on the NextEra Energy merger, and it was itself a take-or-pay: HECO was "obligated to take and pay for, or pay for, if not taken," 43.5 million MMBtu annually. Alongside it sat PUC applications for about \$341 million of unit conversions (\$450 million in 2024\$) and an \$859 million combined-cycle plant at Kahe (\$1.1 billion in 2024\$). When the Commission rejected the merger in July 2016, the utility terminated the fuel agreement and withdrew the conversion and plant applications within days (HEI Forms 8-K, May 18 and July 19, 2016), and its December 2016 plan update dropped LNG entirely; no one has revived the case since. That venture had 29 years of runway to the 2045 mandate. A venture starting today has 19, and faces the full 100 percent requirement rather than the interim milestones. On timing alone, the case should be weaker now than it was then. That the numbers are nonetheless close (Section 4.1) traces to two price shifts, both visible directly in this report's fuel inputs.

Long-term LNG is priced lower against crude than it used to be. Oil-indexed LNG contracts of the 2010s carried slopes of 13–14 percent of Brent; Qatari contracts signed since 2020 average 10–11 percent, with a 2020 minimum of 10.1 percent and a weighted average of 11.79 percent across disclosed contracts (Yusuf, Govindan, and Al-Ansari, *Heliyon*, 2024). The indicative contract in this analysis — 11.8 percent of Brent plus a small fixed charge (Section 1.2) — sits exactly at that modern average: JERA's pricing is the market's current term sheet. Sellers concede those slopes because the market has turned toward buyers. A record ~300 bcm/yr of new export capacity — nearly a 50 percent expansion — arrives by 2030, 70 percent of it from the United States and Qatar, more than the IEA's base case expects demand to absorb (IEA, *Gas 2025*). Meanwhile the markets that anchored the industry are leaving it: Japan, South Korea, and Europe — together more than half of world LNG demand — saw combined imports fall in 2023 with declines expected through 2030; Japan's imports are down 20 percent since 2018 to their lowest level since 2009, and its utilities' resales of surplus contracted cargoes to other countries have nearly tripled (IEEFA, *Global LNG Outlook 2024–2028*: "lackluster demand growth combined with a massive wave of new export capacity is poised to send global liquefied natural gas markets into oversupply within two years"). Suppliers facing that outlook compete hard for the creditworthy 20-year buyers who remain. (The 2026 war has sharpened rather than reversed this picture — demand fell under the supply shock and the U.S. export wave kept arriving; Section 8's closing observation carries the wartime evidence.)

The import infrastructure was already cheap a decade ago. Hawaiʻi Gas ran a global competitive bid in 2014 and published the results in January 2016: a chartered, third-party FSRU moored off Barbers Point, with the entire onshore package — buoy, subsea pipeline, and pipeline extensions to Kalaeloa, Kahe, and Waiau — estimated at $200 million ($260 million in 2024$), an all-in infrastructure adder of $1.20/MMBtu (about $1.60 in 2024$), and the FSRU departing at the end of a 15-year term ("no stranded assets"). JERA's proposed infrastructure today ($460 million; a $1.31/MMBtu regasification adder at larger scale) is the same order of magnitude in real terms. The floating model solved the terminal problem a decade ago.

What moved is the commodity term. The binding bid Hawaiʻi Gas held in mid-2015 implied a delivered formula of about 13.3 percent of Brent (Table 2 of its report, vendored in `sources/`); JERA's indicative formula is 11.8 percent. Run both at reference oil and the all-in delivered price falls from $13.65/MMBtu in 2016 dollars — roughly $17.8 in today's — to $11.43 (Table 1.1): about a third cheaper in real terms, nearly all of it the slope and the general deflation of LNG relative to oil.

LSFO has, meanwhile, become more expensive relative to crude, especially when crude is cheap. The current supply formula carries a slope of about 0.74 on Brent with a large fixed component (Box 4.1), so the fixed component becomes a larger share of the delivered price as crude falls: on the market's low path LSFO still costs about $8.8/MMBtu in 2030 with Brent near $23 (Table 1.1). The absolute saving from LNG holds near $5.2/MMBtu across the whole band, so in proportional terms the gap between the fuels is widest exactly where the earlier LNG case was weakest — in cheap-oil worlds.

What changed is the global market — Hawaiʻi's own runway, a decade shorter, moved the other way. Sellers of a fuel facing lasting decline in their core markets now offer terms attractive enough to make even a late, small, and remote buyer's arithmetic close. Both readings of that fact belong in the record. The terms are better than the ones Hawaiʻi walked away from. And the terms are better *because* the commodity's future is weaker — the same weakness that makes a 20-year take-or-pay commitment, and the offtake certainty it hands the seller (Section 4.6), worth pricing carefully.

4.5 Methodological comparison with HSEO's study

HSEO's May 2026 "Alternative Fuel, Repowering, and Energy Transition Study" (revised) is the most detailed public analysis of the fuel question, and within its frame its conclusion is correct: if a new thermal plant is built, LNG is the cheaper fuel to burn in it. Four features of the study limit what it can say about the broader decision:

The hydrogen/biodiesel asymmetry. HSEO's dispatch tables assign hydrogen ($40.66/MMBtu in 2050) exclusively to the LNG scenario and biodiesel ($63.84/MMBtu) exclusively to the no-LNG scenario, from 2045 onward. The asymmetry enters the headline through a $514M "avoided hydrogen capital" credit — which is most of Alternative 1A's $651M NPV; remove it (Alternative 2A) and the headline falls to $137M. Re-pricing the no-LNG biodiesel at the hydrogen price the LNG side enjoys flips the post-2044 comparison from roughly +$5.8B in LNG's favor to −$0.4B against.

Efficiency versus fuel price. Decomposing HSEO's $1.32B fuel-saving line: about 56 percent is the efficiency gain of a new combined cycle over old steam — available on any fuel — and 44 percent is the LNG price advantage.

Fuel-price tracks. Run through the Brent–LSFO and Brent–LNG relationships derived in the earlier brief (Roberts, 2026), HSEO's LSFO track implies roughly $50–65/bbl Brent while its LNG track implies $70–80 — two different oil worlds in one comparison; their analysis states no explicit Brent linkage. Their fuel-price assumptions lean moderately against LNG relative to ours.

The land cap. The study's solar ceiling is inherited from the utility's planning workbooks — NREL's PV-Alt-1 screen, one of twelve NREL defined (Section 2.6), whose criteria appear only in NREL's tables — and the study neither states those criteria nor tests the cap's sensitivity, though on NREL's own numbers the neighboring screen that admits military lands roughly doubles the ceiling. Their land screen is nonetheless very similar to ours, and their scenarios use only about a quarter of it for utility-scale solar under their pricing assumptions.

The menu. The Puʻuloa engine plant is absent from HSEO's inventory; the no-LNG counterfactual burns oil at scale through 2044 rather than substituting solar and storage. That is the comparison a lifecycle framework can run; it is answering the narrower question. (Full detail and workbook citations: Appendix A.6.)

> **Box 4.1 — the 2024 LSFO contract restructuring.** The August 2024 Second Amendment to Hawaiian Electric's 2022 fuel supply agreement with Par Hawaii (effective June 2025 on final PUC approval) cut the LSFO price slope on Brent to about 0.74 with higher intercept — better price protection for Par when crude is low, and a capped premium when it is high, worth roughly $70–75M/yr against the prior structure since it was signed. It narrows but does not close the LNG fuel gap (Table 1.1).

And the LSFO contract has no take-or-pay structure. An LNG contract's take-or-pay keeps the FSRU amortized (preserving the per-unit advantage, but reducing it), and it forces the dispatch volumes that dilute the system-level benefit. See the earlier UHERO brief Roberts, 2026.

Offshore wind HSEO's scenarios lean on 400 MW of offshore wind, which the study reports the model "preferred over other resources." Its cost basis is a 2023 ATB PPA figure carried over from the state's Decarbonization Report, which itself described its offshore wind cost and potential data as "preliminary and simplified assumptions" in need of refinement. Neither document explicitly addresses the features that dominate offshore economics at Oʻahu: water deep enough to require floating platforms rather than the fixed-bottom foundations behind most existing cost experience, and interconnection across that same deep water. Our model prices offshore wind at capital costs in the range of NREL's floating-platform classes, with no Hawaiʻi cost multiple, and it selects the resource only when utility solar is assumed to cost about twice mainland levels (our 1.7× sensitivity, close to HSEO's 2.154 multiplier); at that point it builds 200–270 MW, near what HSEO's model selects. At 1.8 times the mainland baseline (the 1.5× sensitivity) offshore wind enters only at the margin, about 200 MW in a few of the gas-heavy cells, and the least-cost paths still decline it.

The essential question is whether high solar costs stem from flawed procurement, and whether the same flaws would afflict offshore wind purchases. Nothing pins down a Hawaiʻi multiple for offshore wind (there is no local procurement record to estimate one from, for us or for HSEO), and the mainland baselines both models use rest on shakier ground than solar's: floating offshore wind has far less construction experience behind its cost estimates than utility-scale PV does. If procurement is the essential problem, offshore wind is unlikely to escape it.

The plans, side by side. Figure 4.4 sets our solved generation mixes beside the plans the institutions have published: both scenarios of Hawaiian Electric's Integrated Grid Plan and both cases of HSEO's study, the latter taken from the study's own results worksheets. Every bar is normalized to that plan's own annual generation and includes customer-sited solar, so the mixes compare cleanly even though the underlying demand forecasts differ.

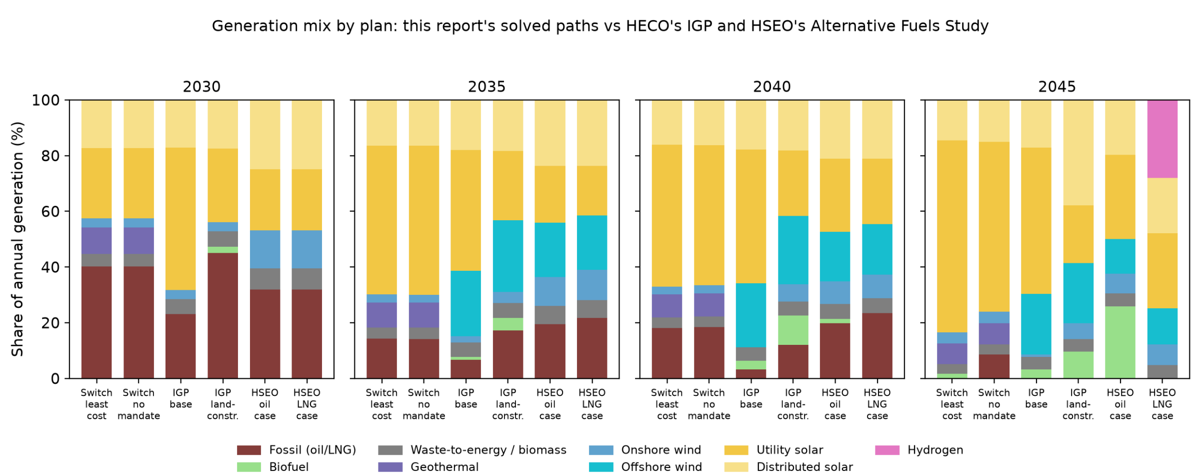


Figure 4.4 — Generation mix at anchor years: this report's least-cost and no-mandate paths (labeled "Switch" after the model; base rooftop trajectory, reference oil) against Hawaiian Electric's Integrated Grid Plan (Figure 2-3 of the May 2023 final report). The IGP plans are named here by scenario —

base and land-constrained — because the utility reversed which one it called "preferred": the May 2023 report gave that label to the base scenario, and the November 2023 Supplemental Response moved it to the land-constrained scenario, which is now the plan of record. The HSEO study's oil and LNG cases are drawn from the study's own results worksheets (vendored in `sources/plan_mix/`). Shares include customer-sited solar; the oil shares quoted in Section 2.7 are of grid supply alone and run a few points higher. In our solves, thermal generation with zero dispatch emissions is counted as biodiesel.

Three contrasts carry most of the information. First, everyone agrees on the destination: every plan reaches nearly zero fossil generation by 2045. The paths differ in pace. The IGP's base scenario — the one the utility did *not* adopt — de-fossilizes fastest early, 23 percent fossil by 2030, against 45 percent in the land-constrained plan of record; our least-cost path holds about 40 percent in 2030 and then moves fast (14 percent by 2035), because the optimizer waits for the solar and storage it buys to get cheap and for the credit window rather than building early at higher cost; HSEO sits between, at about 32 percent in 2030. The pace question matters for LNG: the slower the early clean build, the more fuel there is for a conversion to save (Sections 4.1, 4.7). Second, the instruments differ. The IGP and HSEO plans both lean heavily on offshore wind — 18 to 26 percent of their 2040 generation — the resource our model selects only under the doubled solar costs discussed above. Our least-cost build takes none at baseline costs, substituting more solar and the 100 MW of enhanced geothermal (8 to 9 percent of generation) that neither institution's plan considers. Third, the HSEO LNG case shows the same bridge-shape our conversion scenarios do — LNG peaks near 2,400 GWh a year around 2040, about a fifth of supply, and is gone by 2045 — but it then fills the gap with hydrogen at 28 percent of 2045 generation, an implied endgame far costlier than the solar-and-storage path every other column lands on.

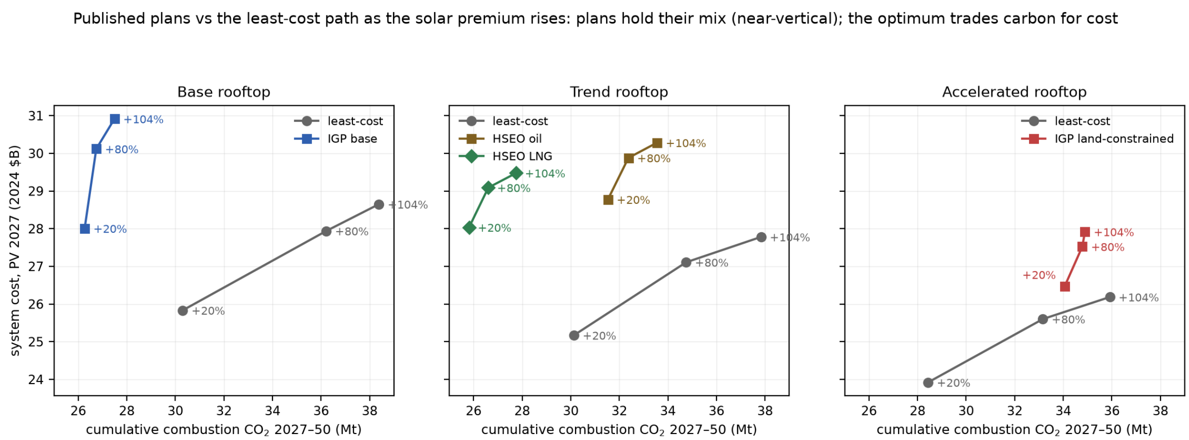
The figure also puts the mandate's role in one picture. Our least-cost and no-mandate paths are indistinguishable through 2040 — the cheapest build runs ahead of the requirement on its own — and split only at the end: without the mandate the model keeps about 9 percent fossil generation in 2045 (10 percent of grid supply) and retires it over the following decade on economics alone. That last step is what the mandate buys, and it costs about $0.25 billion in present value, roughly 0.2 cents per kilowatt-hour (Sections 2.7 and 4.8).

Pricing the plans. The mixes in Figure 4.4 can be priced. For each published plan we solve a scenario whose generation is constrained to that plan's mix: the plan's shares of its own grid supply, rescaled to the grid demand our matching rooftop trajectory leaves the utility to serve, and imposed each planning period as a band on utility solar and on total wind, a band on fossil generation through 2040, and a floor under the plan's firm clean energy from 2045. Dispatch, storage, and everything the plan does not specify stay optimized. The difference between that cell's cost and the least-cost build on identical settings is the plan's price tag: what following the plan costs over building the cheapest system that meets the same requirements, with both sides priced by the same model, the same fuel forecasts, and the same capital costs. Note that the least-cost builds import no LNG, neither a new plant nor the conversions of existing plants that Section 4.7 finds save money (but also impose unquantified risks), so the tags are, if anything, understated. The comparison each plan should be read against is the least-cost column at the same solar premium, because a planner who believes Hawaiʻi utility solar carries a 104 percent premium over mainland costs faces both columns at that price.

Two features of the construction need stating. First, wind is pinned as a total, not by type: the plans carry 257–287 MW of onshore wind, but the county setback ordinance caps onshore at about 150 MW in our framework (Section 2.6), so offshore stands in for onshore the ordinance forbids, and only the plans' combined wind is reproduced. Second, from 2045 the plans lean on firm clean energy — a quarter to a third of grid supply for the HSEO cases — which HSEO's study carries as biodiesel in its oil case and as imported hydrogen in its LNG case. We hold each plan to its firm *quantity* and let the model choose the fuel, for a reason given below; the model chooses biodiesel every time. Appendix A.15 documents the quotas, the contract carve-outs (Kalaeloa's minimum-take PPA sits outside the fossil band), and the tolerance basis.

Plan (vs least cost, same settings)	20% Hawaiʻi premium	104% Hawaiʻi premium
IGP land-constrained (plan of record)	+$2.55B	+$1.73B
IGP base	+$2.17B	+$2.27B
HSEO oil	+$3.60B	+$2.50B
HSEO LNG	+$2.86B	+$1.70B

Table 4.1 — Plan price tags: NPV system cost above the least-cost build at the same utility-solar premium over mainland ATB costs (2024$, EIA-reference oil; the 20 percent premium is the study baseline, Section 2.6).



The tags are large — roughly $1.7 to $3.6 billion, on a least-cost base of about 24 to 29 — and they are not purchasing emissions cuts. At baseline solar costs the HSEO oil plan is the starkest case: it costs $3.6 billion more than least cost and emits *more* — 31.5 against 30.1 Mt of cumulative combustion CO_2 through 2050 — because its early years keep more oil on the grid than the cheapest build would. The land-constrained plan of record shares the pattern at baseline costs: $2.6 billion above least cost and 5.6 Mt more emitted. Where a plan does cut cumulative emissions, the implied abatement price runs $170 to $670 per tonne of CO_2 (the one exception is the land-constrained plan at the high premium, whose cut is only 1.0 Mt and prices near $1,700). The one comparison that narrows with expensive solar is HSEO-LNG against least cost: at the 104 percent premium its tag falls to $1.7 billion, because the plan's offshore wind and firm fuel look better when the alternative's solar is dear.

The oil-versus-LNG pair is the comparison HSEO's study was built to make, and our framework agrees with its direction: the LNG case lands $0.7–0.8 billion cheaper than the oil case at every solar premium, and 5.7 Mt lower. About $0.14 billion of that gap — a fifth — is a fuel-price assumption rather than a physical difference: our forecasts hold LSFO at roughly 1.4 times the delivered price of LNG across the horizon, while HSEO's tracks converge to 1.2 by 2050, so the same plans priced on HSEO's fuel assumptions would sit about $0.14 billion closer (our LNG price includes the import infrastructure charge; the fuel-track discussion above).

The firm-fuel choice, and why we take it away from the plans. HSEO's post-2044 supply rests on a fuel its two cases do not share: biodiesel at $63.84 per MMBtu in the oil case, hydrogen at $40.66 in the LNG case, both externally supplied at assumed prices. Hydrogen at that price presupposes an import supply chain — liquefaction, shipping, storage — that does not exist at scale today, serving an island that would be among its first buyers; we regard that as somewhere between very uncertain and unlikely, and our framework does not assume it. Here hydrogen must be made: electrolyzers, tanks, and fuel cells are available to the model at reference costs, charged as capital, powered by whatever electricity the build can spare (Section 3's hydrogen scenarios build exactly this way, and remain feasible). Offered that choice, the plan cells never make hydrogen: with utility solar and wind pinned to the plans' own levels there is no surplus generation to electrolyze, and hydrogen made from biodiesel-fired electricity returns about a third of the fuel burned, so biodiesel burned directly is always cheaper. Both HSEO plans, priced here, therefore meet their post-2044 firm requirement with the same fuel — which has the useful side effect of making the oil and LNG cases directly comparable, freed of the hydrogen-price assumption that would otherwise do much of the work (the same asymmetry noted in this section's first bullet). Readers who expect cheap imported hydrogen to materialize should read both tags as upper bounds on that world.

Box 4.2 — What the plans' own price tags say. Hawaiian Electric's IGP prices its plans directly (Table 9-1 of the May 2023 report): a revenue requirement of $29.40 billion NPV for the base plan, $30.36 billion for the land-constrained plan of record (+3%), and $33.89 billion for the status quo (+15%). Those totals are not comparable to ours — different demand, different discounting, different scope — but their internal structure is instructive in three ways.

First, the two plans are not directly comparable *with each other* in the IGP's own accounting: the land-constrained plan's utility serves about 26 percent less energy in 2045 (customer DER supplies 4,082 GWh against the base plan's 1,878 on near-equal totals, Tables 2-3/2-4), and the rooftop capital that substitution requires is customer-paid and appears in no revenue requirement. A plan can look cheap by moving spending off the utility's books. Our cross-family cells put both plans on a common rooftop trajectory before comparing, and on common demand the ordering reverses the IGP's: the land-constrained plan of record costs $0.4–0.8 billion more than the base plan and emits 8 to 11 Mt more (both ranges span which trajectory the plans share) — the fossil and firm energy it substitutes for the utility solar it forgoes, priced and emitted.

Second, the transmission line. The base plan carries $9.77 billion (nominal) of transmission capital against the land-constrained plan's $2.35 billion and the status quo's $0.82 billion — roughly $6.0 billion in 2024 dollars, most of it in 2045–2050. Set against each plan's total capital program (Tables 9-2 through 9-4), transmission is 63 percent of the base plan's capital, 29 percent of the land-constrained plan's, and

12 percent of the status quo's. That last figure is the ordinary one — transmission is the smallest component of what U.S. electricity customers pay, about 13 percent of the delivered price against 58 for generation and 28 for distribution (EIA, AEO 2023 Table 8, 2022 values; those are shares of price, not of capital, so the comparison is loose). The cleaner benchmark is internal to the same tables: continuing as-is puts 12 percent of capital into wires, the land-constrained plan 29 percent, and the base plan 63 percent — five times the status quo's share. Even the land-constrained plan, which builds the least of the three, roughly doubles the wires share of a continue-as-is program. (The IGP labels both Tables 9-2 and 9-4 a "Preferred Plan": the term designates the chosen plan *within* each scenario, not a choice between scenarios, and the utility's plan of record is now the land-constrained one.) The IGP's own text prices renewable-zone enablement at $950–1,100 per kW; our solved builds carry solar at $1,511 per kW *including* the generation. Its own non-wires analysis identifies avoidable segments: the Archer cable replacement avoidable with 37 MW of storage or demand response, Ewa Nui reducible from 324 to 175 MW, Wahiawa by 220 MW, and a $3,980M 2040–2045 network expansion whose stability contribution is listed as "Not studied." For scale: Hawaiian Electric's entire Oʻahu net utility plant — every wire, pole, and plant it operates today — stands at $5.96 billion ($9.29 billion gross of depreciation, at historical cost; HEI 2024 10-K, utility-only balance sheet). A regulated utility earns its allowed return on the capital it deploys, so it profits more from building infrastructure than from avoiding the need for it — the incentive economists call the Averch–Johnson effect. A plan that spends a new grid's worth of capital on transmission at least raises the question, though performance-based regulation mutes the incentive, and settling it requires modeling the alternatives — storage, demand response, generation sited to need less wire — priced against the line. At national scale that comparison has been made, in the same model family used here: an optimal U.S. transmission buildout would more than triple interregional capacity yet lowers the cost of a zero-emissions system by only about 7 percent, because storage and generation siting substitute closely for wires (Zheng, Schivley, Fripp, and Roberts, *Applied Energy* 421(15), 2026). Oʻahu is one small zone rather than a continent, so the number does not transfer — but the mechanism is what v2 will test here.

Third, the status quo. HECO prices continuing as-is at 15 percent above its base plan. Our equivalent — solar and wind frozen at contracted projects, conservative rooftop growth, no clean-energy requirement — lands 19 percent above least cost with more than triple the emissions (108 against 30 Mt), or 4 percent above when LNG is allowed in. The direction agrees; the magnitude says the cost of standing still is, if anything, larger in our framework than in the utility's.

A fourth comparison no accounting here captures: the rooftop capital itself. The accelerated trajectory adds about 1,120 MW of customer solar over the base one by 2050 — $2.1–2.4 billion of present-value capital at Hawaiʻi residential prices (Section 2.3) that appears in no plan's revenue requirement and no cell of Table 4.1. Version 2 of this model is being built to close both gaps: transmission expansion optimized jointly with storage, demand response, and generation siting — rather than inherited from the plan being priced — and a full both-sides-of-the-meter accounting of customer capital.

4.6 Considerations the analysis does not capture

Contract structure. Whatever formula the LNG contract carries — indexed to world oil or mainland gas — it is likely written to insure the supplier's return. The proposal does not disclose the contract's term or volume provisions, so the model gives the contract the most favorable treatment that still recovers the supplier's capital: import volumes are re-optimized every period against every alternative (2035 imports range from 5.9 to 23.4 million MMBtu across the oil paths), the delivered price stays at the contract floor no matter how far volumes fall, and the $460 million of import infrastructure is recovered in full over 2030–2044. Any actual contract that recovers that capital — with minimum-take floors, diversion fees, or volume penalties — can only cost the system weakly more than these solutions. A contract more favorable than this would require the supplier to absorb unrecovered capital without pricing that risk. The delivered price this analysis carries ($11.4/MMBtu incl. regasification, roughly half the wartime Pacific spot — JKM $21/MMBtu, July 17, 2026) is the contract-indexed floor; the downside protection in such contracts accrues to the seller. The asymmetry is inherent in the contracting stage, not in any party's conduct: before financing closes, either side can exit an LNG contract at modest cost — JERA's March 2026 termination of its Commonwealth LNG agreement, routine as pre-construction exits are, shows how ordinary that flexibility is at that stage — while a buyer that has built the terminal and unwound its oil logistics has no comparable exit.

Hawaiʻi's position after building the terminal would be different in kind from its position today. The current LSFO supply arrangement with Par Hawaii is a requirements-style contract: Hawaiian Electric buys what it requires; Par carries a supply obligation of 13,500 barrels per day of LSFO at the formula price (a ceiling on Par's Tier 1 supply obligation — Hawaiian Electric's purchases follow its actual requirements); self-supply is permitted above it; and no take-or-pay or minimum-purchase obligation appears in the public contract text or in HEI's disclosed purchase commitments (Second Amendment, August 14, 2024, SEC Exhibit 10.1; some pricing clauses remain redacted). The system can therefore reduce oil purchases as renewables grow; a take-or-pay LNG contract removes exactly that freedom. The contrast extends to term: the Par agreement runs through January 2029 with one-year extensions, while the LNG proposal is built around twenty years of FSRU operation.

Execution and supply chains. The renewable-heavy paths assume the buildout can be executed: by 2035 the no-new-plant path carries roughly 6,200 MWh of bulk storage and about 2,200 MW of utility solar, and by 2050 roughly four to five times the battery capacity of the largest completed solar-plus-storage project in the United States (Edwards & Sanborn, 3,287 MWh). Nothing in the model prices construction logistics, port throughput, specialized labor, or global battery supply chains. But the constraint these numbers point to is process, not physics. Solar with storage is as fast to build as anything on the grid: in the engineering estimates behind EIA's own outlook, utility PV and batteries take about twelve months to construct — only onshore wind is comparable — against 22–30 for gas turbines and combined cycles, five years for coal, and seven for nuclear (EIA, Capital Cost and Performance Characteristics, 2024; vendored in sources/), and solar and storage have supplied the majority of all new U.S. generating capacity in each of the last two years (EIA, Today in Energy, Feb. 2025). Texas, which runs interconnection on connect-and-manage rules (Section 2.4), grew its utility-scale solar fleet from 1.9 to more than 20 gigawatts in five years — its additions in 2024 alone were more than double what this report's least-cost path builds by 2050. Oʻahu's slower pace comes from the procurement and interconnection process this report documents (Sections 2.2, 2.4, 2.8): a multi-year RFP cycle in which the incumbent utility runs the competition and manages the queue while being the party whose generation the entrants displace. Speed serves almost no participant with influence over that process, and the reforms of Section 2.8 are aimed at exactly this. The residual risks that reform cannot reach — battery supply chains, port capacity, labor — cut against every pathway that builds at scale (the JERA path builds most of the same solar, a decade later), and the 1.5×/1.7× solar-cost cases price the world in which they persist.

Employment. The clean-energy path is by far the more labor-intensive; the job-year arithmetic is in Section 8 and Appendix A.9.

The utility's finances. JERA's 500 MW would join a grid already served by independent generators (the planned 99 MW Puʻuloa engine plant, Kalaeloa, H-Power, and smaller producers), leaving Hawaiian Electric's own generation — old units carrying substantial rate base from decades of capitalized upgrades — with little to do. Whether the Commission would continue to allow capital recovery on plants that no longer run is an open question with two uncomfortable answers: continued recovery has customers paying for JERA's contract and idle HECO plant at once; disallowance imposes a write-off on shareholders already exposed at Waiau. Section 8 weighs whose loss that would be and what follows from it; the immediate point is that the question should be resolved openly and in advance, before any contract is signed.

The refinery. The LNG path displaces Hawaiian Electric's LSFO purchases nearly completely and quickly; the clean-energy path displaces them gradually. What rides on that difference — the economics of Par's product slate (the mix of fuels the refinery makes), direct jobs, in-state fuel supply, and the biofuel joint venture — is set out in Section 8.

4.7 If LNG comes: the cheapest configurations use existing plants

The proposal bundles three separable decisions — import LNG, build a new plant, and size it at 500 MW. Separating them changes the picture, and the three decisions respond to oil prices in different ways.

Switching fuel and buying efficiency are distinct economic propositions. The value of conversion is the price gap between the two fuels, times the volume burned. Because gas is priced at about 11.8 percent of Brent and the fuel oil Hawaiʻi buys tracks Brent at nearly the same per-unit rate, that gap stays close to \$6.60 per million Btu across the whole range of oil prices we test (Appendix A.14). The value of a new, more efficient plant is the fuel a better heat rate saves, which is worth more when fuel is expensive. Section 4.1 traces how the two respond differently as oil prices move.

One condition attaches. The near-constant fuel gap follows from oil-indexed pricing. A contract indexed to a gas hub instead, such as Henry Hub plus liquefaction, would let the gap move with the oil-to-gas price ratio, and conversion economics would then depend on oil prices after all. HECO's own LNG program a decade ago planned exactly this. The 2014 Power Supply Improvement Plan assumed Kahe 1–6 and Waiau 5–10 "converted to use LNG beginning in 2017," with the independent Kalaeloa plant converted at Hawaiian Electric's expense; in May 2016 Hawaiian Electric asked the PUC to approve about \$341 million for unit conversions (four sites across three islands) and \$117 million for LNG shipping containers, withdrawing both requests that July when the NextEra merger collapsed (PSIP 2014; HEI Forms 8-K, May 18 and July 19, 2016). The Kalaeloa combined cycle sits adjacent to the proposed FSRU landing at Campbell Industrial Park. We test these configurations directly: the import terminal is built, existing units are permitted to burn gas (at zero conversion cost — an upper bound; actual conversion capital would reduce the saving), and no new plant of any kind is built. One case converts Kalaeloa alone; a second extends conversion to the units closest to HECO's own former program — Kahe 5 and 6 and the CIP combustion turbine.

Configuration (reference oil)	System cost ($B)	vs no-new-plant
No new fuel plant, no LNG	25.83	—
FSRU + Kalaeloa conversion, no new plant	25.45	−0.39
FSRU + Kalaeloa, Kahe 5 & 6, CIP CT conversions, no new plant	24.79	**−1.05**
...same, net of the full 2016 conversion-program charge ($0.45B)	*25.24*	***−0.60***
...same, also charging the entire 2016 onshore package ($0.26B)	*25.50*	*−0.34*
JERA 500 (bare-EPC), no conversions	26.37	+0.54
JERA 500 (bare-EPC) + Kalaeloa conversion	25.92	+0.09

At JERA's cost quote, the conversion configurations beat building the new plant. Kalaeloa alone saves $0.39 billion where the new plant adds $0.54: the model routes about 200 million MMBtu of LNG through Kalaeloa's existing units — running them at 80–90 percent capacity factor into the early 2030s and 60–80 percent through 2044 — with no new construction at all. Extending conversion to Kahe 5 and 6 and the CIP turbine makes the no-new-plant configuration the cheapest LNG arrangement tested; on the credited basis the bare-EPC plant does not save at all (+0.54). And the terminal needs no mandate to be used this way: in a variant where LNG import is offered as an option rather than forced, the model activates the terminal and the Kalaeloa conversion on its own, reaching the same saving (−0.39, identical to the forced case within tolerance). At the quoted fuel price, the terminal pays for itself through conversions alone.

Because conversion capital is set to zero, each saving doubles as a break-even budget: the most that conversion, refurbishment, remaining-life, contract-renegotiation, and gas-delivery costs could total before the configuration stops paying. Against the no-LNG baseline the budgets are large — about $1,800 per kilowatt, both for Kalaeloa's 220 MW and for the additional 370 MW of Kahe and CIP capacity — each a substantial fraction of the cost of building a new plant outright (and about a fifth higher in as-spent dollars, since capital spent near 2030 discounts against these 2027 present values). Real costs are far smaller. Delivery geography favors these particular units — Kalaeloa and the CIP turbine sit adjacent to the proposed FSRU landing at Campbell Industrial Park, and Kahe, roughly ten miles up the coast, needs only a pipeline extension that has been priced before: Hawaiʻi Gas's competitively bid 2016 onshore package (Section 4.4) came in at $200 million ($260 million in 2024$). Conversion capital has a 2016 benchmark too: the $341 million ($450 million in 2024$) program of HECO's withdrawn May 2016 request, above. Both sit an order of magnitude inside the budgets. (The terminal, mooring, and regasification infrastructure is already charged in these runs; plant laterals and conversions are not.)

These 2016 estimates permit a direct net-of-capital assessment without re-running anything, exactly as the EGS cost sensitivity is handled: a capital charge against the solved saving. The table's net rows charge the **entire** 2016 conversion program at face value against our three-unit Oʻahu configuration alone, although it covered far more capacity (Kahe 1–6 and Waiau 5–10 among others). The full conversion set still nets **−$0.60 billion**. The stricter bound adds the entire 2016 onshore package on top, even though the JERA runs already charge $460 million of import infrastructure that covers most of the same functions; the saving is still **−$0.34 billion**. Both charges are conservative by construction: the 2016 program's $450 million covered roughly 1,300 MW of conversions (≈$0.35M per MW), so charging it all against our 590 MW set implies about $0.76M per MW, roughly twice the program's own rate, and over $1.2M per MW with the onshore package added. Every direction of approximation here overstates conversion cost. On the credited basis the central ranking is not fragile: "Kalaeloa conversion beats building JERA's plant" rests on a margin of $0.73 billion at bare-EPC capital — about $3,300/kW of converted capacity — which realistic conversion costs are unlikely to consume. What remains fragile is the fine ordering among the conversion configurations themselves, some of which differ by less than the solve tolerance. The configuration also has precedent: it is nearly the shape Hawaiʻi Gas proposed in 2016 — a temporary chartered FSRU, gas to Kalaeloa, Kahe, and Waiau through pipeline extensions, no new plant, and the vessel gone at the end of the contract.

Three caveats temper the result further: no per-unit Oʻahu conversion figures were ever published, so the $341 million is a program total rather than a project estimate, and the units most likely to exceed their share of the charged allowance are the oldest — Kahe 5 and 6 are 1970s-era steam plants whose refurbishment and remaining-life costs are the least predictable part of the package. Conversion is effectively one-way, since dual-fuel combustors pair gas with distillate rather than with residual oil, so converted units would give up their LSFO capability. And ownership and contract status shape who captures the saving — Kalaeloa is an independent producer whose amended PPA (executed 2021, approved November 2022) runs ten contract years from 2023, roughly the same window the LNG tier occupies, and a biofuel-capable repowering of the plant was in active contract negotiation as of July 2026 (Hawaiian Electric release, July 9, 2026) — a path a gas conversion would compete with directly; the Kahe and CIP units are HECO's own. The finding that survives the caveats: the fuel benefit and the new plant are separable, and the proposal's value, if any, lies in the fuel — which strengthens the case for evaluating the terminal, the plant, and the contract as distinct decisions rather than a single bundle. Separability carries a procurement corollary: if Hawaiʻi proceeds with LNG in any form, the terminal and its services should be put out to competitive bid rather than accepted as one proposer's package — Hawaiʻi Gas's 2014 global invitation shows that a competitive process for exactly this service is feasible here, and the current buyers' market (Section 4.4) is the right moment to run one.

The 2016 bid also shows the contract shape that fits the mandate: a chartered FSRU whose term ends as the 100 percent requirement binds, leaving no stranded asset — and its onshore package, with the delivery pipelines the conversion pathway needs included, came in about 40 percent below the $436 million (the $460 million headline figure deflated to 2024$) of import infrastructure in JERA's proposal. Its per-unit adder ran higher ($1.60/MMBtu in 2024$ against JERA's $1.31) because its volumes were smaller, and both figures assume zero overruns. The conversion cells above already carry this contract shape — the infrastructure charge is levied only while the LNG tier operates, 2030–2044 (Appendix A.8) — so their savings are computed under the lease-to-the-mandate structure; repricing the terminal at the 2016-bid level would add roughly $0.15–0.2 billion more. A leased terminal on that template, feeding converted plants until 2044, is the configuration that captures the fuel benefit while retaining the State's exit.

4.8 If the clean-energy mandate were abandoned

The strongest case for LNG arises if Hawaiʻi walks away from the 2045 requirement. We solve that case directly: the RPS is removed, LNG may run past 2044, and the model chooses plant size and fuel volumes freely.

No-mandate configuration (reference oil)	System cost ($B)	vs no-mandate baseline
No new fuel plant (no gas available)	25.58	—
LNG unrestricted (model's choice)	25.43	−0.15
JERA 500 forced — bare-EPC / +20%	25.60 / 26.13	+0.02 / +0.55

Without the mandate, the model builds 500 MW of gas capacity, imports 15–21 million MMBtu of LNG per year through 2050, and the LNG advantage is $0.15 billion against the no-gas baseline. Even with the mandate gone, *forcing* the JERA bundle does not pay: at the vendor's own bare-EPC quote it roughly breaks even (+$0.02 billion), and at the +20 percent sensitivity it costs $0.55 billion. Sunshine keeps most of the market either way: the no-mandate system still builds about 2,200 MW of utility solar (54 percent of the mandated build) with gas fully available, and 3,740 MW (92 percent) without it.

The more striking number is how little is at stake. With no gas option on the menu, dropping the rule saves just $0.25 billion (25.83 against 25.58) — about 0.19 cents per kilowatt-hour; the larger figure arises only because abandonment also unlocks unrestricted gas: **abandoning the mandate saves about $0.41 billion over twenty-four years — roughly three tenths of a cent per kilowatt-hour — when the replacement is a 500 MW gas plant with an import terminal.** The configurations in this table exclude the conversion of existing plants, which Section 4.7 shows is the cheaper use of LNG. Solving the no-mandate case with conversions on the menu completes the picture. Without the mandate, the model converts Kalaeloa, Kahe 5 and 6, and the CIP turbine and reaches $24.12 billion; the with-mandate conversion configuration reaches $24.79 billion, so on a like-for-like menu the mandate itself costs about $0.67 billion, roughly half a cent per kilowatt-hour. Offered a new plant on top of conversions, the no-mandate model declines the 2030 proposal entirely, adding only about 250 MW around 2045 as the converted units age out, worth about $0.03 billion — within the solve tolerance. Most of the apparent gain from abandonment therefore comes from the conversion decision, which the mandate permits:

converting saves about $1.05 billion with the requirement kept in place. These comparisons set conversion capital to zero on both sides, so it cancels; all of them assume LNG at the contracted price (about $11.4 per MMBtu delivered) for the duration, while the July 2026 spot price is roughly double that. Rooftop growth is part of why the stake shrank: distributed solar and storage reduce the energy any new plant would serve, and the no-mandate gas build is half what it was on a gross-load basis. For comparison, letting solar-and-storage deployment costs persist at today's procurement-implied level costs $2.11 billion (Section 2.1; the 1.5× sensitivity, i.e. roughly 1.8× the mainland ATB benchmark once the baseline's 20 percent Hawaiʻi premium is included — approximately the effective level implied by the contract prices approved since 2024, Section 2.3). Walking away from the clean-energy commitment is worth half of what fixing procurement is worth. And the ranking of levers survives the mandate's removal: with no clean-energy requirement at all, cheaper solar deployment remains the largest cost reducer, because solar carries most of the load in every world. The money is in fixing procurement — with the mandate or without it.

Two readings follow. For LNG proponents: the proposal's economics improve several-fold in a no-mandate world. The case for the proposal should state that dependence. For policy: what the mandate buys — a fully clean grid five to ten years sooner — costs about two tenths of a cent per kilowatt-hour, and the decisions that dominate bills lie elsewhere.

4.9 Emissions and the pace of decarbonization

Counting only combustion on Oʻahu — the accounting most favorable to LNG, with no upstream methane — the two paths run close: 30.0 Mt for JERA (bare-EPC), 30.5 for the +20% case, against 31.3 for no-new-plant over 2027–2050, so the LNG path is about 0.8–1.3 Mt lower on combustion CO_2. LNG displaces oil early (about 0.8 Mt/yr cleaner around 2030) and displaces solar and geothermal in the middle years (about 0.6 Mt/yr dirtier around 2035), with the RPS forcing both paths to zero by 2045 (Figure 4.1). The credited base case pulls both totals well below the earlier no-credit figures, as cheaper storage and geothermal displace more oil.

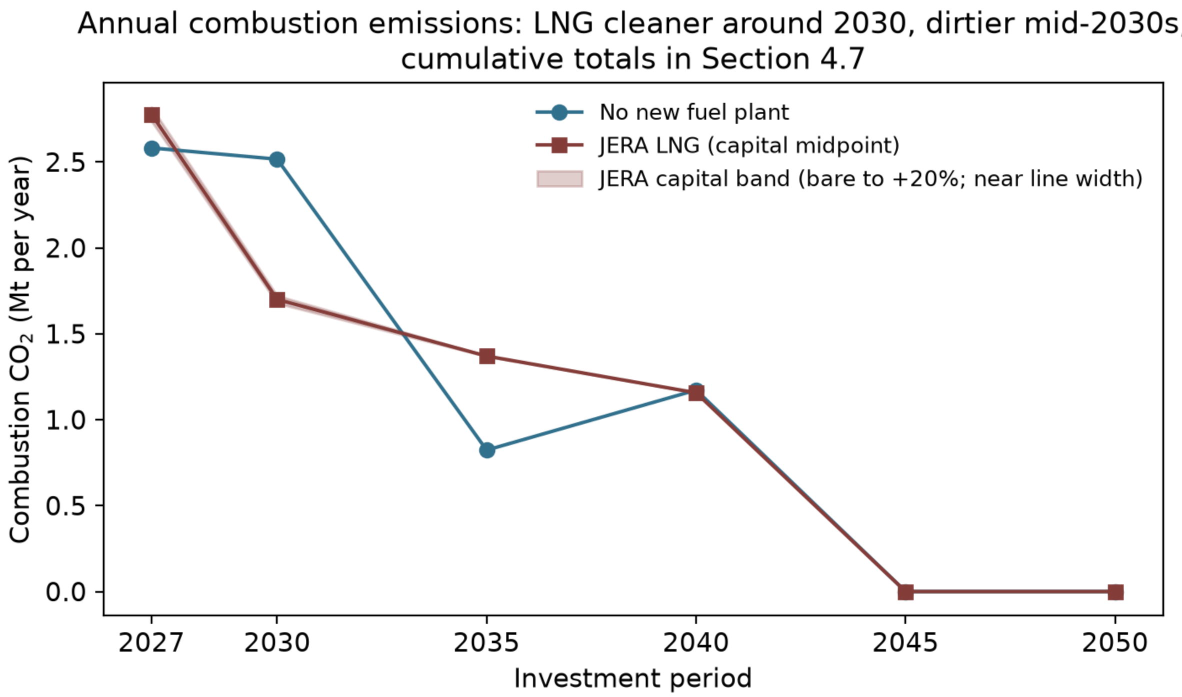


Figure 4.1 — Annual combustion CO_2 on the no-new-plant and JERA LNG pathways. Combustion CO_2 (million tonnes per year) by investment period at reference oil prices. The JERA line is the capital midpoint; the shaded band spans bare-EPC to +20 percent capital and is close to line width. Combustion only; upstream methane is treated in the text. Both paths reach zero by 2045 under the clean-energy mandate.

The clearer difference is the pace of the transition. In 2035 the no-new-plant path generates 83 percent of Oʻahu's electricity from renewables; the JERA path, 58 percent — a 25-point gap (by the model's own renewable-share metric) that narrows through the 2040s. The LNG path defers roughly a decade of clean-energy deployment, and its cumulative-CO_2 parity depends on the mandate forcing the same endpoint.

Upstream methane overcomes LNG's combustion edge, by an amount that depends on a debatable question of incidence. Natural gas is mostly methane, and some share leaks from wells, gathering, processing, liquefaction, and shipping. Appendix A.10 carries the calculation: the JERA path imports about 246 million MMBtu of LNG over the horizon, roughly 4.7 million tonnes of methane throughput. Each percentage point of supply-chain leakage adds about 1.4 Mt CO_2-equivalent at a 100-year warming potential, or about 3.9 Mt at the 20-year potential — against a combustion gap of 0.8–1.3 Mt in LNG's favor. LNG's greenhouse advantage therefore reverses at leakage above roughly nine-tenths of one percent (100-year basis) or one-third of one percent (20-year basis) — thresholds at or below every published measurement of U.S. supply chains (Sherwin et al. 2024: the lowest basin measured, Appalachia, sits at 0.75 percent and the production-weighted average at 2.95 percent).

How far above depends on whose gas is counted. Counting only the literal cargoes Hawaiʻi would buy — plausibly sourced from lower-leakage suppliers — the addition may sit at the modest end; this is the accounting in HSEO's study, whose scenarios also assume LNG displaces oil and imported fuels rather than solar. The economically meaningful accounting asks what production expands when global LNG demand grows by one buyer. U.S. exports are the growing margin of world LNG supply (EIA projects ~30 percent growth by 2027), drawn from the Haynesville and from Permian associated gas. Measured U.S. supply-chain leakage is well above the tie-breaking thresholds: Sherwin et al. (2024, *Nature*), from nearly one million aerial site measurements, report basin-level loss rates from 0.75 percent (Appalachia) to 9.6 percent (New Mexico Permian), with a production-weighted average of 2.95 percent across the six regions measured. On the literal accounting the LNG path's greenhouse effect is somewhat worse than the clean path's; on the marginal accounting it is much worse. Appendix A.10 tabulates the range at 1, 3, and 6 percent leakage under both warming potentials.

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5. Reliability under a renewables-dominant grid

The headline findings depend on the system meeting load every hour, including hours when sun and wind are scarce. The model solves for the cheapest mix that delivers reliability, timepoint by timepoint.

5.1 How reliability is tested

Reliability in the model means three things at once: energy balance at every timepoint; operating reserves at every timepoint (spinning contingency at 5 percent of load plus half the largest online unit, non-spinning at the same level, regulation at 1 percent, providable by committed thermal, quick-start units, and batteries with the required energy margin; solar and wind are excluded from reserve provision); and unit commitment (minimum up/down times, ramp limits, minimum loads) on every thermal unit. A configuration that fails any test at any timepoint is infeasible. Because solving every hour of 2027–2050 is computationally prohibitive, each investment period is represented by thirteen sample days at two-hour resolution, following the Switch-Hawaiʻi design (Fripp 2020; Imelda, Fripp & Roberts 2024): twelve days chosen by K-means clustering on the 2007–2008 record, plus the single most difficult day of that record — November 22, 2008, low sun, weak trades, evening peak — found by dispatching a candidate system against every hour of both years at a $5,000/MWh penalty on unserved load, and carried at full weight in every scenario thereafter (Appendix A.5).

5.2 The easy day and the hard day

The 2035 annual peak (≈1,271 MW, hot summer evening) is straightforward: hot days are sunny days, so peak demand and peak solar coincide; batteries charge through the midday surplus and discharge through the evening. The binding day is the low-renewable one. On the November 22 profile, solar output falls to roughly a quarter of the peak day's, wind to a third, and the system carries the day with the thermal fleet run harder, the new plant (in trajectories that build one) near-continuous, and storage shifting the reduced midday solar into the evening. Figure 5.1 shows both days for the no-new-plant base case.

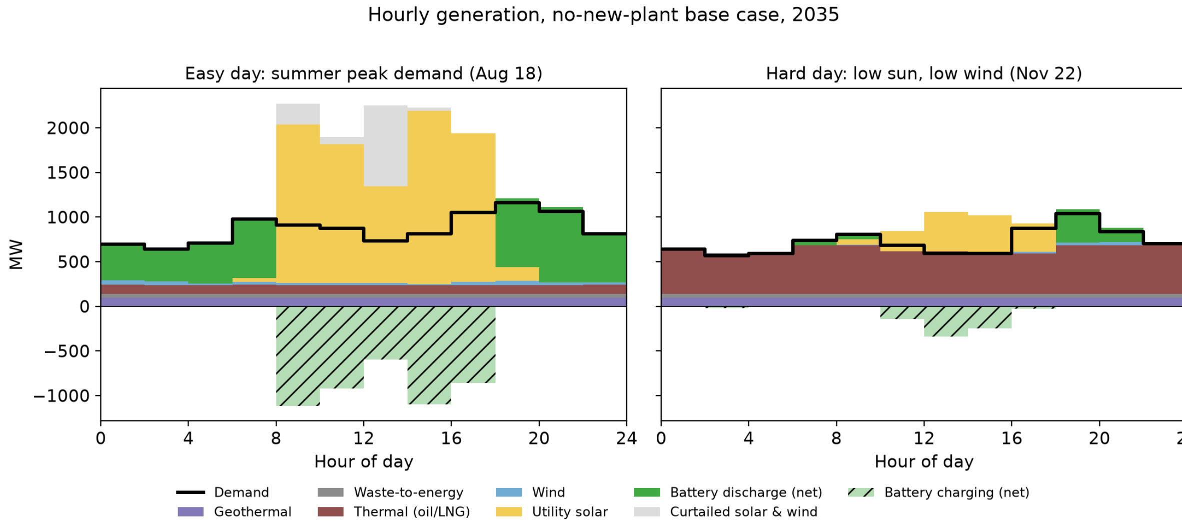

Figure 5.1 — Hourly dispatch on the two 2035 sample days, no-new-plant base case. Stacked generation (MW) by two-hour model timepoint on the summer peak-demand day (August 18) and the low-sun, low-wind day (November 22), rendered as in the results explorer. The black step line is demand; the hatched bars below zero are net battery charging; the gray band is curtailed solar and wind, potential output above what was dispatched. Battery flows are shown net: in surplus hours the optimizer may charge and discharge simultaneously, a costless way to shed surplus, which the net view removes; the round-trip losses of that disposal are counted as curtailment. Generation sources stack in the legend's order.

What firm capacity actually has to cover. A common claim is that a renewables-dominant grid needs enough backup thermal capacity to meet peak demand. The model, and the demand data behind it, show otherwise. What firm generation has to cover is closer to the highest *daily-average net load* — demand minus solar and wind, averaged over the day — because batteries move energy within the day from surplus hours to deficit hours. In 2035 the annual peak demand is about 1,271 MW, but the largest daily-average net load across all sample days is about 610 MW, less than half of it, and it falls on the low-renewable November day (right panel), where the existing thermal fleet runs at a nearly flat ~600 MW while storage and the day's smaller solar output handle the peaks. Three things drive the gap between peak demand and the net load that sizes firm capacity: batteries smooth within-day variation, so firm generation follows the daily average rather than the momentary peak; low-sun, low-wind days tend to be cooler, lower-demand days (the hard day's peak demand, ~1,040 MW, is the lowest of the year, not the highest); and even the cloudiest days still deliver meaningful solar. This metric is conservative for Oʻahu, where storage could in principle draw on neighboring days, which the sample-day design does not allow (Section 5.4). The build carries ≈2,100 MWh of bulk storage (modern-plant trajectory) to ≈6,200 MWh (no-new-plant) by 2035, reaching ≈13,700–18,100 MWh by 2050. (Storage totals are loosely pinned: they move by up to ~40 percent between the 0.25% and 0.1% solutions while total cost moves under 0.2 percent, so read them as indicative of scale.) Because each sample day starts from a reset battery state, the configuration meets back-to-back recurrences of the worst day without inter-day banking.

One offsetting effect appears in the demand data. As rooftop systems have grown, grid demand on very low-sun days has begun to run above what installed capacity alone would predict —

about 11 MW at 2021–2024 penetration — because households that self-supply on ordinary days draw from the grid on dark ones (Appendix A.11). The effect offsets roughly a sixth of the low-sun demand relief described above and grows with the distributed fleet; the sample-day tests in this section use net loads that carry the full distributed profile, so the mechanism is inside the model rather than an unmodeled risk.

5.3 What the test does and does not cover

The design enforces feasibility on the historical worst day but does not test multi-day events more severe than that record, generator or fuel-supply contingencies, or correlated storm damage; those are follow-on work (Section 8). Two omissions run conservative for the renewable-heavy paths: no inter-day storage carryover, and no real-time pricing or system-wide demand response — which prior work finds reduces high-renewable system costs six to twelve times more than fossil-system costs (Imelda, Fripp & Roberts 2024). We are currently building the input data needed to test resiliency over many years.

A grid built on solar, wind, and batteries needs a different kind of resilience planning. Traditional planning guards against the failure of large generators at the hour of peak demand. It sizes reserves to the biggest unit and tracks each unit's outage rate. A renewable grid spreads generation across thousands of small pieces: panels, inverters, battery containers, rooftops. No single failure among them matters much. The main risk becomes weather. Bad weather hits the whole island at once, can last for days, and can be seen coming days ahead. The hard hours also move, from the hot afternoon to the evening and the hours around dawn (Sections 5.2 and 6.3). The limit that matters becomes stored energy rather than generating capacity. Operators manage the charge in the batteries, fill them before expected stress, and schedule maintenance in the hours with room to spare. The new weak point is a long stretch of cloudy, windless days that drains the batteries. The builds in this report already carry the recorded version of that stress. Every solved system serves demand and reserves through the hardest day of the two-year record, each sample day starts from the batteries' overnight reserve, and any build here could therefore repeat its worst day indefinitely. Plant failures have little reason to correlate with clouds or wind, so treating them as independent shocks is reasonable. On that basis the chance of coming up short is about 0.01 to 0.1 hours per year (Section 6.3), twenty-five to five hundred times inside the strictest standard in use, Great Britain's three hours per year. Forecasting adds a further layer. Hard hours are visible days ahead, batteries can be filled before them, and after one plant fails, stored energy can be positioned before the next tight evening. Demand response would add more, and this report leaves it out entirely. Fuel supply still matters for the remaining oil- and biofuel-burning plants. Two limits remain: the weather record covers two years (a longer meteorological record contains a 16-day run of low wind), and a storm can couple failures. v2 tests both, with many weather years in sequence, random unit failures, and maintenance schedules chosen inside the model, and it will pin the numbers down more precisely. The answer already in hand is clear: the reliability of these systems exceeds the world's strictest standards by an order of magnitude or more.

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6. The Waiau Repower decision

The project is approved (D&O 42411, March 2026); the live question is scope. Two findings and one open question bear on it.

6.1 System cost and who pays

Forcing the repower into the build raises system cost by $1.40 billion at reference oil ($1.38–1.49 across oil paths). The model prices the project at Hawaiian Electric's stated construction cost ($1.155B; $4,545/kW) — the resource cost of building it — while the Commission's recoverable-cost cap (the $847M approved bid, $931.7M absolute ceiling — Section 1.1) determines who pays: roughly $220–310 million falls to shareholders. The plant runs at 51 percent capacity factor in 2030, falls to 27–32 percent through the late 2030s, and drops below 1 percent from 2045 as renewables and more efficient units displace it.

6.2 The price, and the technology comparison

A modern combined-cycle plant on the existing fuel supply prices at roughly $2,900/kW on this report's basis (Section 4.3), about a third below the repower's $4,545/kW, with a heat rate near 6,900 Btu/kWh against the repower's simple-cycle ≈9,500. A reviewer with long Hawaiʻi regulatory experience pressed both halves of that comparison, and both points belong in the record. The machines do different jobs. Simple-cycle turbines start and ramp fast and suit peaking and contingency duty; a combined cycle suits sustained operation. Batteries reduce the value of that speed. Storage responds faster than any turbine, supplies contingency and regulation reserves in the model, and every solved build carries it at scale — 2,100 to 6,200 MWh by 2035, rising to 13,700 to 18,100 MWh by 2050 (Section 5.2). The model also prices flexibility directly: ramp limits, minimum loads, minimum up and down times, and start-stop apply to every thermal unit (Section 5.1), so the worth of a fast fleet is in the solved costs. The role distinction sharpens the price question. Simple-cycle capacity is normally the cheaper machine per kilowatt: Lazard's June 2025 review carries $1,150–1,450/kW for a new-build gas peaker and $1,200–1,600 for a combined cycle (its market-quote commentary and any Hawaiʻi premium would raise both). The repower prices simple-cycle capacity at $4,545/kW, three to four times the peaker benchmark and above every combined-cycle figure in the same review. The system-cost gap tells the same story: no-new-plant beats the repower by $1.40 billion, and no bundle rescues it — every Waiau-containing configuration inherits the penalty (Table ES.1).

6.3 Unit size and resiliency: the open question

The strongest argument for the repower is its shape, and it stands outside the cost comparison. Six small units spread outage risk; one large unit concentrates it. On a grid Oʻahu's size, unit size is a reliability variable in its own right, because the largest single contingency sets the reserve everything else must hold. The model credits part of this. Contingency reserves scale with the largest unit online (Section 5.1), so a fleet of small units carries a smaller reserve burden than a fleet anchored by a large one, and quick-start turbines help supply those reserves; the $1.40 billion penalty is measured with those credits in place.

The model leaves failure itself unpriced: no unit in v1 ever suffers a forced outage (Section 5.3). The omission is large. Hawaiian Electric's 2021 and 2022 Adequacy of Supply filings carry forward-looking equivalent forced-outage rates (EFORd, the probability a unit is unavailable when called on) of 7 to 25 percent across its fleet, with four-year actuals reaching 39 percent; Table 6.1 gives the unit detail. Correlated failures compound the per-unit rates: the January 2024 rolling outages came from losses across a single weather system, and the August 2024 Kalaeloa pipe rupture removed all 208 MW of the island's largest independent plant at once (V2.md documents the record). Reviewers also recall a neighbor-island turbine out of service for more than two years in the 1990s, with months of rolling blackouts while it was shipped abroad for repair — the kind of rare, long outage that many small units protect against.

Table 6.1 — Oʻahu's firm fleet: size, age, efficiency, and outage rates. Capacity, in-service year, retirement, full-load heat rate (MMBtu/MWh), and modeled derates (scheduled/forced) are from the public model inputs; retirements match the utility's announced deactivation schedule, and years beyond 2050 fall outside the model horizon. Filed EFORd is the forward-looking rate in the 2021 and 2022 Adequacy of Supply filings (Table 3), with four-year actuals in parentheses; the filings report Kahe 1–4 and Waiau 7–8 as a single actuals group ("grp"). Bracketed values are the IGP's August 2021 planning assumptions for units the AOS tables omit — independent producers never appear in them. Heat rates are full-load; average operating rates run higher (Section 4.3). Cogeneration heat rates are net of steam credit, and the large scheduled derates on H-Power and the refinery cogens are availability calibrations carried from the Switch-Hawaiʻi inputs.

Unit(s)	MW	In service	Age 2026	Retires	Full-load HR	Modeled derate	Filed EFORd (actual)
Kahe 1	78	1963	63	2033	10.1	0 / 0	10% (grp 14.4%)
Kahe 2	78	1964	62	2033	9.9	0 / 0	10% (grp 14.4%)
Kahe 3	82	1970	56	2037	9.7	0 / 0	10% (grp 14.4%)
Kahe 4	87	1972	54	2037	9.9	0 / 0	10% (grp 14.4%)
Kahe 5	128	1974	52	2046	9.7	0 / 0	7% (12.3–13%)
Kahe 6	129	1981	45	2046	10.1	0 / 0	7% (12.3–13%)
Waiau 7	78	1966	60	2029	10.6	0 / 0	10% (grp 14.4%)
Waiau 8	78	1968	58	2029	10.2	0 / 0	10% (grp 14.4%)
Waiau 9 (CT)	51	1973	53	2055	13.1	0 / 0	8.2% (10–11.3%)
Waiau 10 (CT)	51	1973	53	2055	12.5	0 / 0	8.2% (10–11.3%)
CIP CT	113	2009	17	2070	11.7	0 / 0	4.2% (4.2%)
Airport DSG	8	2013	13	2074	10.2	0 / 0	[5%]

Unit(s)	MW	In service	Age 2026	Retires	Full-load HR	Modeled derate	Filed EFORd (actual)
Schofield (6 engines)	49	2018	8	2048	8.8	0 / 0	2%
Kalaeloa CC (2 GT + 1 ST)	220	1989–91	35–37	2055	8.6	0 / 0	[1.5%]
H-Power (waste-to-energy)	86	1989	37	2055	—	50% / 0	[3%]
Refinery cogens (2)	33	1982–90	36–44	2045	~4.7	12–17% / 0	—
Puʻuloa engines (11 × 9)	99	2027	new	2057	8.5	3% / 2%	—
Candidate: Waiau Repower (6 × 42)	252	—	—	—	9.9	3% / 2%	—
Candidate: JERA CC (4 × 125)	500	—	—	—	6.9	5% / 0	—
Candidate: LSFO CC comparator (4 × 125)	500	—	—	—	7.2	5% / 3.5%	—

Table 6.1 takes stock of the fleet those filings describe. The island is long on firm units now: about 1,350 MW of thermal capability across some two dozen units, rising to about 1,450 MW when the Puʻuloa engines arrive in 2027, against a peak the model puts near 1,270 MW in 2035 and a binding daily-average net load of about 610 MW (Section 5.2). Age and outage risk concentrate together, and the retirement schedule sheds the worst first: the units with the worst filed rates — Waiau 3 and 4, at 25 percent forward in the 2022 filing with actuals to 39 percent — were deactivated in 2024 and are not in the model; Waiau 7 and 8 (10 percent forward) retire in 2029, Kahe 1 and 2 in 2033, Kahe 3 and 4 in 2037. The fleet the high-renewable future inherits

is the younger, lower-rate tail: Kalaeloa, the CIP turbine, Schofield, Pu‘uloa, and the newest Kahe units.

The modeled-derate column also states our own treatment plainly, and it is inconsistent in a way v2 will correct: existing utility units are never derated, the candidate plants carry 2 to 3.5 percent forced-outage derates, and the JERA plant carries a 5 percent scheduled derate but no forced-outage derate. None of these matches the filings. Zero derating flatters the existing fleet, and the missing forced-outage derate flatters the JERA plant relative to its own LSFO comparator.

A pilot re-solve prices the missing derates. We applied the filed rates of Table 6.1 to every unit — the utility's forward rates for its own plants, the IGP planning rates for the independents, and 3.5 percent to the JERA plant — and re-solved nine headline cells at the 0.1 percent tolerance (inputs, scenario lists, and results are in the repository: gen_info_efor.csv, results/EFOR_PILOT.csv). No total cost moved by more than $12 million on systems of $24 to $28 billion. Every headline comparison reproduced: at reference oil the JERA band is +$0.53 to +$0.96 billion derated against +$0.54 to +$0.96 billion published, the Waiau penalty $1.39 billion against $1.40, and the conversion saving $1.04 billion against $1.05. The result matches the mechanics: a derate prices average energy availability, the fleet carries slack in energy and capacity alike, and the questions that matter for resiliency are the coincidence and correlation questions taken up below.

Some of the recent stress is scheduling as much as hardware. The 2021 filing attributes rising rates partly to units "operated in ways for which they were not designed" as cycling duty grows, and its actuals adjustment cites maintenance disrupted through the pandemic. The hour that matters has also moved: Section 5.2 shows the binding net-load peak sits in the evening, away from the gross peak that maintenance calendars were built around, and v2 treats maintenance timing as a decision co-optimized with the outage draws (V2.md).

Timing separates the near-term options. Solar and storage construct in about a year (Section 4.6); a new thermal unit waits in a multi-year turbine queue (Section 4.2). Additions started now can be online before Waiau 7 and 8 retire in 2029. The repower arrives after them. For the next five years, the insurance available on time is storage and solar added to an existing fleet held in good repair.

Pricing outages would cut hardest against large single units, and the 500 MW proposal concentrates more unit risk than anything else on the menu. The same pricing would credit configurations made of many small pieces: small thermal units, and batteries, which come in smaller pieces still and which every build already carries. Two questions follow, and v1 cannot answer them. Would an outage-aware security criterion return any of the repower's $1.40 billion? And if resiliency is the product, are six new turbines the least-cost way to buy it, against the alternatives the model already builds — batteries, the Pu‘uloa engine plant, and existing units held as reserve? v2 is designed to answer both: unit-level outage draws in a chronological simulation, a loss-of-load screen on every solved build, and a stress day with the largest unit out while the second largest is on maintenance (V2.md).

Sizing the tail. The event that matters is a coincidence: several units forced out at once, in one of the few hours with a small surplus. Both pieces can be sized from data already in hand. In the solved no-new-plant base case, the committed spinning stack exceeds its requirement — which itself always covers the largest online unit plus a variability adder (Section 5.1) — by a median of about 560 MW, with hours weighted by how many days of the year each sample day represents. The cushion falls below 50 MW in about 7 percent of hours — the weighting barely moves the share, because the thin hours land on ordinary evenings, and rare days contribute little — and they cluster tightly: the evening ramp and the hours around dawn, the same sliver of the day every time. Behind the spinning stack stand a non-spinning requirement of the same size, storage energy, and units available but not committed.

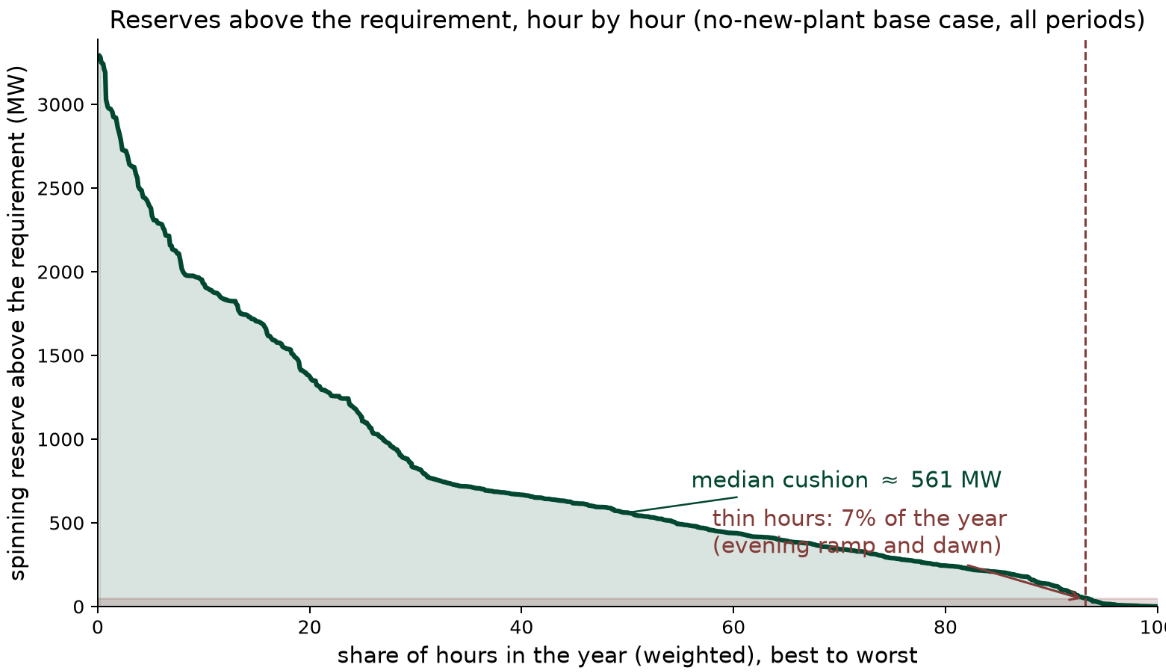

Figure 6.1 — Spinning reserves above the requirement, hour by hour. Each point is one modeled timepoint of the no-new-plant base case, sorted from largest cushion to smallest and weighted by the number of days its sample day represents. The requirement covers the largest online unit plus a variability adder (Section 5.1). The shaded band marks cushions under 50 MW: about 7 percent of hours, at the evening ramp and around dawn. Built by `build/fig_reserve_cushion.py` from the solved outputs.

On the outage side, treat each unit as independently unavailable at its Table 6.1 rate. In the 2035 fleet the expected capacity forced out at any moment is about 60 MW. The chance that outages exceed one large unit (130 MW) at a given moment is about 13 percent — that is the event the reserve requirement is built to absorb, and it absorbs it by construction. Exceeding two large units (230 MW) has a probability near 2.4 percent, and three (330 MW) near 0.3 percent. Multiplying by the weighted thin-hour share: a second-large-unit outage landing on a thin evening hour comes out to roughly fifteen hours a year, and a three-unit coincidence to about two hours a year. Those hours test the second layer of defense described above. Lost load requires that layer to fail too.

Conventional standards give the yardstick. North American planners build to a loss-of-load expectation of 0.1 days per year, and the utility's grid plan has adopted that target for its probabilistic tests. European standards run 3 to 8 expected unserved hours per year: Great Britain uses 3, Ireland 8. Hawaiian Electric's older guideline was 4.5 years per day, about 0.22 days per year. Lost load in this system requires roughly 430 to 530 MW out at once during a thin hour, enough to consume the cushion, the largest-unit reserve, and the non-spinning layer together. The independent-failure arithmetic puts that coincidence at 0.006 to 0.12 hours per year, one to two orders of magnitude inside every standard above. Correlated failure would have to raise the multi-unit odds by a factor of thirty or more to close the gap to the strictest standard. The January 2024 event says the factor exceeds one. The v2 simulation measures it.

Two biases run in opposite directions, and both are large. Independence understates correlated failure — the January 2024 event was multiple units and dimmed solar in one weather system, and the August 2024 Kalaeloa outage took 208 MW with a single pipe. Counting every broken unit overstates harm, because in a high-renewable system most of the fleet is idle most of the time, and a unit that fails while idle costs nothing. Anticipation overstates it further: forced outages persist for days and are known to the operator, tight evening hours are forecastable well ahead, and storage can be charged into a foreseen tight window. The solved schedule already positions batteries within each day, and once a unit is down, the system re-commits and re-charges around it — so a second failure usually meets a grid already braced for the first. A derate prices average unavailability. Coincidence is the question, and simulation is the tool for it. Monte Carlo draws whole outage histories — units, durations, repairs — against years of synchronized weather and demand, counts the hours load is actually lost, and lets maintenance move to the hours that can afford it. v2 implements exactly this (V2.md). We specifically invite comment on this question — how the resiliency value of many small units should be quantified, and against which alternatives — before version 1 locks.

6.4 For the proceeding

Every Waiau-containing bundle is more expensive than its Waiau-free counterpart by $1.38–1.49 billion (Table ES.1). The substantive questions open to the parties are whether the approved scope remains the least-cost way to meet the need it was approved to address, and whether the proceeding remains open enough to substitute a configuration that meets that need at lower cost. If the repower's value is resiliency, the proceeding should say so and price it against the alternatives that provide it (Section 6.3). The recoverable-cost gap and its incidence (Section 4.6, "utility finances") sharpen the stakes on both sides.

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7. Open questions and indicators worth watching

Questions the analysis raises for the proceedings. What in the procurement, permitting, and interconnection process accounts for the soft-cost gap, and which parts can Act 266 implementation reach? What would it take to learn whether Oʻahu's EGS resource lands on the favorable cost trajectory before the 2034–36 demonstration window closes? Should the plant, the terminal, and the fuel contract in the JERA proposal be evaluated as separable decisions (Sections 4.7–4.8 make the case that they are)? Is the Waiau scope still the least-cost answer to the need it was approved for, and how should the resiliency value of many small units be weighed against batteries, the Puʻuloa engine plant, and existing units held in reserve (Section 6.3)? And how should LSFO-contract renewals be evaluated so the current structure — a requirements contract without volume lock-in (verified against the public contract text; Section 4.6) — is not surrendered inadvertently?

Indicators. Brent realizations against the reference path; Pacific LNG contract terminations and renegotiations (the Commonwealth precedent); delivered costs of gas plants contracted into the current turbine market (a check on the proposal's quote); Fervo Cape Station's delivered cost and schedule; Stage-4 RFP pricing against the 21–23 cents per kilowatt-hour of the contracts approved since 2024; Act 266 implementation milestones; and the Pacific spot-LNG-versus-LSFO spread — as of July 17, 2026 the JKM marker sits at \$21/MMBtu, above delivered LSFO, a war premium worth watching as it unwinds.

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8. Conclusions and perspective

What the analysis establishes. On system cost, under current law the JERA proposal costs modestly more than building no new fuel plant. Taking the midpoint of JERA's own cost range — their bare-EPC estimate and their +20% sensitivity, which restores the customs, insurance, design-allowance and contingency items the estimate itself says it excludes — the JERA bundle is about \$0.75 billion more expensive over twenty-four years, and more expensive in every oil-price case tested. The bundle earns back roughly half its up-front capital in fuel savings. In an open, competitive market, no firm would make such an investment; regulated electricity is not such a market. Per kilowatt-hour delivered, the difference is about half a cent — small against a bill of thirty-plus cents, so compensating benefits in reduced emissions, energy security, or local economic spillovers might justify the investment. But on these measures a new combined-cycle plant, even at JERA's competitive price, looks worse, not better. A case might remain for LNG paired with retrofits of existing plants. The savings there are real, but they would be unnoticeable on most customers' bills — maybe a half-cent per kWh — and the off-model implications are as likely to reduce the benefits as increase them. Clearer findings from the analysis are that the Waiau Repower is uneconomic under every oil price tested (+\$1.4 billion); if any new plant is built it should be smaller than JERA's 500 MW; solar-and-storage procurement reform is worth several times any fuel decision; and, under current law, Enhanced Geothermal is in the least-cost build with meaningful value and a bounded downside, provided the technology is accepted by the community.

When the cost gap is small, the decision rests on structure — and the structures are not symmetric. The no-new-thermal path is an option-rich position: it commits to nothing irreversible, its inputs (solar, storage, geothermal) keep getting cheaper on every documented trajectory, and its "fuel" — procurement reform — is within the State's own control. The LNG path is an option-poor position: a two-decade commitment to a single supplier, under a confidential contract, tied to infrastructure with one use. Its advertised price should be read as a floor: such contracts are written to insure the supplier's return (Section 4.6). The bare-EPC cost quote in JERA's proposal carries the same asymmetry: it explicitly excludes contingency, insurance, customs, and design allowance, categories that history says run over (three of four electricity projects exceed their estimates; Sovacool, Gilbert & Nugent 2014). The asymmetry is easiest to see in timing: before construction, contract exits are routine and cheap (JERA's own Commonwealth termination, Section 4.6), while Hawaiʻi, having built the terminal and unwound its oil logistics, could not mirror that exit. None of this is priced in the model, and all of it weighs against the same side.

On emissions, the combustion accounting is neutral — and the tie does not survive the gas field. Counting only what is burned on Oʻahu, the LNG path and the clean-energy path produce similar cumulative CO_2 through 2050: LNG's efficiency and lower carbon intensity displace oil early (roughly −0.8 Mt/yr around 2030), but the plant also displaces solar and storage that would otherwise have been built, leaving the island's power about 25 percentage points less renewable in 2035 (about +0.6 Mt/yr) — a decade of deferred clean energy, with the ledger closing only because the RPS forces both paths to 100 percent by 2045. Combustion, though, is only part of the ledger. Natural gas is mostly methane, a far more potent greenhouse gas over the decision-relevant decades, and some of it leaks — from wells, gathering lines, processing, liquefaction, and shipping. How much depends on where the gas comes from, and here the debate splits on a question of incidence. If one counts only the *literal source* — the specific cargoes Hawaiʻi would buy, plausibly from relatively low-leakage suppliers — the upstream penalty may be modest. But the economically meaningful question is the *marginal source*: when global LNG demand rises by one buyer, which production expands to meet it? U.S. exports are the growing margin of world LNG supply, drawn from the Haynesville and from Permian associated gas (EIA 2026), and measured U.S. leakage rates are far above the tie-breaking thresholds — 0.75 to 9.6 percent by basin, 2.95 percent production-weighted (Sherwin et al. 2024, *Nature*, ~1M aerial site measurements). On that incidence, the upstream penalty ranges from material to severe. We do not put a single number on it; the range is wide and reasonable people will weigh the incidence question differently. The direction, though, is clear: any nonzero leakage breaks the combustion tie against LNG, and the marginal-source reading breaks it badly.

The costs the model does not price fall disproportionately on one side. Three sit outside the capacity-expansion frame and deserve weight in the decision. *Local employment.* The clean-energy path is the labor-intensive one: on the coefficients documented in Appendix A.9, the solar-and-storage buildout supports on the order of 29,000–43,000 local construction job-years and 500–1,000 permanent positions by 2050, against roughly 1,500–3,000 construction job-years and 50–80 permanent operating positions for the JERA bundle — international FSRU and gas-trading operations are commonly staffed by experienced international crews, and how much of JERA's operation would be hired locally is unresolved in the proposal. *The finances of the utility itself.* JERA's 500 MW would arrive on a grid that already carries a fleet of independent generators — Puʻuloa, Kalaeloa, H-Power, and smaller IPPs. Add the JERA plant and the island's firm-capacity need is met almost entirely by non-utility plants: Hawaiian Electric's own generation — old units, but carrying substantial rate base from decades of capitalized upgrades and refurbishments — would be rendered idle: in the JERA scenarios HECO's own fleet runs at 0.3 percent capacity factor from 2030 onward, against 26 percent in 2030 falling to 4 percent by 2035 on the no-new-plant path. Whether the Commission would continue to allow capital recovery on plants that no longer run is an open question with no good answer: continued recovery means customers pay twice — for JERA's contract and for idle HECO plants; disallowance means a write-off that could be devastating to HECO shareholders, on top of the Waiau exposure documented in Section 6 — and a utility whose investors have just absorbed a stranding event pays more for capital on everything it builds thereafter, including the wires and grid modernization every pathway needs. The no-new-thermal path retires the same fleet, but gradually, on its depreciation schedule, with the existing units earning their keep as reserve capacity through the transition.

A note on standpoint. The stranding question looks different depending on whose welfare is counted. From the standpoint of total cost, disallowed capital recovery destroys no resources: the plants are built, the money spent; disallowance changes only who bears a sunk cost. From the standpoint of Hawaiʻi residents specifically, the distinction matters more: Hawaiian Electric's shareholders are predominantly institutional and largely out-of-state (~73 percent institutional per 2025 13F aggregations), so a write-down of legacy rate base would function, in substantial part, as a transfer from external investors to local ratepayers — one large enough, on some readings, to offset much of the new capital the transition requires. We do not advocate that outcome, and two considerations weigh against welcoming it: a utility whose investors absorb an unplanned stranding pays more for capital on everything it builds afterward, a premium that returns to ratepayers through the wires and grid investment every pathway needs; and cost recovery on prudently incurred investment is the regulatory bargain that makes private capital willing to fund public infrastructure at all. HEI's post-settlement condition sharpens the point. The wildfire settlement fixed the company's liability at $2 billion, and it made the first of four payments in April 2026 (HEI Q1 2026 results). Moody's upgraded the utility on that progress, but only to Ba1 — still below investment grade — and the stock, though stabilized, remains far below its pre-fire level. The Legislature has authorized a cap on liability for future catastrophic wildfires (Act 258 of 2025, applying only to fires that destroy more than 500 structures, with no cap on liability for death or injury), but the Public Utilities Commission is still writing the implementing rules and has so far recommended against a companion recovery fund (PUC, December 2025 study; June 2026 rulemaking). A company in this position has little capacity to absorb another shock, whether from a fire or from stranded capital. The point of raising it is narrower: the treatment of legacy generation capital is a critical aspect of the LNG decision, its incidence is contested, and it should be settled openly and in advance by the Commission, before either party discovers it holds the loss. *The refinery.* Displacing Hawaiian Electric's LSFO demand, which the LNG path does immediately and nearly completely, changes the slate economics of the State's only refinery, with consequences for roughly 660 direct jobs (Par Hawaiʻi's stated statewide workforce), in-state jet-fuel and gasoline supply, and the Hawaiʻi Renewables joint venture that is one of the few in-state pathways for the biofuel volumes every 2045 scenario needs (Section 4.6). None of these three considerations is decisive alone, but all three point the same direction, and the cost comparison leaves no LNG saving to offset them.

A final observation on the moment this decision arrives in. The war in Iran has stress-tested the world's LNG market in real time, and the results bear directly on how Hawaiʻi should read the offer in front of it. The closure of the Strait of Hormuz in March 2026 took roughly a fifth of the world's LNG shipments off the water and damaged the Ras Laffan complex; loadings from Qatar and the UAE fell 35 billion cubic meters year-on-year between March and June (IEA, *Gas Market Report Q3-2026*). Spot prices did what a supply shock does — European TTF up 32 percent year-on-year to about $16/MMBtu, Asian spot averaging 45 percent higher (IEA), with the JKM marker at $21/MMBtu on July 17, 2026, up more than 60 percent on the year. And yet the deeper response ran the other way. Oil peaked near $120 a barrel against wartime forecasts of $150–200 (Borenstein, Energy Institute, June 2026), and global gas demand is *falling* in 2026 — by about half a percent, the third annual decline this decade (IEA). The demand side absorbed the shock, and much of the absorption looks permanent: Japan's monthly imports slid from 6.2 million tonnes in January to 4.0 in May, 15 percent below a year earlier, with coal imports up 14 percent (Ministry of Finance data via LNG Prime); China's demand fell about 4 percent March through June; Europe's 2026 demand is down more than 2 percent — driven, the IEA notes, by renewable expansion as much as by price; and Asia-Pacific LNG demand is now in its second consecutive annual decline (278 → 268 → 257 million tonnes, 2024–2026), a retreat Wood Mackenzie characterizes as "structural responses rather than purely tactical ones." The oil market makes the same point more sharply. World oil demand is expected to fall by about a million barrels per day this year, the first decline since 2020 (IEA, July 2026). In sixty years of data, declines like this have come only in and around recessions: every prior year of falling world oil demand saw world growth at or below 2.6 percent, and the deepest declines came with growth near or below zero. The IMF expects 2026 growth near 3 percent (Figure 8.1). Rationing without substitutes would have caused a recession, so a decline this large in a growing economy suggests energy users are finding alternatives. Where inventory data are public, the LNG decline looks like reduced consumption rather than delayed purchases: Japan cut imports while holding stocks above their five-year average. Not all of it will stick. Europe's decline is mostly deferred buying, and Qatari volumes return when the Strait reopens. A supply glut within a year of a Hormuz settlement would not surprise us, though no one knows when that settlement will come.

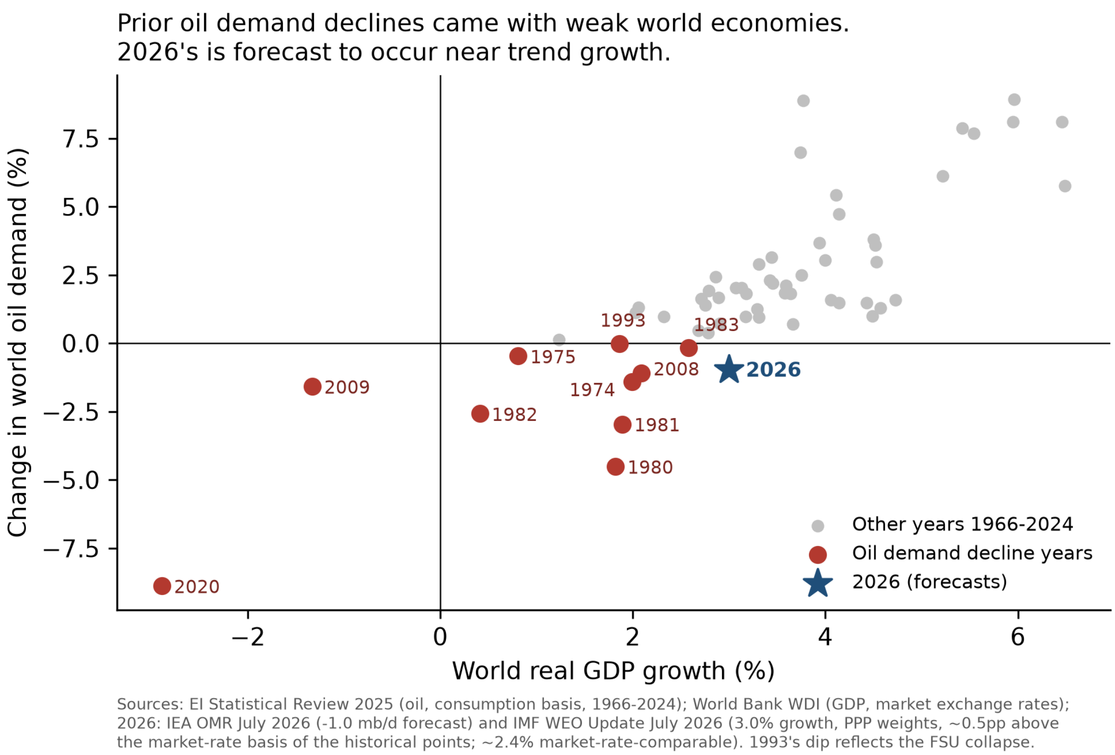

Figure 8.1 — World oil-demand changes and world GDP growth, 1966–2026. Each point is one year: the percent change in world oil demand (vertical) against world real GDP growth (horizontal). Red points mark the years demand fell; the star is 2026, combining the IEA's July 2026 demand forecast (−1.0 mb/d) with the IMF's July 2026 growth projection (about 3 percent). Demand history is the EI Statistical Review (consumption basis); GDP is World Bank data. Prior declines

cluster at weak or negative growth; 2026's sits near trend.

Nor is the retreat austerity: a record 664 GW of solar was installed worldwide in 2025 as the global fleet passed three terawatts (SolarPower Europe), and the war has accelerated the substitution where it hurts most — in March 2026, the war's first full month, China's clean-technology exports doubled to 68 GW, with shipments to Africa up 176 percent month-on-month and 55 countries setting all-time purchase records, while a survey of Philippine solar installers found weekly installations up 70 percent since the conflict began (AP, May 13, 2026; Ember trade data via Newser). The fuel-price shock is driving the very buyers abandoning LNG toward its substitute. Meanwhile the American supply wave arrives on schedule regardless — record export months in spring 2026, with U.S. net gas exports forecast to grow nearly 30 percent by 2027 as Corpus Christi Stage 3, Golden Pass, Port Arthur, and Rio Grande ramp (EIA, April 2026) — capacity that is permanent because the steel is already in the ground. Against that wave, the IEA puts the war's cumulative supply losses through 2030 at about 140 bcm — roughly 15 percent of the new export capacity expected over the same window. When the dust settles, the world faces the pre-war arithmetic with fewer buyers: a record supply expansion meeting demand that fell under stress and, in its largest markets, shows no sign of returning. Today's high spot prices are a war premium on a commodity in lasting retreat. That is the context for reading a 20-year contract offer at 11.8 percent of Brent: the terms are attractive relative to history because they are the seller's hedge against a falling market. None of this decides Hawaiʻi's question by itself. But it sets the negotiating posture. If the State proceeds with LNG in any form, it is a buyer in a deepening buyers' market: it should put the terminal out to competitive bid (Section 4.7), treat the offered slope as an opening bid rather than a final one, and bargain for terms that share the downside the seller is hedging.

The bigger picture is the one this report began with. The largest economic lever on Oʻahu's electricity costs is the cost of deploying solar and storage, which sits roughly \$2.1–2.8 billion above what the fundamentals support for reasons — procurement cycles, interconnection queues, permitting, land policy — that are within Hawaiʻi's own power to fix. With that reform delivered, the no-new-thermal path is plausibly the least-cost path outright, and it comes with insulation from fuel-price shocks in a world that keeps supplying them. Hawaiʻi imports every barrel of oil it burns and would import every cargo of LNG. Sunlight is the only major energy input produced at home. The weather-driven variability of a renewables-dominant grid, which the model meets hour-by-hour on the hardest day in the historical record, is by comparison a solved engineering problem. With the right political will, Hawaiʻi can likely decarbonize faster, and more cheaply relative to its conventional generation, than any other state. Within the error of our tools, the analysis suggests that the cheapest version of Hawaiʻi's future and the cleanest one are virtually the same.

What we recommend. (1) Treat solar-and-storage procurement reform (Act 266 implementation, interconnection throughput, land policy) as the most consequential energy decision before the Commission and the Legislature. It dominates everything else in this report; get the pricing mechanism right and it could be transformative. (2) It probably doesn't make sense to build a new power plant of any kind beyond the anticipated 99 MW plant at Puʻuloa, but LNG might be entertained further in conjunction with conversion of existing plants. The stakes here are smaller than some believe, and some of the tradeoffs are subtle. There would be little downside to collecting competitive bids for leased facilities and the best possible contract terms, and then making a decision. (3) Revisit the Waiau Repower scope to whatever extent Docket 2025-0211 allows; it is the clearest negative-value item in the cost analysis (the resiliency argument for its six small units is Section 6.3's open question, and quantifying it is v2 work). (4) Fund the Enhanced Geothermal demonstration pathway; the option is cheap and the payoff asymmetric.

What would change our minds. We state these so readers can hold us to them: (1) sustained delivered-LNG prices well below the contract floor we model with clear terms that would ensure delivery of contracted fuel even under major disruptions to world fuel markets (like today); (2) solar procurement reform failing, so that even with simplified PPA terms, streamlined interconnection, and universal purchase at real-time avoided cost of any clean energy provided to the grid by any customer or third party seller, solar cannot be procured in the necessary quantities for prices less than double those typical in other states; (3) detailed clarification of which lands are available for solar and which lands are not, showing the quantity of inventory this report assumes (not necessarily the specific parcels counted) is overstated. The companion land use repository is the place to report information about specific parcels or sets of parcels.

All code, data, and analysis in the report are public. We invite anyone — including those who disagree — to change our minds with better evidence, and we commit to publishing whatever the numbers say.

What we will do next, and an invitation

v1 — this repository

- **Source-vendoring completion**: the Lazard CCGT table, the EGS resource screen, the rooftop-potential derivation, the Par contract structure, HEI ownership shares, and the 2021–2022 Adequacy of Supply filings and IGP outage-assumption tables behind Table 6.1.
- **A solar-premium sweep** presented as a curve (the solved pv15/pv17 LNG cells give two points; the curve fills the range).
- **A supplement figure on the LNG demand shift** — pre-war and wartime demand paths (IEA, Wood Mackenzie) against the U.S. export-capacity ramp (EIA), documenting the buyers'-market context in Section 8's closing observation.

v2 — the next, more refined model

- **A zonal model of the Oʻahu grid** — from single-node to multi-zone, to price the transmission upgrades a large buildout requires and the offsetting value of distributed generation and storage.
- **Slope screening extended and refined** — the Flat/Moderate/Steep cost tiers (0–15/15–20/20–30% slope at ×1.00/×1.05/×1.10), which all reference-land scenarios already carry, extended to the land-constrained inventory and refined to 5-percentage-point slope bins. Refine land screens with more detail about specific parcels as information is collected.
- **Solar refinements.** Account for aspect and capacity-factor adjustments for more highly sloped parcels. Possibly consider alternative agrivoltaic systems (e.g., vertically situated double-sided panels, which suit some types of crop production and capture early and late sunlight), and fixed panels to increase capacity per acre. Allow each parcel to have

different solar configuration: the present model offers only tracking arrays, so a fixed-tilt option, which uses less land but mixes poorly with farming and suits smaller sites, is not available to it.

- **Onshore wind re-screened under current Honolulu rules.** The model caps total onshore wind at 150 MW — the existing plants (Kahuku, Kawailoa, Nā Pua Makani) plus modest headroom — a blunt stand-in for the county setback restrictions that have effectively halted new projects. The cap binds: every solved scenario builds to it. v2 replaces the cap with a parcel-level wind screen under the current ordinance, parallel to the solar screens, so the wind resource is priced rather than assumed away.
- **Land requirement by vintage.** Acres per megawatt is now a single number applied to the whole horizon. Panel efficiency is still improving, so a project built in 2045 should need less land than one built in 2030. Making the requirement vary with install year, alongside the fixed-tilt option above, matters mainly in land-constrained cases, where the screen binds and the density assumption changes what gets built rather than only what gets reported.
- **Multi-day and climate-stress reliability.** Develop synchronized wind, solar, and demand data for many years to better assess reliability and optimal capacity expansion under extreme or unusual conditions and any needed capacity adjustments. The longer series links the high-resolution 2007–2008 record to NREL's NOW-23 meteorological dataset (Covelli et al. 2024 use NOW-23 directly and count an average of 93 low-wind days a year on Oʻahu, with a worst run of 16 straight days).
- **Real-time pricing and inter-day storage.** Consider sample weeks instead of sample days, or full 8760, and gains from variable pricing under selected scenarios.
- **Parking-lot and other canopy structures.** Assess solar potential for canopy structures over parking lots, on buildings, and over public and private walkways.

The model, inputs, code, and every number here are public. We invite specific, sourced challenges to any input or finding — from Hawaiian Electric, HSEO, JERA, and every other reader. The findings can be replicated from the code and data provided, and assumptions adjusted to run new scenarios. If you lack the computing resources or solver, make a request here: we will solve any reasonable, feasible scenario and publish what we find. To be reflected in version 1, comments should arrive by September 15, 2026 (tentative); later suggestions inform v2.

Appendix A — technical notes

A.1 Dollar-year and valuation convention

All figures in this report are real 2024 U.S. dollars, discounted to a 2027 present-value date at the 3 percent real social discount rate. Each cost is rebased to 2024$ from its own source year (ATB 2024 from 2022$; the JERA proposal from ~2026$; fuel series from their pipeline year), so the model carries one dollar unit throughout and no post-hoc scaling is applied. Capital is amortized at the 6 percent regulated cost of capital over asset life; the resulting payment streams are discounted at 3 percent. These conventions and every constant behind them are in the repository conventions file, and `verify_claims.py` re-derives each headline input from the vendored sources on a bare clone.

Cents per kilowatt-hour. Per-kilowatt-hour figures are a present-value ratio: the present value of the cost difference over the present value of delivered electricity, both discounted to 2027 at the same 3 percent rate (discounting only the dollars would understate the per-unit cost). Delivered energy — served load of about 6.8 TWh per year rising to 7.4 by 2050 — has a present value of 135.6 TWh over 2027–2050 (about 198 undiscounted), so a $1 billion cost difference equals about 0.74 cents per kilowatt-hour. Numerator and denominator come from the same solved run.

A.2 Capacity-expansion modeling versus lifecycle cost analysis

A capacity-expansion model solves jointly for what to build and how to dispatch it, letting resources substitute across classes (solar for gas, storage for peakers, geothermal for baseload). A lifecycle cost analysis compares specified configurations within a class, holding the rest fixed. Both are internally consistent; they answer different questions. HSEO's LCCA answers "which fuel is cheaper in a given new plant"; this report's framework answers "whether to build the plant at all." The findings do not contradict each other.

A.3 Discount rates: 3 percent social, 6 percent regulated

The 3 percent real social rate values total welfare over the horizon and is the model's objective-function rate. Section 6 additionally reports cost-recovery arithmetic at the utility's ~6 percent regulated return (real — a conventional nominal rate would add expected inflation, about 2 percent, to this number), because who-pays questions are rate-setting questions. A stream amortized at 6 percent and re-discounted at 3 percent has a present value above its overnight cost; that is why forcing a project into the build can cost more in NPV than its capital alone, as we find for the Waiau retrofit.

A.4 Fuel share of generating cost, year by year

At the final 0.1% solve (no-new-plant, reference oil), fuel is the following share of total annual system cost: 43 percent (2027 period), 38 percent (2030), 19 percent (2035 and 2040), 5 percent (2045), 4 percent (2050). These shares use total system cost as the denominator: the fuel bill starts near two-fifths of costs and nearly vanishes as the mandate binds.

A.5 Sample-day design and the most difficult day

Each investment period is represented by 13 days at 2-hour resolution. Twelve are K-means representatives of the 2007–2008 solar/wind/load record, weighted to reproduce period totals. The thirteenth is the record's most difficult day, found by a one-time production-cost dispatch of a candidate system against every hour of 2007–2008 with unserved energy priced at $5,000/MWh; the day the system came closest to failing — November 22, 2008 — is added at full weight to every subsequent run. The design enforces feasibility on the historical worst day; it does not test events outside the record (Section 8). Storage state-of-charge resets between sample days, and demand-side flexibility (≈10 percent of load reschedulable within-day, plus partially flexible EV charging) operates within but not between days — both conservative for the renewable-heavy paths.

A.6 HSEO workbook mechanics

The decomposition behind Section 4.5, with workbook and page citations: the $1.32B "Fuel Cost Savings" line (LCCA Calculator, 2030–2044 LNG window); the $514M "Avoided Hydrogen Capital Costs" credit and its role in Alternative 1A's $651M NPV versus 2A's $137M; the hydrogen/biodiesel scenario assignment ($40.66 vs $63.84/MMBtu in 2050, ~3,100–3,640 vs ~2,930–3,410 GWh/yr from 2045); the ~56/44 efficiency-versus-price split of the fuel saving; the implied-Brent inconsistency between the LSFO and LNG tracks; and the Engage-priced full-scenario fuel gap ($8–11B undiscounted, ~70 percent of it from the post-2044 fuel assignment); all figures are HSEO's own.

A.7 EGS demonstration arithmetic

A 10 MW demonstration costs about $100M gross at the reference capital trajectory ($10M/MW), $147M at the high; at a 50 percent federal cost share, Hawai'i's net exposure is about $50M / $74M. At an 85 percent capacity factor the plant delivers ≈74.5 GWh per year — $5.2M per year at a conservative 7 cents per kilowatt-hour, a 30-year present value of $102M at the 3 percent social rate ($72M at 6 percent, Appendix A.3). That energy value covers the reference exposure at either rate, and the high exposure fully at 3 percent and mostly at 6. The downside is bounded near zero to modest tens of millions, against the $0.56–1.0 billion system saving if the resource proves out (Section 3).

A.8 JERA capital and infrastructure treatment

The plant is priced from the proposal (p.30: $1,510M/500 MW, exclusions quoted in Section 4.2; p.29: the +20 percent case) and solved at both bounds; the $460M import infrastructure is recovered once, through the fuel-supply tier's fixed charge, verified equivalent to amortizing the infrastructure at 6 percent over the tier life. HECO's 2016 planning figure ($4,050/kW nominal, 152 MW unit) is treated as corroborating context for small-unit costs only.

Part-load fuel curve. The plant burns fuel at 6.92 MMBtu/MWh at full output, an assumption the fleet record confirms; below full output its average rate rises. The curve is derived from EPA continuous-emissions monitoring records for 408 mainland combined-cycle combustion turbines in the F-class capacity band (140–260 MW nameplate, in service 2002 or later), selected through the EPA–EIA crosswalk and fitted on steady operating hours over 2022–2024. The fleet's realized full-load heat rate is 6.895 MMBtu/MWh, within 0.4 percent of the figure the model already carried; its median no-load fuel share is 10 percent and its median minimum stable load 52 percent of maximum. Applied to the model's 125 MW blocks, that gives a no-load draw of 86.9 MMBtu/h, an incremental rate of 6.225 MMBtu/MWh, and average rates of 7.62 at minimum load, 7.15 at three-quarters, and 6.92 at full. Minimum load is set at 50 percent (62.5 MW per block), just below the fleet median and inside its interquartile range; JERA's proposal states no block-level turndown figure, and its flexibility claims rest on the simple-cycle portion of the hybrid, which the four-block commitment structure already represents. Selection criteria, per-unit fits, and provenance are in `sources/epa_cems/`.

A.9 Local-employment coefficients

Construction and O&M job-year ranges per MW by technology, from Wei, Patadia & Kammen (2010); Rutovitz et al. (2015, 2025 update); NREL JEDI; IRENA annual reviews; USEER; EPA power-sector methodology — local labor only, with manufacturing excluded as imported. Utility PV 5–7 construction FTE-yr/MW and 0.10–0.20 permanent FTE/MW; batteries 0.2–0.4 FTE-yr per MWh; CCGT 2–4 and 0.06–0.10; FSRU + terminal 500–1,000 one-time job-years and 20–30 permanent. Applied to the trajectories these give the ranges in Section 4.6. The bands are wide by construction; they exclude induced spending (the H-CGE channel of Coffman et al. 2022).

A.10 Upstream methane calculation

The JERA path imports ≈246 million MMBtu of LNG over 2027–2050 (model dispatch). At ≈19.3 kg CH_4 per MMBtu, that is ≈4.74 Mt of methane throughput. CO_2-equivalent added by supply-chain leakage = throughput × leak rate × GWP:

Leak rate	GWP_{100} = 30	GWP_{20} = 82.5
1%	1.4 Mt	3.9 Mt
3%	4.3 Mt	11.7 Mt
6%	8.5 Mt	23.5 Mt

On the current-law base case the combustion ledger gives LNG a ≈1.3 Mt edge (30.0 Mt JERA bare-EPC versus 31.3 no-new-plant), implying break-even leakage of ≈0.9 percent (100-year basis) or ≈0.3 percent (20-year basis) — at or below every measured U.S. basin, so on the marginal-source reading the LNG path is behind on total greenhouse effect.

Break-even leakage depends on the comparator. A cleaner alternative leaves LNG a smaller combustion edge to defend, so the threshold falls. Each row prices the LNG configuration against the no-new-thermal least-cost path carrying the same Waiau decision, at reference oil on the base rooftop trajectory (`analysis/assemble_methane_breakeven.py`).

pathway	imports (M MMBtu)	CH_4 (Mt)	edge (Mt)	100-yr	20-yr
JERA 500 MW, bare-EPC	246	4.7	1.3	0.88%	0.32%
JERA 500 MW, +20% capital	250	4.8	0.8	0.57%	0.21%
JERA 375 MW, bare-EPC	213	4.1	1.1	0.88%	0.32%
JERA 375 MW, +20% capital	214	4.1	0.9	0.75%	0.27%
Conversion, optimized	195	3.8	2.3	2.06%	0.75%
Conversion, no new plant	197	3.8	2.2	1.94%	0.71%
Conversion, HECO configuration	356	6.9	1.9	0.93%	0.34%

The same calculation inside HSEO's own comparison. HSEO sets its LNG case against its oil case rather than against a least-cost clean path, and that oil case burns oil at scale through 2044. LNG's combustion edge over that comparator is therefore much larger — about 5.7 Mt at every solar premium, since both plan cells hold their mixes fixed as solar costs move — and the break-even leak rate correspondingly higher: 3.3 percent on the 100-year basis, nearly flat across the 20, 80, and 104 percent solar premiums (1.2 percent on the 20-year basis). Measured U.S.

supply chains average 2.95 percent, production weighted. So on the comparison most favorable to it, HSEO's LNG emissions advantage survives average U.S. gas — narrowly, and only on the century timescale. Against a least-cost clean path the break-even is 2.5 percent at the study's baseline solar premium — below the U.S. production-weighted average, so the edge does not survive average gas there — and clears it only if solar is very expensive (4.7 to 5.8 percent at the 80 and 104 percent premiums). (Plan cells here are the Section 4.5 pricing cells: mixes pinned to each plan, firm energy delivered as biodiesel in both cases, so the comparison is free of the study's hydrogen-price asymmetry.)

Leak-rate measurements: Sherwin et al. 2024 (*Nature* 627, 328–334): basin loss rates 0.75% (Appalachia) to 9.63% (New Mexico Permian), production-weighted 2.95% across six measured regions. GWP values: IPCC AR6 WG1, Table 7.15 (fossil-origin CH_4: $GWP_{100} = 29.8$, $GWP_{20} = 82.5$); the table uses 30/82.5, and the thresholds are insensitive to the rounding.

A.11 Distributed solar and storage: empirical identification

The distributed-resource assumptions in this report are estimated from three data sources: hourly system demand from FERC Form 714 (Hawaiian Electric planning area, 2006–2024, via the PUDL compilation), hourly solar radiation from the NREL NSRDB (GOES-aggregated v4, island mean over 264 Oʻahu grid cells), and cumulative installed distributed PV and battery capacity compiled from permit and interconnection records (793 MW of PV and roughly 250 MWh of storage at mid-2025; the storage total is a calibration and less certain than the PV total).

A unit-basis caveat applies to every capacity figure here. Neither the permit-record series nor Hawaiian Electric's published totals states a rating basis (DC nameplate versus AC); EIA's utility-level data, which is AC, runs about 12 percent below the matched residential figures — consistent with the reported series being DC nameplate at a typical inverter loading ratio. The ambiguity does not affect the report's projections, because every coefficient is estimated per *reported* megawatt on the same series used to project net load — the basis cancels. It matters only when comparing our megawatts with outside sources, which needs the basis stated on both sides. A pre-battery calibration pins the physical interpretation: on 2013–2019 data, when net metering put nearly all rooftop generation on the meter and batteries were negligible, day-to-day radiation variation identifies a grid-load reduction of 0.79 MW per reported MW at reference irradiance (s.e. 0.03; analysis/18_prebattery_effective_mw.py) — the expected usable AC output of a DC-nameplate megawatt after tilt, aspect, soiling, and inverter conversion. The battery-era 0.61 additionally nets the behind-the-meter wedge and post-net-metering tariffs; the two bracket the tariff regimes.

Two measurement problems come first. The FERC 714 hourly series carries clock misalignments coinciding with its reporting-format changes: 2006–2012 reads one hour early and 2021–2024 one hour late relative to 2013–2020. We detect and correct these on hours distributed technologies cannot affect: the pre-dawn ramp (5–7 am, sun down, batteries at overnight reserve) shifts +0.95 hours in a single step at the 2020–2021 format boundary and stays constant while battery capacity more than doubles — a reporting artifact, removed by a uniform one-hour roll and validated against astronomical solar noon (the midday net-load trough aligns within a few tenths of an hour). Calibrating on battery-inert hours matters: detecting the shift from the evening peak would absorb the real battery reshaping into the correction. Second, 4 am anchors every day: its load carries almost no loading on installed PV or batteries (+0.001 MW per MW; −0.05

MW per MWh), so subtracting it removes demand drift (efficiency, EVs, the economy) without touching the distributed signal.

Rooftop PV and battery effects are then identified from weather variation rather than from installation trends, because installed PV, installed storage, and EV adoption all grew together after 2020 and cannot be separated on trend alone. Day-to-day radiation is exogenous and orthogonal to those trends. Contemporaneous radiation interacted with installed PV identifies the PV response: a reduction in grid load of 0.61 MW per MW installed at midday, consistent across two independent estimators. Same-day midday radiation interacted with installed battery capacity, predicting load after sunset (when PV output is zero), identifies the battery: a sunnier midday charges batteries fuller and lowers that evening's grid load ($t = -4.9$). The estimate implies 0.45 MWh delivered to the evening per installed MWh per day, distributed 19:00–22:00 with peak weight at 20:00. That figure passes the physical bound of 0.62 MWh (round-trip efficiency of 0.86, a 20 percent outage reserve, 90 percent usable capacity), a check the trend-based estimator fails by a factor of five, which is what motivated the weather-based design.

The estimating equation, for load in hour h of day d with the quarter q(d) carrying the installed stocks:

$$L(h,d) - L(4am,d) = \delta(h, season) + b(h)\cdot[GHI(h,d) \times PV(q)] + c(h)\cdot[GHI(midday,d) \times Batt(q)] + f(T(h,d)) + \varepsilon(h,d)$$

where L is metered system load (MW, clock-shift corrected), GHI is island-mean solar radiation (W/m^2), PV(q) and Batt(q) are cumulative installed rooftop capacity (MW) and battery energy (MWh), $\delta(h, season)$ are hour-by-season fixed effects, and f(T) is a quadratic in temperature. The PV response is read from b(h) at midday; the battery response from c(h) at 19:00–22:00, where PV output is zero and the only channel from midday sunshine to evening load is storage. Full derivation, code, and diagnostics: the repository's analysis/ directory.

The gap between what rooftop systems physically generate (about 4.4 MWh per day per MW) and the grid-load reduction they produce (about 3.4) is 24 percent of generation, stable across 2018–2024. This wedge is demand that exists only behind the meter, consumption induced by the solar itself plus storage losses, and it never crosses the meter in either direction. Island-wide it grew from roughly 180 to 290 GWh per year over 2018–2024. Net-load projections in this report net out only the grid-visible fraction.

One further pattern matters for reliability. On very low-radiation days, grid load now runs about 11 MW above what a linear netting model predicts (2021–2024, within-year estimate), where the same test on 2013–2016 data shows the opposite sign. Households that self-supply most of the time draw from the grid when the sun does not appear, and the effect grows with the installed fleet. This offsets roughly a sixth of the demand relief that low-sun days would otherwise provide and is accounted for in the reliability discussion. Remaining caveats: the storage-capacity series scales all per-MWh results, air-conditioning correlates with radiation despite temperature controls, and no cross-island placebo is possible because FERC 714 covers only O'ahu.

A.12 Net-load construction from distributed projections

Projected net load is built from gross load in three steps. First, distributed PV is netted per timepoint using the model's own site-level DistPV capacity factors, so that distributed output, utility-scale solar output, and demand move with the same weather realization in every hour the optimizer sees. This matters for curtailment and firm-capacity sizing: a cloudy timepoint has low rooftop and low utility solar together. A check of whether installs' actual locations change this (weighting the 264 radiation cells by installed MW in each zone) moves the effective capacity factor by under one percent, so the model's island-level profile is used unchanged. Second, the full trajectory is netted at those capacity factors: the pre-2020 stock keeps the PV-only midday shape, and capacity added since carries the battery-reshaped profile. Third, that reshaped profile moves 40 percent of midday output into the evening hours, energy-conserving, a shift calibrated to the reshaping observed in the metered record (A.11's battery estimates identify the same behavior; the 24 percent wedge of A.11 describes where the meter sits relative to generation, and enters the trajectory calibration rather than the netting itself). In the alternative representation used for comparison, distributed PV and batteries are dispatched by the optimizer against gross load, with the fleet's capital charged to the system. The two representations differ in capital accounting as well as behavior, so their raw cost difference does not measure the value of scheduling; the paired experiment of Section 2.7, pinned inside a single representation, does.

Three installed-capacity trajectories are run. The conservative trajectory grows from 800 MW (2027) to 1,000 MW (2050); in gross-build terms this approximately continues the realized 2020–2024 installation rate, because the 2012–2016 build wave retires within the horizon. The trend trajectory reaches 1,560 MW, about 1.5 times the recent realized build rate; we regard it as the most realistic projection absent a policy change to rooftop-solar pricing (Section 2.7). The accelerated trajectory, constructed as twice the trend increment (2,120 MW by 2050), represents the response to unlimited sellback at avoided cost as discussed in the body; its new installs pair 2 MWh of storage per MW of PV, the configuration of a typical 6.5 kW residential system with one 13.5 kWh battery. Federal tax treatment supports the storage-heavy, third-party-owned character of that growth: residential-owned credits ended after 2025 and third-party solar loses eligibility for systems placed in service after 2027, while third-party-owned batteries retain the Section 48E credit through 2033 (Arnold & Porter 2025; RSM 2025).

A.13 Rooftop adoption versus Hawaiian Electric's forecasts

The trajectories can be placed against the utility's own forecast record. Hawaiian Electric's 2016 Power Supply Improvement Plan projected Oʻahu distributed PV reaching 770 MW in 2034; the installed base reached 765 MW at the end of 2024 — a decade ahead, within eight years of publication. Projected additions over the forecast's first eight years were 27 MW per year against 41 realized; its steady phase (2021–2045) projected 11 MW per year against 42 realized over 2020–2024, with no deceleration in the record. (The digitized PSIP series anchors to the realized 2016 base within 0.7 percent.) The utility's current IGP forecast is closer — a forward rate near 29 MW per year — though still below the realized rate; its published series bundles territories and storage, preventing a clean Oʻahu-only comparison. Against this record, the conservative trajectory's net growth (8.7 MW per year) sits below even the PSIP steady-phase rate that adoption exceeded several times over, and the accelerated trajectory assumes a gross rate twice the recent realized one. Full series and provenance: `analysis/appendix_forecast_vs_actual.csv`, `analysis/APPENDIX_forecast_evidence.md`.

A.14 Oil-price cases from futures and options

The report uses four oil-price cases (Figure A.14). The reference is the EIA-anchored path used throughout earlier drafts, kept because readers and agencies expect it. The market's central expectation, the Brent futures strip, runs 25 to 50 percent below that reference in every period, and the strip is the second case.

How much weight the reference deserves is worth stating carefully. Oil prices are the least accurate quantity EIA publishes. Across its own retrospective of AEO1994 through AEO2022 Reference cases, the average absolute error on the constant-dollar imported crude oil price is 45.6 percent, against roughly 9 percent for energy consumption and electricity prices (EIA 2022). We do not claim a general upward bias, because the sign of that error follows the price cycle rather than being fixed: EIA ran low through the 2000s run-up and high afterward (EIA 2009; Alquist, Kilian and Vigfusson 2013), and past bias has not predicted future bias (Kaack et al. 2017).

The more specific concern is with the far end of the projection. Our reference path rises in real terms, from about $90 per barrel in 2027 to about $107 by 2050, while the futures market prices a decline to the low $50s. A rising real path is what theory once predicted: under the Hotelling rule, the rent on a depletable resource should grow at the rate of interest. The evidence for that trend in resource prices is weak. Reviewing 34 empirical studies, Livernois (2009) concludes the data do not strongly support the rule's predictions. Tests on long price series find them difference-stationary, meaning they behave like random walks without a deterministic upward trend, so past prices carry no information that the level will turn up (Berck and Roberts 1996). Technical progress in extraction has repeatedly outrun depletion. This bears directly on our own construction: we hold the market band flat in real terms beyond the last listed contract for the same reason, and consistency requires saying that the escalator in the reference path is a modeling convention rather than an empirical regularity. Three narrower empirical statements point the same way. The path we carry sits about 15 percent above EIA's own AEO2025 Reference Brent. The AEO2015 through AEO2019 Reference cases over-projected the real imported crude price in 18 of 25 comparisons at one-to-five-year horizons. And EIA's long-horizon oil-price forecasts do not beat a simple no-change benchmark over the two-to-eight-year range where the LNG decisions sit (Bernard et al. 2018). We therefore treat the reference as a plausible high-side case rather than a best estimate.

We carry the futures strip not because futures forecast better than EIA does, but because they are the price at which the market will actually transact, set by participants with money at risk. The forecasting evidence is mixed. Alquist, Kilian and Vigfusson (2013) find futures no better than a no-change forecast at short horizons and clearly inferior at multi-year ones, while Reeve and Vigfusson (2011) find they beat a random walk by a considerable margin when spot and futures diverge sharply, which describes the steeply backwardated curve we observe. What no method reliably beats is the no-change forecast itself, which is why we hold the band flat in real terms beyond the last listed contract rather than extending a trend. One further caveat belongs on the futures path. It is a risk-neutral price, not a physical expectation, so part of the gap between it and the EIA reference may be a risk premium rather than a difference in beliefs. Estimates of that premium disagree on sign and magnitude, and Baumeister and Kilian (2016) find no statistically significant average premium at any horizon and that correcting for one worsens real-time forecasts, so we apply no correction and simply note the distinction. For the band we follow EIA's own practice of inverting NYMEX option prices for implied volatility, a method also used by the Federal Reserve Board and the Bank of England (Ryan and Lidderdale 2009). The low and high cases are the market's 10th and 90th percentiles, computed from the futures strip and option-implied volatility: each contract's percentile is the futures price times $\exp(\pm 1.2816 \cdot \sigma \sqrt{T})$, with σ taken from quoted implied volatilities where they exist and held at 0.30 beyond the last liquid tenor (the one assumed number in the construction; sensitivities are reported in the method note). Nominal futures convert to real 2024 dollars using market inflation expectations, the TIPS breakeven curve, matched to horizon. Percentiles are fitted smoothly across contract dates, held flat in real terms beyond the last listed contract (January 2035) on the view that oil prices are near a driftless random walk at long horizons, and averaged over each model period.

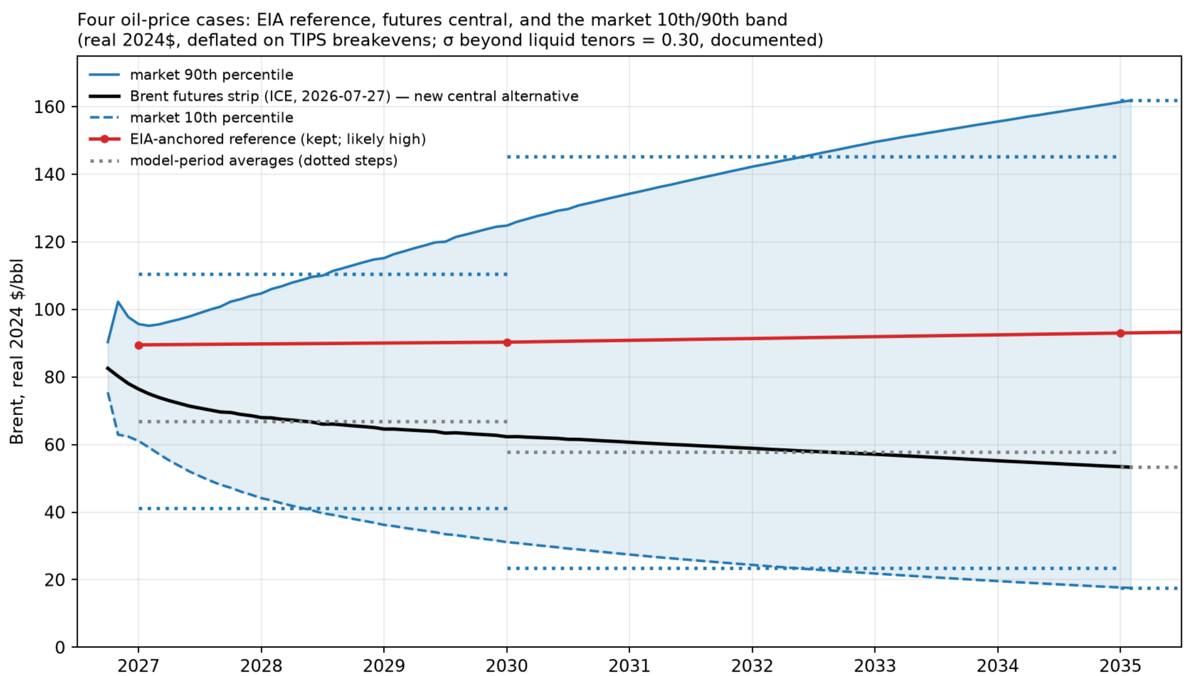

Figure A.14 — The four Brent price cases used throughout the report. Real Brent (2024$ per barrel): the EIA reference (deflated to 2024$), the Brent futures strip (the central market case), and the market 10th and 90th percentiles derived from the implied volatility of Brent options, converted to real dollars with TIPS breakeven inflation and held flat in real terms beyond the last liquid contracts. The model uses period averages of these paths, translated to LSFO and LNG prices through the report's fuel-price linkages. Quote dates, raw data, and calculations are vendored in build/market_band and sources/market.

Period averages, real 2024 dollars per barrel of Brent: the 10th percentile runs \$41 (2027–29), \$23 (2030–34), then \$18; the futures path \$67, \$58, then \$53; the EIA reference \$90 to \$107; the 90th percentile \$110, \$145, then \$162. Two properties are worth noting. The band is asymmetric around the futures path in dollars but symmetric in logs; option markets show a strong upside (call) skew at short tenors today, but that is war pricing concentrated in the next several months, and the smile is unobservable at planning horizons, so we use at-the-money volatilities and say so. And option-implied distributions are risk-neutral: they embed insurance premia, so the band is somewhat wider than a pure probability band, which is the conservative direction for stress-testing. Every market quote used, its source, and its date (2026-07-27), along with the reconstruction script, are in the repository (sources/market/, build/market_band/).

Two properties of the fuel mapping deserve note, since Brent itself never enters the model. Brent shocks pass through the link estimated in the authors' earlier fuel-price brief, which puts low-sulfur fuel oil at 0.7388 times Brent plus $37.30 per barrel, and prices LNG on the indicative contract at 11.8 percent of Brent. Those two pass-through rates are nearly identical in energy terms, about 0.118 dollars per MMBtu for each dollar per barrel of Brent, so the spread between the two fuels is almost the same in every oil case: $6.56 per MMBtu at the market 10th percentile in 2027 against $6.61 at the 90th. The per-unit conversion saving, from burning gas instead of oil in existing plants, therefore barely depends on the oil price; what moves total conversion savings across oil worlds is the volume burned and the clean investment cheap gas defers (Section 4.1). The second note is a limit on the linear link outside its estimation range. At the low end of the band the fitted intercept dominates, implying fuel oil at nearly three times the crude price, which overstates the delivered price in a crude collapse and so understates how cheap fuel becomes. The contract formula for LNG, applied literally, likewise carries no price floor, while a real contract almost certainly would. Both approximations make new gas capacity look better in the low case than it probably would be. A further limit applies to holding any contract formula fixed for twenty years. Agerton (2017) shows that oil-indexed LNG contracts get renegotiated when the formula price and the prevailing spot price diverge, and that these revisions appear as statistically identifiable structural breaks in the pricing relationship. Contract terms are not a physical constant, and the party with the weaker outside option tends to absorb the adjustment.

Earlier drafts used the EIA AEO 2025 case spread for the low and high paths; those inputs and results are archived (*_aeo.csv, aeo_archive/) for comparison. The AEO-cased fan opened too slowly to represent near-term oil-price risk, which is concentrated exactly where the LNG decisions live.

A.15 Plan pricing methodology

The Section 4.5 price tags come from scenarios whose generation is constrained to each published plan's mix. This appendix documents the constraint set and its carve-outs.

Sources and rescaling. IGP plan mixes come from the November 2023 Supplemental Response Tables 2-3 (the land-constrained plan of record, which that document labels "preferred") and 2-4 (the base scenario, which it labels "alternate") — the utility's plan of record rather than its RESOLVE model output; HSEO mixes from the study's own results worksheets. Customer-sited solar rows are excluded and each plan's remaining categories are converted to shares of its grid supply, then multiplied by the served demand of the rooftop family whose customer-solar assumptions that plan tracks (land-constrained on accelerated, base on conservative, HSEO on trend). A plan is thereby held to its mix, not to its demand forecast.

The quota set, per planning period 2030–2050. Utility solar: a two-sided band at 0.98–1.02 times the plan's level. Wind: the same band on combined onshore-plus-offshore, with a floor (no ceiling) under offshore alone — the plans carry 257–287 MW of onshore against the county-setback screen's 150 MW, so a ceiling on offshore specifically would forbid the only substitute for onshore the model may not build. Fossil: a 0.95–1.05 band in 2035 and 2040, a floor only in 2030 (see carve-outs), nothing from 2045, where the clean-energy requirement binds instead. Firm clean energy: from 2045, a floor at 0.98 times the plan's combined biofuel-plus-hydrogen level, counted as hydrogen fuel-cell output plus generation in the multi-fuel thermal fleet — which at the 100 percent requirement is necessarily renewable fuel. There is no ceiling on firm energy; it is the one category left open, and cells that retire thermal capacity late can exceed the plan's level (the IGP cells run 600–900 GWh of biodiesel above their plans' own 2050 levels for this reason — the residual our grid cannot place elsewhere surfaces here). 2050 quotas hold each plan's 2045 shares, rescaled to 2050 demand; the 2027–2029 window is unpinned because the plans publish no anchor there.

Carve-outs. Kalaeloa is exempt from the fossil quota (`--plan-quota-fossil-exempt`): the plans close it around 2033 while our model carries its power-purchase minimum-take as a floor dispatch cannot go below, so its plant is excluded from the constrained set and its generation stripped from the IGP fossil targets (KPLP rows). The refinery cogens are must-run baseload in this framework, identical in plan and reference cells, and sit outside the quota for the same reason. Biofuel *quantities* before 2045 are unconstrained.

What the tag means. Because the firm floor holds each plan to the firm clean energy it says it needs, the tag prices the plan as published — not the cheaper build a model would choose if allowed to substitute solar for the plan's biofuel. Dispatch, storage sizing, and every unnamed category remain optimized, so the tag is still generous to the plan wherever the plan is silent. The design was validated two ways: an elastic variant of the quota module (slack on every row, minimizing total violation) proves each quota set feasible before any cell is solved, and a fidelity audit confirms solved cells reproduce their plans' pinned categories within the band. Cross-family cells (the Box 4.2 head-to-head) apply one plan's shares to the other plan's rooftop trajectory, so the two IGP plans can be compared serving the same demand; a cell read against the wrong trajectory's reference mixes demands that differ by a quarter and is not a price tag.

Tolerance. Plan cells and their least-cost references are all solved to 0.1 percent MIP gap. Because the quota bands leave the dispatch some freedom, a cell's cumulative emissions are more sensitive to the solve tolerance than its cost is; the figures here are from the 0.1 percent solutions.

Appendix B — references

Vendored sources carry a sha256 in `SOURCES.md`; items marked (*vendored*) are held in `sources/` and hashed there. Paywalled journal articles are cited with DOI only and are not redistributed.

Peer-reviewed literature

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Additional peer-reviewed citations whose exact locators the authors confirm at release: Bernard, Kaack, and Ryan & Lidderdale (oil-price forecast accuracy, detailed in `analysis/EIA_FORECAST_ACCURACY.md`); Coffman et al. (2022, §A.10); Fripp (2020) and the Switch-Hawaiʻi design; Grue et al. (2020, HECO distributed-solar characteristics, archived in this repository); Majer et al. (2012, induced-seismicity protocol, §3).

Government, agency, and institutional reports

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- Hawaiʻi Gas (2016). "The Facts About LNG for Hawaiʻi." January 2016. (*vendored*)
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sources/eia_price_components/); *Annual Energy Outlook 2025* (*vendored*); *Annual Energy Outlook Retrospective Review* (2022); *Today in Energy*, April 16, 2026.
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- Hawaii News Now (2024). Coverage of the January 8 rolling outages and the July 31–August 2 Kalaeloa outage, Oʻahu.
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- Engineering News-Record (ENR), *POWER*, *Honolulu Star-Advertiser*, *Utility Dive*, *The Narwhal*, and *Maui Now*, as cited in Sections 4.2 and 4.4 for the public cost record and the 2016 FortisBC contract termination.

Companion work and data

- Roberts, M. J. Oʻahu solar-and-wind land-use study. https://github.com/mikejrob/solar-wind-landuse.
- Data sources (FERC Form 714 via PUDL; NREL NSRDB; EPA CEMS; permit and interconnection records) are documented in Appendix C and `sources/`.

Appendix C — data and reproducibility

This report is release pre-v1.03: the single-node (copper-plate) model with the rebuilt distributed-solar treatment of Appendices A.11–A.12, issued as a preliminary version open for comment (requested by September 15, 2026, tentative). After the comment period, responses, and revisions, the report will be locked as version 1 — the version of record — and subsequent changes will be limited to documented errata (v1.01, v1.02, ...). Suggestions beyond errata will be directed to v2, the regional (zonal) grid model described in V2.md. An earlier working paper, since withdrawn, preceded this series; the changes relative to it are documented in docs/CORRECTIONS.md.

The model is Switch 2.0.9 with CPLEX, solved on the University of Hawaiʻi's Koa cluster. The public repository contains the complete inputs, the build scripts that regenerate them from vendored primary sources, the scenario definitions, the solve scripts, and `verify_claims.py`, which re-derives every headline input from the vendored sources and fails loudly on any mismatch. Scenario results are aggregated in `results/RESULTS_SUMMARY.csv` (0.1 percent optimization tolerance, with a handful of degenerate cells at 0.15 percent documented in docs/HARD_CELLS.md).